Ramsey Alignment and the Postwar Tax System

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How far is the tax system from optimal, and how much does it matter? I develop a sufficient statistic that answers both questions: the sine of the angle between the welfare and revenue gradients of the tax system. When the two are aligned, the system is Ramsey-optimal. Applying the framework to the postwar U.S. tax system, I find that the income tax system was heavily misaligned in the 1950s, when a one percentage point reform in the optimal direction would have improved welfare by 0.5% of GDP, but reached near-optimality by the 1980s. Reductions in effective corporate tax rates drove essentially all of the convergence. This result rests on measuring the full matrix of elasticities of taxable income in dynamic general equilibrium; partial equilibrium estimates understate the gains by an order of magnitude. However, extending the framework to include tariffs reveals persistent remaining misalignment, with the optimal direction pointing toward somewhat higher tariffs, consistent with recent shifts in tax policy.

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1 Introduction

Since the end of World War II, policymakers have enacted sweeping reforms to the tax system. Many such reforms have been motivated by dynamic general equilibrium forces, such as the effects of taxation on capital accumulation and wages. But reform is only warranted if the tax system is far from optimal. How far it is, and how much the gap costs, remains an open question. I develop a sufficient statistics metric that answers it.

I start from a simple insight: in any model with a unique equilibrium, the Ramsey optimum features alignment between the government's welfare and revenue objectives. In practice, that means taxes are set such that the marginal welfare cost per dollar of revenue raised is equalized across tax instruments. This condition is captured geometrically by the degree of alignment between the welfare gradient and the revenue gradient, which capture the marginal welfare cost and fiscal consequences of each instrument, respectively. The cosine of the angle θ between the gradients is a sufficient statistic for the proximity of the tax system to the Ramsey optimum, equaling one when the system is optimal, while the sine of that angle measures the share of welfare gains available from revenue-neutral reform. At the optimum when $\sin\theta = 0$ and $\cos\theta = 1$, the marginal value of public funds (MVPF) is equalized across instruments, meaning that the framework is an aggregate diagnostic for the MVPF (Finkelstein and Hendren 2020).[1]

To get at the economics behind the geometry, consider the canonical neoclassical model in which a government may tax the labor and capital income of a representative household with time-separable isoelastic preferences. In that model, the fiscal externality of capital taxes is so large that the optimal tax on capital is zero. That emerges in part because the supply of capital is perfectly elastic, which makes both the cross-base and the own-base elasticity of capital income large. By contrast, the labor tax has small dynamic and cross-base spillovers. Away from the optimum, a positive capital tax makes the gradients diverge and $\cos\theta$ falls below one. In the Chamley (1986) or Chari, Nicolini, and Teles (2020) models, $\sin\theta = 0$ exactly when the tax on capital is zero, and the tax burden falls on labor.

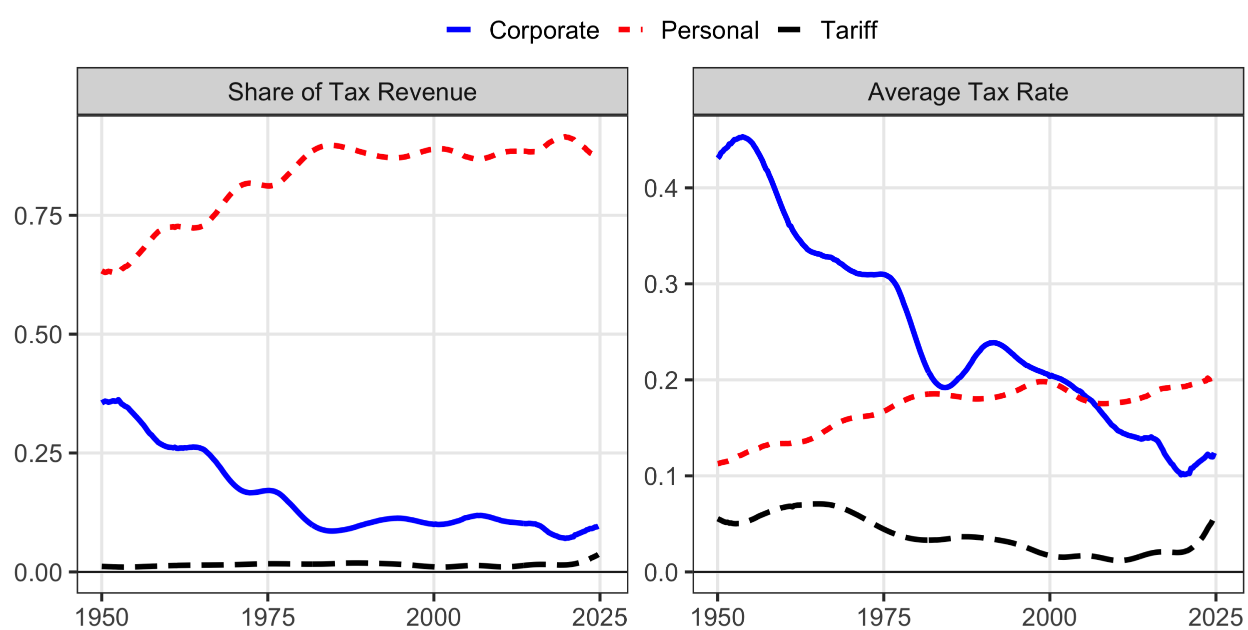
Distance from the optimum tells policymakers how much is at stake. A second question is whether their reforms moved in the right direction. Building on the classical theory of marginal tax reform Dixit (1975, 1979) and Tirole and Guesnerie (1981), I show that the optimal revenue-neutral reform is the orthogonal projection of the welfare gradient onto the revenue-neutral hyperplane. I extend this result to dynamic economies where the revenue gradient incorporates intertemporal spillovers and cross-base fiscal externalities. Comparing any actual reform to this optimal direction, the cosine of the angle ϕ between them measures whether policymakers moved

1. However, the alignment metrics remain well-defined even when individual instruments are near or beyond the peak of their Laffer curves, whereas the MVPF becomes infinite or negative.

toward or away from efficiency. When $\cos\phi = 1$, the reform moves in the right direction; when $\cos\phi = -1$ it moves in the opposite. In the Chamley (1986) setting, marginal reductions in the corporate tax rate away from the optimum always feature $\cos\phi = 1$. The empirical question is the degree to which the estimated revenue gradient exhibits this pattern in practice, and whether historical reforms moved toward or away from alignment.

The postwar transformation of the federal tax system provides a natural setting for this framework. At the end of World War II, corporate and personal income taxes together accounted for nearly all of federal revenue, while the tariff share of federal tax revenue was negligible. Until 2018, almost all reform happened on the income tax margin, with the corporate share falling from nearly 50 percent to below 15 percent and personal taxes absorbing the difference. Since then, tariff increases in 2018 and 2025 brought the customs share of tax revenue to nearly ten percent—well above the typical postwar share of 2-3 percent. Those developments prompt two questions. First, was the shift from corporate to personal taxation welfare-improving? Second, how big of a role should tariffs play in the federal tax system?

Figure 1: Changes in Postwar Sources of Revenue and Tax Rates



Note: The right panel assumes that tariffs, personal taxes, and corporate taxes make up 100% of tax revenue. In reality, they sum to just over 90%. All series are trend components from the two-sided Christiano and Fitzgerald (2003). I strip out all components with a frequency less than forty quarters.

The answer to both questions depends on measuring the welfare and revenue gradients. The welfare gradient follows from a straightforward application of the envelope theorem. Under equal incidence weights, the welfare gradient is simply the vector of present-valued tax bases adjusted for the domestic incidence of the tax (Finkelstein and Hendren 2020). In that benchmark, the welfare gradient is observable from national accounting data. By contrast, the revenue gradient is more challenging: it requires estimating how total government revenue responds to tax changes, accounting for behavioral responses across all bases and over time. I estimate these dynamic general equilibrium elasticities using narrative tax shocks, cumulating discounted impulse responses from local projections and vector autoregressions. Static taxable income elasticities estimated from cross-sectional data miss both dynamic responses and cross-base spillovers. My estimates embed both.

To evaluate whether it was welfare-improving to shift the burden of taxation away from corporations, I identify the dynamic own- and cross-base elasticities of taxable income for labor and corporate income using exogenous narrative personal and corporate tax shocks from Mertens and Ravn (2013b) and Cloyne et al. (2022). I find that corporate income is highly elastic with respect to both personal and corporate tax shocks, while labor income is not. Because the fiscal externalities from both taxes fall disproportionately on the corporate base, the alignment metric indicates the system was heavily misaligned when corporate rates were high. Indeed, I find that $\sin\theta$ fell from around 0.6 in the early 1950s to nearly zero by the 1980s, indicating the tax system moved from substantial misalignment to near-optimality. The welfare stakes were large: in the early 1950s, a 1 percentage point reform in the optimal revenue-neutral direction would have improved welfare by approximately 0.5% of GDP. By the 2020s, the same reform would yield only 0.05% of GDP. As a result, every major income tax reform since 1980—including the Economic Recovery Tax Act of 1981, the Tax Reform Act of 1986, and the Tax Cuts and Jobs Act of 2017—has a welfare contribution on this margin indistinguishable from zero. The trajectory from misalignment to near-optimality is robust to a wide variety of tests.

The increase in measured alignment is driven entirely by dynamic, general equilibrium channels. Partial equilibrium elasticities surveyed by Vergara and Swonder (2025) would understate the increase in efficiency by an order of magnitude, because they generate small fiscal externalities and therefore find little to gain from reform even when corporations bore a large share of the tax burden. Under partial equilibrium, the postwar tax system appears roughly efficient throughout. Under general equilibrium, it was massively misaligned in the 1950s and gradually corrected. These are qualitatively different stories about postwar fiscal policy. The general equilibrium elasticities explain why the tax reforms of the early postwar era were welfare-improving.

But did policy drive the improvement? A decomposition attributes changes in alignment to two channels: tax wedges, which policy directly controls, and revenue composition, which reflects both policy and structural shifts in the economy. Tax policy changes account for essentially all of the convergence. Reductions in effective corporate tax rates—through investment credits, accelerated depreciation, and statutory rate cuts—lowered the fiscal externality on corporate taxation until it equalized with the personal tax margin. Shifts in the composition of the tax base, including the rise of pass-through entities, contributed little. The practical implication is that there is little left to gain from reforming the income tax mix.

But if there is little left to gain on the income tax margin, gains may remain on others. To evaluate whether there are gains from adding tariffs to the mix, I estimate the three-by-three matrix of taxable income elasticities by adding narrative tariff shocks from Besten and Känzig (2026). The inner corporate-personal tax elasticity block is unaffected by this addition, and I find that there are modest spillovers from tariffs to personal and corporate income while the own-elasticity of tariffs is large. Compared to the two-instrument baseline, the trajectory of $\sin\theta$ is virtually unchanged until 1980, but $\sin\theta$ remains persistently around 0.2 thereafter. That implies a one percentage point change in the optimal reform direction would be worth 0.15 percent of GDP today. The optimal reform direction has pointed toward higher tariffs since the 1970s. That motivates a timely question: would there be gains to swapping tariff revenue for corporate income tax revenue, which is exactly what policy did in 2025? Existing theory supports this possibility: Alessandria et al. (2025) finds positive welfare gains to positive tariffs if they pay for investment subsidies, similar to what actually happened in 2025. I find that such a swap would reduce remaining misalignment by approximately 25 percent, though the posterior uncertainty is substantial. What is robust is that the income-tax-only analysis misses a margin on which the postwar tax system appears persistently misaligned, and the direction of improvement consistently points toward somewhat higher tariffs.

Related Literature Existing approaches to evaluating tax composition face a fundamental trade-off between capturing general equilibrium forces and avoiding parametric disagreement. Structural general equilibrium models capture both cross-base spillovers and dynamic accumulation, but require parametric commitment that leads to fundamentally different conclusions. For example, a subclass of neoclassical models would conclude the postwar shift toward personal taxation was optimal (Chari, Nicolini, and Teles 2020), while overlapping generations models reach the opposite conclusion (Conesa, Kitao, and Krueger 2009). Even within the neoclassical framework, the optimal tax on capital—and hence the optimal direction of reform—depends sensitively on preference specifications and heterogeneity (Straub and Werning 2020). Sufficient statistics approaches avoid this model dependence by working directly with causal elasticities (Chetty 2009; Saez and Stantcheva 2018; Hendren 2016; Kleven 2021). However, existing implementations typically rely on static own-base elasticities estimated from micro data (Saez 2001; Hendren and Sprung-Keyser 2020), missing both the cross-base spillovers and dynamic accumulation that are first-order for evaluating tax composition.[2] Both approaches typically characterize global optima, which implicitly define optimal reform directions from any given baseline.[3] Their parametric disagreements over where the optimum lies therefore translate into disagreements over which direction to reform.

The theoretical foundations for this paper lie in the classical literature on marginal tax reform. Dixit (1975) showed that whether a revenue-neutral tax swap improves welfare depends on how demand responds to the price changes it induces, with the answer encoded in the matrix of compensated demand derivatives. Guesnerie (1977) characterized the full set of welfare-improving swaps, and Tirole and Guesnerie (1981) identified the best one: the reform that reduces welfare cost fastest while holding revenue fixed.[4] Ahmad and Stern (1984) took the framework furthest empirically, applying it to Indian commodity taxes. But they still needed to estimate how demand for every good responds to every price, and their implementation was necessarily static. I extend the projection result to dynamic economies with cross-base fiscal externalities, and require far less structure to do so: the welfare gradient reduces to incidence-weighted tax bases by the envelope theorem, and the revenue gradient requires only present-value elasticities of each tax base with respect to each tax rate, estimated from causal variation in tax rates without specifying preferences or production. The alignment metrics address a remaining limitation: when many welfare-improving reforms exist, they summarize how much scope for improvement remains in a single scalar.

The empirical implementation builds on the marginal value of public funds literature's insight that fiscal externalities can be estimated from causal evidence (Hendren 2016; Finkelstein and Hendren 2020). I estimate these fiscal externalities using narrative identification, constructing present-value revenue gradients from impulse responses to tax shocks. This builds on McKay and Wolf (2023) and Caravello, McKay, and Wolf (2024), who show that impulse responses are sufficient to construct Lucas-critique-robust counterfactuals under alternative policy rules; my application is narrower, evaluating whether actual tax changes moved toward or away from the locally optimal direction rather than simulating counterfactual policies. The geometric approach—testing whether welfare and revenue gradients are aligned—also parallels Barnichon and Mesters (2023), who evaluate monetary policy by testing whether the gradient of the loss function with

2. Other limitations arise even within the Diamond and Mirrlees (1971) model from which the sufficient statistics approach stems. Bhandari, Borovička, and Yao (2025) point to robustness concerns, echoing Mankiw, Weinzierl, and Yagan (2009) that the Diamond and Saez (2011) result depends critically on the unknown distribution of types. Saez, Slemrod, and Giertz (2012) argue, as this paper does, that it is critical to account for spillovers between instruments—in their case, between capital gains and labor income.

3. Vergara and Swonder (2025) incorporate cross-base fiscal externalities from corporate taxes on the labor base in an optimal corporate tax formula, but their framework is static and characterizes the global optimum rather than evaluating marginal reforms from the status quo.

4. Diewert (1978) and Dixit (1979) extend the framework to many-consumer economies.

respect to policy shocks is zero; both frameworks detect suboptimal policies from sufficient statistics without requiring a fully specified structural model.

Roadmap. The remainder of the paper proceeds as follows. Section 2 develops the projection framework for static revenue-neutral reform, showing that aggregate gradients are sufficient statistics and deriving the geometric solution. Sections 3 and 4 implement the framework empirically for corporate and personal taxes, estimating present-value revenue and welfare gradients from narrative shocks and evaluating the complete history of postwar federal reforms. Section 5 adds tariffs and Section 6 concludes.

2 The Geometry of Reform

This section formalizes the projection framework for evaluating multi-instrument tax reforms. The framework shows that the optimal direction of revenue-neutral reform depends on two sufficient statistics—the welfare gradient and the revenue gradient—which can be estimated from reduced-form evidence without specifying a full structural model. Despite the generality of the setup, the framework yields a simple geometric characterization: the optimal reform is the orthogonal projection of the welfare gradient onto the revenue-neutral hyperplane, and system efficiency is measured by a single scalar: the alignment between welfare and revenue objectives. This alignment metric is simply the cosine of the angle between the two gradients.

Section 2.1 introduces notation and derives the welfare and revenue gradients. Section 2.2 establishes the projection formula and shows the revenue gradient is a sufficient statistic. Section 2.3 develops scalar metrics that measure proximity to the Ramsey optimum and the scope for revenue-neutral reform. Section 2.4 specializes to the two-instrument case used in the empirical application and demonstrates how partial equilibrium approaches can prescribe reforms in the opposite direction from the optimum.

2.1 Setup: Taxes, Bases, and Gradients

Policy environment. Let the government choose tax rates $\{\tau_{i,t}\}_{i=1,\dots,n}^{t=0,\dots,T}$ across n instruments and $T+1$ periods. Stack these into a policy vector:

$$\boldsymbol{\tau} := (\tau_{1,0}, \dots, \tau_{n,0}, \tau_{1,1}, \dots, \tau_{n,T})^\top \in \mathbb{R}^{n(T+1)}.$$

Tax bases $B_{i,t}(\boldsymbol{\tau})$ depend on the entire policy sequence, and they respond to tax rates for standard economic reasons: behavioral responses grow with rates, and revenue may eventually decline. Importantly, there are spillovers across bases and across time. A change in $\tau_{i,t}$ affects not only base i at date t but also other bases at other dates. For example, a corporate tax cut today raises investment, growing the capital stock and expanding future labor tax bases through production complementarity. The government discounts its stream of revenue at β, so present-value revenue is:

$$R(\boldsymbol{\tau}) := \sum_{t=0}^{T} \beta^t \sum_{i=1}^{n} \tau_{i,t} B_{i,t}(\boldsymbol{\tau}). \tag{1}$$

I assume that aggregate tax base responses are well-behaved.

Assumption 1. *$R(\boldsymbol{\tau})$ is continuously differentiable in $\boldsymbol{\tau}$ in a neighborhood of $\bar{\boldsymbol{\tau}}$.*

This is an assumption about aggregate equilibrium responses, not individual behavior. It requires the equilibrium to be unique and to vary smoothly with policy. Standard neoclassical, heterogeneous agent, and overlapping generations models all satisfy this condition. The key exclusion is economies with discontinuous equilibrium responses, such as models with multiple equilibria where small tax changes trigger discrete jumps. Given differentiability, define the present-value elasticity

$$\varepsilon_{ik} := \sum_{s=0}^{T} \beta^s \frac{\partial \ln B_{i,s}}{\partial \ln(1 - \tau_{k,0})} \tag{2}$$

measuring the cumulative discounted response of base i to a reform of tax k. Entries with $i = k$ capture own-base responses; entries with $i \neq k$ capture cross-base spillovers. Both include effects that accumulate over time—through capital deepening, human capital investment, or anticipatory adjustments—as well as contemporaneous responses. The object is well-defined for any reform persistence; Section 3 describes how to normalize estimates from transitory shocks into permanent-equivalent elasticities, bridging a gap between the public finance literature, which typically defines elasticities with respect to permanent reforms, and the macroeconomics literature, which estimates impulse responses to shocks that decay over time.

The revenue gradient. The revenue gradient $r := \nabla_{\boldsymbol{\tau}} R \in \mathbb{R}^{n(T+1)}$ measures how present-value revenue responds to marginal tax changes. Consider raising $\tau_{i,t}$ by a small amount. There are three effects on present-value revenue: a mechanical revenue gain from taxing the existing base, an own-base behavioral loss as base i shrinks, and spillovers to other bases. Formally, define discounted bases $\tilde{B}_{i,t} := \beta^t B_{i,t}$. Then:

$$r_{i,t} = \tilde{B}_{i,t} \left[1 - \frac{\tau_{i,t}}{1 - \tau_{i,t}} \varepsilon_{ii} - \Lambda_{i,t} \right] \tag{3}$$

where $\Lambda_{i,t}$ aggregates all cross-base spillovers:

$$\Lambda_{i,t} = \frac{1}{1-\tau_{i,t}} \sum_{k \neq i} \tau_{k,t} \frac{\tilde{B}_{k,t}}{\tilde{B}_{i,t}} \varepsilon_{ki}. \tag{4}$$

The three components have clear economic interpretations. The mechanical effect (+1) is how much revenue would rise absent any behavioral response. The own-base effect (ε_{ii}) captures how base i contracts when its tax rises—including both the contemporaneous response and accumulated future losses from reduced investment or human capital. The spillover term ($\Lambda_{i,t}$) captures how other bases respond—for instance, a corporate tax increase that erodes the labor base through production complementarity. Because the present-value elasticities already aggregate dynamic responses, the revenue gradient formula has the same structure whether spillovers operate within a period or across decades. These spillovers can be decomposed by when revenue effects occur:

$$r_{i,t} = \underbrace{\beta^t \frac{\partial R_t}{\partial \tau_{i,t}}}_{\text{Contemporaneous}} + \underbrace{\sum_{s>t} \beta^s \frac{\partial R_s}{\partial \tau_{i,t}}}_{\text{Persistence}} + \underbrace{\sum_{s<t} \beta^s \frac{\partial R_s}{\partial \tau_{i,t}}}_{\text{Anticipation}}. \tag{5}$$

The contemporaneous term captures revenue effects at date t itself. The persistence and anticipation terms capture intertemporal spillovers: how tax changes at t affect revenue at other dates. Both are embedded in $\Lambda_{i,t}$. Two examples illustrate the economic channels underlying these spillovers.

Example 1: Human capital accumulation (persistence, own-base). Consider an increase in the labor tax at $t = 0$. Workers respond by reducing human capital investment (education, training), as the after-tax return to skill accumulation has fallen. This shrinks the labor tax base $B_{L,t}$ in all future periods:

$$\frac{\partial B_{L,t}}{\partial \tau_{L,0}} < 0 \quad \text{for all } t > 0. \tag{6}$$

The persistence terms in equation (5) capture these effects. Because the revenue losses accumulate over time, the present-value revenue gradient $r_{L,0}$ is smaller than the contemporaneous effect alone would suggest. Badel, Huggett, and Luo (2020) and Kleven et al. (2025) show that accounting for human capital accumulation substantially raises the long-run own-base elasticity of labor income relative to static micro estimates, underscoring why present-value elasticities are the appropriate object for evaluating tax reform.

Example 2: Capital accumulation (persistence, cross-base). A corporate tax increase at $t = 0$ reduces investment, shrinking the capital stock over time. If capital and labor are complements in production ($F_{KL} > 0$), the reduced capital stock lowers the marginal product of labor (wages), eroding the labor tax base in future periods:

$$\frac{\partial B_{L,t}}{\partial \tau_{K,0}} < 0 \quad \text{for all } t > 0.$$

The present-value revenue gradient for the capital tax includes this cumulative cross-base effect:

$$r_{K,0} = \beta^0 \frac{\partial R_0}{\partial \tau_{K,0}} + \sum_{t=1}^{T} \beta^t \left(\frac{\partial R_{K,t}}{\partial \tau_{K,0}} + \frac{\partial R_{L,t}}{\partial \tau_{K,0}} \right),$$

where the second term in the sum captures the cross-base spillover. This channel—production complementarity linking corporate taxes to labor revenue—is the primary supply-side motivation for corporate tax cuts (Lucas 1990).[5]

Welfare. Let $W(\boldsymbol{\tau})$ denote the money-metric social welfare cost of the tax system, measured in present-value dollars. Higher W is worse. We measure welfare by the present-value compensating variation aggregated across agents using social welfare weights $\{\omega_g\}$.

Assumption 2 (Welfare regularity)**.** *$W(\boldsymbol{\tau})$ is continuously differentiable in $\boldsymbol{\tau}$ in a neighborhood of $\bar{\boldsymbol{\tau}}$.*

The welfare gradient $g := \nabla_{\boldsymbol{\tau}} W \in \mathbb{R}^{n(T+1)}$ has a simple characterization. By the envelope theorem applied to individual optimization, the first-order welfare cost of raising $\tau_{i,t}$ is the mechanical burden on taxpayers, discounted to date zero:

$$g_{i,t}(\bar{\boldsymbol{\tau}}) = \beta^t \tilde{\gamma}_{i,t} B_{i,t}(\bar{\boldsymbol{\tau}}). \tag{7}$$

Each piece has an economic reason. The discount factor β^t appears because the burden lands at date t. The base $B_{i,t}$ appears because that is what the tax is levied on. The incidence share $\tilde{\gamma}_{i,t} := \sum_g \omega_g s_{g,i,t}$ appears because that is who bears it, where $s_{g,i,t}$ is group g's share of the incidence from instrument i at date t.

The key to this result is that it applies the envelope theorem to individual optimization, not social optimization. Each agent takes the tax sequence as given and optimizes over their lifetime—labor supply, savings, human capital, organizational form. At that private optimum, the agent's behavioral response to a tax perturbation has zero first-order welfare effect. When there are multiple distorted margins, behavioral responses to $\tau_{i,t}$ do change tax payments on other bases

5. A third channel is anticipation: forward-looking agents adjust behavior today in response to announced future reforms, so that $\partial B_{K,t} / \partial \tau_{K,5} \neq 0$ for $t < 5$. These effects mean the present-value revenue cost of a reform exceeds the contemporaneous calculation. Mertens and Ravn (2012) exploit announcement timing to estimate these effects directly.

and at other dates. But these are transfers between the agent and the government: they affect revenue (entering through $\Lambda_{i,t}$ in the revenue gradient) without creating additional welfare costs beyond the mechanical burden. The welfare gradient captures who bears the tax; the revenue gradient captures what the behavioral responses cost the fisc. The separation is exact to first order.

This separation requires that the taxes in the framework are the only wedge between private prices and social resource costs (Hendren 2016). When behavioral responses interact with other distortions—taxes outside the system, non-tax wedges like market power or externalities, or binding constraints—the welfare gradient requires additional terms beyond the tax bases.[6] The framework as implemented treats federal income taxes as the relevant distortions, maintaining the same assumption as the MVPF literature (Hendren 2016; Hendren and Sprung-Keyser 2020). This assumption is most defensible for the early postwar period, when state and local taxes were smaller and the economy was relatively closed, and becomes more tenuous as other tax margins grow in importance.

Under equal-incidence weights—$\omega_g = 1$ for all g, so that the planner values a marginal dollar of income equally regardless of the recipient—$\tilde{\gamma}_{i,t} = \sum_g s_{g,i,t} = 1$ for domestic taxes and the welfare gradient simplifies to discounted tax bases:

$$g_{i,t} = \beta^t B_{i,t}. \tag{8}$$

This is directly observable from national accounts data. Equal-incidence weights coincide with utilitarian welfare ($W = \sum_g u(c_g)$) only when utility is linear in income; with concave utility, utilitarian weights are $\omega_g = u'(c_g)$, which places greater weight on lower-income claimants. The equal-incidence benchmark is appropriate here because the framework evaluates the composition of revenue between two broad bases—personal and corporate income—not the distribution of burdens within either base. Section 4.2 shows results are nearly insensitive to the relative weight ω_C on corporate income claimants, the margin along which this choice has bite. It also cleanly separates the empirical challenge: the welfare gradient g can be constructed from baseline data, while the revenue gradient r must be estimated.

6. For example, behavioral responses to federal income tax changes create fiscal externalities on state and local government budgets; a corporate tax cut that stimulates production in a polluting industry has welfare costs beyond the mechanical tax burden; and agents facing binding borrowing constraints are not at interior optima, so behavioral responses can have first-order welfare effects.

The marginal value of public funds. Pairing the welfare and revenue gradients yields the marginal value of public funds for each instrument-date pair:[7]

$$\text{MVPF}_{i,t} = \frac{g_{i,t}}{r_{i,t}} = \frac{\tilde{\gamma}_{i,t}}{1 - \frac{\tau_{i,t}}{1-\tau_{i,t}}\varepsilon_{ii} - \Lambda_{i,t}}.$$

Note that this definition embeds welfare weights in the numerator through $\tilde{\gamma}_{i,t}$. In the notation of Hendren and Sprung-Keyser (2020), who define the MVPF as a purely positive object—willingness to pay over net cost—the optimality condition is that welfare-weighted MVPFs $\omega_g \cdot \text{MVPF}_{i,t}$ are equalized across instruments. With welfare weights absorbed into g, the equivalent condition here is simply that MVPFs are equalized across instruments and dates (Appendix H.1 derives this formally). When MVPFs differ, which is the typical case empirically, there exists scope for welfare-improving revenue-neutral reform. Section 2.2 characterizes the optimal direction of such reforms.

2.2 Policy as Projection

The problem of reform arises when MVPFs differ across instruments. When this occurs, there exist directions of tax adjustment that improve welfare while holding total revenue fixed. The planner's local problem is to identify the reform direction $d\boldsymbol{\tau}$ that maximizes the first-order welfare gain subject to revenue neutrality:

$$\max_{d\boldsymbol{\tau}} \quad -g(\bar{\boldsymbol{\tau}})^\top d\boldsymbol{\tau} \quad \text{subject to} \quad r(\bar{\boldsymbol{\tau}})^\top d\boldsymbol{\tau} = 0. \tag{9}$$

The welfare gradient g measures how marginal tax changes affect social welfare; the revenue gradient r measures how they affect government revenue, accounting for all behavioral responses including cross-base spillovers. This constrained optimization problem has a simple geometric solution: the optimal reform direction is the orthogonal projection of the welfare gradient onto the revenue-neutral hyperplane.

Lemma 1 (Gradient projection)**.** *Let $\bar{\boldsymbol{\tau}}$ be a baseline with welfare gradient $g(\bar{\boldsymbol{\tau}})$ and revenue gradient $r(\bar{\boldsymbol{\tau}}) \neq 0$. The optimal revenue-neutral reform direction is*

$$d\boldsymbol{\tau}^* = -\alpha \frac{P_{\perp r} g}{\|P_{\perp r} g\|}, \tag{10}$$

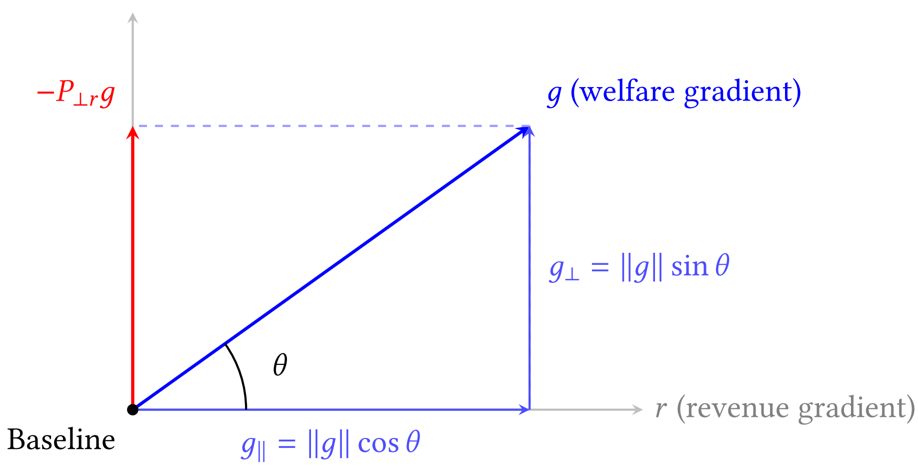
where $\alpha > 0$ is the reform magnitude in percentage points and $P_{\perp r} := I - r(r^\top r)^{-1} r^\top$ is the projection

7. See Mayshar (1990), Hendren and Sprung-Keyser (2020), and Bastani (2025) for a more detailed analysis of the MVPF.

operator onto the hyperplane orthogonal to r. The formula characterizes the optimal direction of reform; α is a free parameter governing the step size, not an optimized quantity.

The derivation is in Appendix H.2. Tirole and Guesnerie (1981) construct a process of revenue-neutral reform based on gradient projections for linear commodity taxes. Dixit (1979) independently derives the same geometric structure—the optimal local welfare improvement is the projection of the welfare gradient onto the feasible set—in many-consumer economies. The contribution here is twofold. First, the revenue gradient incorporates intertemporal spillovers and cross-base fiscal externalities, so the projection aggregates dynamic general equilibrium responses that those static frameworks do not capture. Second, Section 2.3 extracts scalar alignment metrics from the projection that measure how far the tax system is from optimal—an object neither framework provides.[8]

Figure 2: Geometric Decomposition of Optimal Policy



Note: The welfare gradient g (blue) decomposes into a parallel component $g_{||} = ||g|| \cos\theta$ along the revenue gradient r and an orthogonal component $g_\perp = ||g|| \sin\theta$ perpendicular to r. The optimal revenue-neutral reform direction is $-P_{\perp r} g$ (red), which points opposite to the perpendicular component of welfare cost.

Figure 2 illustrates this decomposition. The welfare gradient g (blue) splits into a parallel component $g_{||} = ||g|| \cos\theta$ along the revenue gradient r and an orthogonal component $g_\perp = ||g|| \sin\theta$ perpendicular to r. If we could move freely along g—ignoring the revenue constraint—we would achieve the steepest welfare improvement. However, moving along g would change total revenue, violating revenue-neutrality. The projection operator resolves this tension by extracting

8. The projection framework nests several classical results as special cases. At a Ramsey optimum, $g = \mu r$ and the projection vanishes: no revenue-neutral reform can improve welfare. Chamley's (1986) zero capital tax emerges when the present-value revenue gradient for capital collapses to zero, driving $\text{MVPF}_K \to \infty$. Barro's (1979) tax-smoothing result emerges when the government's borrowing rate equals the social discount rate, equalizing MVPFs across time as well as instruments. These are not separate derivations; they are limiting configurations of the same two gradients. Appendix J develops these connections. Appendix I extends the projection to deficit-financed reforms.

the perpendicular component, yielding $P_{\perp r} g$ (red) as the optimal revenue-neutral reform direction. Because the revenue gradient is the only behavioral object in the projection formula, welfare conclusions follow from the estimated gradients regardless of the specific channels through which spillovers operate.

2.3 Alignment Metrics

The geometry in Figure 2 depends entirely on the angle θ between the welfare and revenue gradients. Existing characterizations of tax optimality focus on MVPF equalization, which answers a binary question—equal or not—without measuring how costly the deviation is. The angle θ fills this gap: $\Delta W = \alpha \|g\| \sin\theta$ converts the MVPF gap into the welfare gain available from a reform of size α in the optimal direction, an object that comparing MVPFs across instruments cannot deliver. Define system alignment as:

$$\cos\theta = \frac{g^\top r}{\|g\| \, \|r\|} \in [-1, 1]. \tag{11}$$

This is the standard inner product formula for the angle between vectors.

Proposition 1 (System alignment and revenue-neutral potential)**.** *The efficiency of the tax system is characterized by two scalar metrics:*

1. ***System alignment**: $\cos\theta$ measures proximity to the Ramsey optimum. When $\cos\theta = 1$, welfare and revenue objectives are perfectly aligned and the system is optimal. When $\cos\theta = 0$, the gradients are orthogonal and inefficiency stems purely from tax composition. When $\cos\theta < 0$, raising taxes increases welfare cost but reduces revenue.*
2. ***Revenue-neutral potential**: $\sin\theta$ measures the share of potential welfare gains achievable through revenue-neutral reform. The first-order welfare gain from a reform of size α (in percentage points) in the optimal direction is*

$$\Delta W \approx -\alpha \|g\| \sin\theta. \tag{12}$$

Proof. The optimal reform direction is $d\tau^* = -\alpha \frac{P_{\perp r} g}{\|P_{\perp r} g\|}$, where α is the reform magnitude in percentage points. The first-order welfare change is $\Delta W = g^\top d\tau^* = -\alpha \|P_{\perp r} g\| = -\alpha \|g\| \sin\theta$. □

Because this is a first-order approximation, it requires only the gradients evaluated at the baseline $\bar{\boldsymbol{\tau}}$—not how elasticities or fiscal externalities vary with the magnitude of reform. Those second-order effects (how r itself shifts as $\boldsymbol{\tau}$ moves) would matter for evaluating large discrete reforms but are absent from the local welfare calculation by construction.

For intermediate values, the metrics have quantitative bite. When $\cos\theta = 0.5$, for example, $\sin\theta \approx 0.87$—nearly nine-tenths of potential welfare gains are achievable revenue-neutrally. Negative values ($\cos\theta < 0$) indicate that raising taxes increases welfare cost but reduces total revenue; this occurs when cross-base spillovers dominate own-revenue effects. Appendix Table H.1 summarizes. While $\cos\theta$ measures how far the system is from optimal, $\sin\theta$ measures what can be done about it. When $\cos\theta \geq 0$—as holds throughout the postwar United States—the two metrics are monotone transformations of each other, so either fully characterizes alignment. Later, the empirical analysis reports $\sin\theta$ because it has the natural scaling: zero at the optimum, increasing in the scope for improvement, and mapping directly to welfare gains through equation (12).

The revenue-neutral potential $\sin\theta$ reflects a fundamental decomposition: the welfare gradient g splits into a component parallel to the revenue gradient (which revenue-neutral reform cannot address) and a perpendicular component (which it can). Because $\sin\theta$ depends only on the perpendicular component, it is maximized when the gradients are orthogonal ($\theta = 90^\circ$) and declines symmetrically toward zero as θ approaches either 0° or 180°. The limiting case $\sin\theta = 0$ is the geometric counterpart of the optimality criterion in Saez and Stantcheva (2016): at the optimum, no budget-neutral perturbation increases the weighted sum of gains and losses, precisely because the welfare gradient has no component perpendicular to the revenue gradient. Whether the system is moderately misaligned on the right side ($\theta = 60^\circ$) or the wrong side ($\theta = 120^\circ$) of the Laffer curve, the scope for revenue-neutral improvement is the same.

Together, $\cos\theta$ and $\sin\theta$ fully characterize the efficiency of tax composition—but they also reveal what composition cannot address. The perpendicular component $\|g\|\sin\theta$ measures the welfare gains available from revenue-neutral reform. The parallel component $\|g\|\cos\theta$ measures the welfare stake in the level of taxation—gains that require changing total revenue, not just its mix. As the system approaches alignment ($\cos\theta \to 1$), the perpendicular component vanishes and the entire welfare gradient lies along the revenue direction. The framework cannot say whether taxes should be higher or lower—that requires comparing the cost of revenue to the value of spending—but it can quantify how large that question is. It also requires only local information. Assumption 1 asks that the revenue function be differentiable at the baseline $\bar{\boldsymbol{\tau}}$; it does not require that the elasticities embedded in the gradients remain stable under counterfactual policies. If a reform is large enough to change the elasticities, the welfare gain formula (12) ceases to be exact—but the alignment diagnostic remains valid as a characterization of the current system. The framework answers “is the current tax mix efficient?” and “which direction should the next marginal reform go?,” not “what would welfare be at the global optimum.” This locality allows the analysis to remain agnostic about whether elasticities are policy-invariant, sidestepping the parametric disagreements that divide structural models computing global optima.

Beyond measuring system efficiency, the framework evaluates whether historical reforms moved in the right direction. For an observed reform $d\boldsymbol{\tau}^{\text{actual}}$, we compute its directional alignment with the optimal reform $d\boldsymbol{\tau}^{*}$:

$$\cos\phi = \frac{(d\boldsymbol{\tau}^{\text{actual}})^{\top} d\boldsymbol{\tau}^{*}}{\|d\boldsymbol{\tau}^{\text{actual}}\|\,\|d\boldsymbol{\tau}^{*}\|}. \tag{13}$$

This statistic inherits the same interpretation as $\cos\theta$. When $\phi \approx 0^{\circ}$ ($\cos\phi \approx 1$), the actual reform moved nearly parallel to the optimal direction. It was well-designed given the economy's position. When $\phi \approx 90^{\circ}$ ($\cos\phi \approx 0$), the reform was orthogonal, neither helping nor hurting. When $\phi > 90^{\circ}$ ($\cos\phi < 0$), the reform moved opposite to the welfare-improving direction, making the system less efficient. Together, $\cos\theta$ (system position) and $\cos\phi$ (reform direction) provide a complete characterization of tax policy.[9]

The projection formula also enforces the classic intuition of MVPF equalization. Rewriting equation (10) component-wise:

$$d\tau_i^{*} \propto -g_i + \left(\frac{r^{\top}g}{r^{\top}r}\right) r_i = -r_i \left(\frac{g_i}{r_i} - \frac{r^{\top}g}{r^{\top}r}\right).$$

Define $\text{MVPF}_i = g_i/r_i$ and the revenue-weighted average $\overline{\text{MVPF}} = (r^{\top}g)/(r^{\top}r)$:

Corollary 1 (MVPF equalization)**.** *The optimal reform direction satisfies*

$$d\tau_i^{*} \propto -r_i(MVPF_i - \overline{MVPF}). \tag{14}$$

The projection automatically lowers taxes on instruments with above-average MVPFs and raises taxes on instruments with below-average MVPFs. That MVPFs are equalized at the optimum is a standard implication of the Ramsey first-order conditions (Jacobs 2018; Bastani 2025). The geometric derivation adds a quantitative dimension: when MVPFs are *not* equalized, the angle θ measures how costly the deviation is, converting the MVPF gap into the welfare gain $-\Delta W = \alpha\|g\|\sin\theta$ available from reform in the optimal direction—an object that comparing MVPFs across instruments cannot deliver. The advantage is sharpest when an instrument is near or beyond the peak of its Laffer curve: the MVPF becomes infinite or negative, rendering pairwise comparisons undefined, while $\cos\theta$ remains well-defined and continuous.

9. The reform evaluation statistic $\cos\phi$ is related to the welfare effectiveness index of Raimondos-Møller and Woodland (2014), who use the cosine between a proposed tariff reform and the welfare gradient to rank reforms in a small open economy. The system-level diagnostic $\cos\theta$, measuring alignment between welfare and revenue gradients as separate objects, is new.

2.4 Specializing to Two Instruments

The empirical application focuses on personal (L) and corporate (C) income taxes. In this case, the alignment condition reduces to a simple requirement: the marginal fiscal externality from each tax must be equal. This externality depends on three observable quantities—tax wedges, revenue composition, and behavioral elasticities including cross-base spillovers—which jointly determine how far the system is from optimal and what drives changes in alignment over time.

Fiscal externalities and alignment decomposition. Corollary 1 shows that the optimal reform equalizes MVPFs across instruments. Specializing to personal (L) and corporate (C) taxes under equal dollar weights ($g_{i,t} = \tilde{B}_{i,t}$), MVPF equalization requires $r_L/\tilde{B}_{L,t} = r_C/\tilde{B}_{C,t}$. From equation (3), this ratio equals

$$\frac{r_i}{\tilde{B}_{i,t}} = 1 - \eta_i \varepsilon_{ii} - \Lambda_{i,t}$$

where $\eta_i = \tau_i/(1-\tau_i)$ is the tax wedge on instrument i. The spillover term $\Lambda_{i,t}$ from equation (4) involves base ratios, which can be converted to revenue ratios using $\tilde{B}_{j,t}/\tilde{B}_{i,t} = (R_j/R_i)(\tau_i/\tau_j)$. Substituting:

$$\Lambda_{L,t} = \eta_L \cdot \frac{R_C}{R_L} \cdot \varepsilon_{CL}, \qquad \Lambda_{C,t} = \eta_C \cdot \frac{R_L}{R_C} \cdot \varepsilon_{LC}.$$

Collecting terms, define the *effective elasticity* for each instrument:

$$\tilde{\varepsilon}_L = \varepsilon_{LL} + \frac{R_C}{R_L}\varepsilon_{CL}, \qquad \tilde{\varepsilon}_C = \varepsilon_{CC} + \frac{R_L}{R_C}\varepsilon_{LC}.$$

The effective elasticity captures both the own-base response and the revenue-weighted spillover from the other base. Then

$$\frac{r_i}{\tilde{B}_{i,t}} = 1 - \eta_i \tilde{\varepsilon}_i \equiv 1 - A_i$$

where $A_i = \eta_i \tilde{\varepsilon}_i$ is the *marginal fiscal externality*—the total revenue leakage (as a share of the mechanical revenue gain) from raising tax i. MVPF equalization reduces to $A_L = A_C$: alignment requires equalizing marginal fiscal externalities across instruments.

This representation connects to the classical Ramsey intuition. The inverse elasticity rule states that optimal tax rates should be inversely proportional to elasticities. With cross-base spillovers, the relevant object is not the own-elasticity ε_{ii} but the effective elasticity $\tilde{\varepsilon}_i$, which incorporates the fiscal externality imposed on other bases. A tax instrument with large positive spillovers (high ε_{ji}) has a higher effective elasticity than its own-base response would suggest, calling for a lower tax wedge.

Corollary 2 (Decomposition of alignment)**.** *In the two-instrument case, changes in system align-*

ment can be attributed to three policy levers: (i) tax wedges η_i, (ii) revenue composition R_C/R_L, and (iii) behavioral elasticities ε_{ij}. Holding elasticities fixed, alignment improves when tax wedges and revenue shares adjust to equalize fiscal externalities $A_i = \eta_i \tilde{\varepsilon}_i$ across bases.

The corollary decomposes alignment changes into observable components. Consider an economy that moves from misalignment ($A_L \neq A_C$) toward alignment ($A_L = A_C$). This convergence can occur through three channels. First, *tax wedges*: if one instrument has an excessive fiscal externality ($A_L > A_C$), reducing the wedge η_L directly lowers A_L. Second, *revenue composition*: the revenue ratio R_C/R_L enters the effective elasticities, so shifts in the tax base mix change both $\tilde{\varepsilon}_L$ and $\tilde{\varepsilon}_C$. Third, *behavioral elasticities*: if the underlying elasticities ε_{ij} change over time—due to globalization, financial innovation, or changing industrial structure—the alignment condition shifts even with fixed policy. Section 3 quantifies the contribution of each channel to postwar alignment improvements.

Why general equilibrium matters. The effective elasticity $\tilde{\varepsilon}_i$ incorporates cross-base spillovers through the terms ε_{CL} and ε_{LC}. What happens if a planner ignores these and uses only the own-base elasticity? Define the PE revenue gradient by setting $\Lambda_{i,t} = 0$:

$$r_{i,t}^{\mathrm{PE}} = \tilde{B}_{i,t}\left[1 - \frac{\tau_{i,t}}{1-\tau_{i,t}}\varepsilon_{ii}\right]. \tag{15}$$

This corresponds to using only the own-base elasticity ε_{ii}, ignoring how each instrument affects the other's base. If capital and labor are complements in production, a higher corporate tax reduces investment, shrinking the capital stock and lowering wages, which erodes the personal tax base. Vergara and Swonder (2025) formalize this wage-incidence channel in an optimal corporate tax formula, showing it creates a strategic complementarity between corporate taxes and in-work transfers. The present-value elasticities here capture this channel, among others. This spillover means the true revenue gradient for corporate taxes is smaller than the PE estimate:

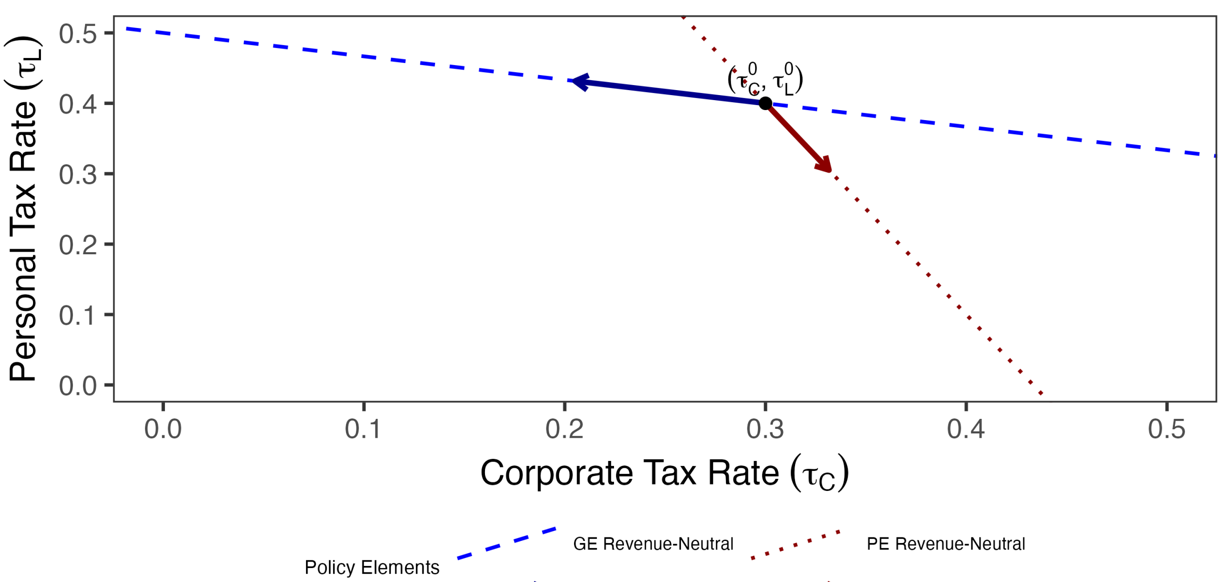
$$r_C^{\mathrm{GE}} = \tilde{B}_{C,t}\left(1 - \eta_C\varepsilon_{CC} - \Lambda_{C,t}\right) < r_C^{\mathrm{PE}} = \tilde{B}_{C,t}\left(1 - \eta_C\varepsilon_{CC}\right),$$

where $\Lambda_{C,t} > 0$ captures the cross-base spillover. The PE planner thinks corporate taxes raise more revenue than they actually do.

Figure 3 illustrates the consequences. The GE revenue-neutral line (blue) and PE revenue-neutral line (red) have different slopes because PE ignores spillovers. The projection formula tells the planner to project the welfare gradient onto the revenue-neutral line—but the PE planner projects onto the wrong line. The result: GE prescribes lowering corporate taxes and raising personal taxes, while PE prescribes the opposite. When cross-base spillovers are large, partial equilibrium analysis can reverse the optimal reform direction entirely.

The example illustrates a general point: the alignment metric $\cos\theta$ is only as good as the revenue gradient used to compute it. The gap between partial and general equilibrium is one dimension along which the revenue gradient can go wrong. The other is dynamics. Most existing MVPF calculations use static partial equilibrium—only own-base contemporaneous responses from micro data. The framework requires the opposite extreme: dynamic general equilibrium elasticities that cumulate cross-base spillovers over time. Between these sit two intermediate cases: dynamic PE, which cumulates own-base responses over time but misses cross-base fiscal externalities, and static GE, which captures contemporaneous cross-base spillovers but misses their compounding through capital accumulation. The ranking between these intermediate cases is genuinely ambiguous—it depends on whether cross-base spillovers or persistence is the larger channel empirically. In the personal/corporate application, the cross-base channel dominates: ε_{CL} is an order of magnitude larger than the own-base terms, and it is this fiscal externality that drives most of the early misalignment. But this ranking need not hold in other settings. Section 3 estimates dynamic general equilibrium elasticities directly, capturing both channels simultaneously.

Figure 3: The Geometric Failure of Partial Equilibrium



Note: Starting from baseline policy (black dot), the partial equilibrium reform (red arrow) projects onto the PE revenue-neutral line (red dashed) and prescribes raising corporate taxes. The general equilibrium reform (blue arrow) projects onto the correct revenue-neutral line (blue dashed) and prescribes lowering corporate taxes.

3 Estimation of General Equilibrium Dynamic Elasticities

As an application of the geometric framework, I evaluate the alignment of the U.S. federal tax system with the Ramsey optimum. Throughout the postwar era, personal and corporate income taxes accounted for over 90% of federal tax revenue, with tariffs contributing a negligible share. Figure 1 shows that until quite recently, most reform happened on the income tax mix, with corporate rates and revenue share falling dramatically and personal taxes rising modestly to pick up the burden. That motivates a retrospective look at the income tax mix, and a prospective analysis of whether other sources of revenue, like tariffs, may have a larger role to play in the federal tax system. This section and Section 4 take up the retrospective. Section 5 takes up the prospective.

3.1 From Theory to Estimation

The key empirical object is the revenue gradient r_j from equation (3), which captures how an exogenous change in tax rate j affects present-value government revenue, accounting for dynamic adjustments and general equilibrium spillovers across tax bases. Specializing to the two-instrument case from Section 2.4:

$$r_j = \tilde{B}_j \cdot \left[1 - \frac{\tau_j}{1-\tau_j}\varepsilon_{jj} - \frac{\tau_k}{1-\tau_j}\frac{\tilde{B}_k}{\tilde{B}_j}\varepsilon_{kj}\right], \tag{16}$$

where $\tilde{B}_j$ is the discounted base and each present-value elasticity ε_{ij} measures the cumulative discounted response of base i to a permanent change in tax j, as defined in equation (2). The welfare gradient under utilitarian weights is:

$$g_j = \tilde{B}_j, \tag{17}$$

since the common annuity factor cancels when computing alignment metrics. The welfare gradient is directly observable from national accounts data. The empirical challenge is estimating the four elasticities ε_{ij} that enter the revenue gradient.

I estimate these elasticities from impulse responses of log tax bases to shocks in log retention rates $\ln(1-\tau_j)$. Cumulating and discounting the impulse responses yields the present-value base response; dividing by the cumulative retention rate persistence $\kappa_j := \sum_{s=0}^{H} \beta^s \cdot \mathrm{IRF}_s(\ln(1-\tau_j))$ converts these into permanent-equivalent elasticities. This normalization is exact when base responses scale linearly with shock size, so that the impulse response to a shock of size a is a times the unit-shock response. The baseline uses $\beta = 0.9926$ (approximately 3% annual discount rate) and $H = 20$ quarters, following the five-year horizon in Mertens and Montiel Olea (2018).

3.2 Data and Specification

The baseline specification is a quarterly VAR with four variables:

$$Z_t = \left[\log\left(1 - \text{APITR}_t\right), \log\left(1 - \text{ACITR}_t\right), \ln B_t^{PI}, \ln B_t^{CI}\right],$$

where APITR_t and ACITR_t are the average personal and corporate income tax rates, and B_t^{PI} and B_t^{CI} are the corresponding tax bases. The sample covers 1947Q1–2019Q4 with four lags. Average rates are the theoretically required object: the framework defines revenue as $R_i = \tau_i B_i$, so the revenue gradient requires variation in average rates rather than statutory marginal rates.

The four-variable system is not a parsimony choice but a theoretical requirement. The revenue gradient requires the total derivative of tax bases with respect to tax rates, including effects operating through output, investment, and wages. Adding variables like GDP or government spending would partial out these channels, estimating a conditional elasticity rather than the total elasticity the theory requires. Exogeneity comes from the narrative identification strategy described in Section 3.3, not from the conditioning set. Appendix E develops this argument in detail and shows that a seven-variable specification following Mertens and Ravn (2013a) yields qualitatively similar but less precise estimates.

Following standard practice in this literature, the corporate base is BEA corporate profits, which includes both C-corporations and S-corporations; the implications of this measurement choice are discussed in Section 3.5. As a robustness check, I also estimate the model using adjusted bases that reallocate S-corporation income to the personal base (Appendix F). Appendix A provides detailed variable definitions and data sources.

3.3 Identification

The narrative approach provides the exogenous variation needed to estimate the revenue gradient. I follow the identification strategy developed by Romer and Romer (2010) and extended by Mertens and Ravn (2013a) to distinguish personal and corporate income taxes.

Narrative instruments. Romer and Romer (2010) read the Congressional Record and Treasury documents to classify major federal tax legislation from 1945 to 2007 according to the motivations policymakers articulated at the time. They code a reform as exogenous when the legislative record indicates policymakers pursued objectives unrelated to the contemporaneous business cycle—such as long-run deficit reduction, ideological commitments, or tax simplification—rather than short-run stabilization. Mertens and Ravn (2013a) refine the series by separating personal and corporate tax changes and excluding reforms implemented more than 90 days after announcement to avoid anticipation effects. I extend their series through 2019 using the classification in Cloyne et al. (2022). Each reform receives a quantitative measure equal to the projected revenue impact scaled by the lagged tax base, as announced at the time of enactment. This scaling delivers an approximation to the change in the average tax rate on that instrument. The resulting series are denoted m_t^{PI} for personal income taxes and m_t^{CI} for corporate income taxes.

From instruments to structural shocks. The narrative measures serve as instruments for latent structural tax shocks rather than entering as regressors directly. Let Z_t denote the vector of endogenous variables. A reduced-form VAR produces residuals u_t, which relate to structural shocks ε_t via $u_t = A_0\varepsilon_t$, where ε_t are mutually orthogonal with unit variance and A_0 is the contemporaneous impact matrix. Following Mertens and Ravn (2013a) and Cloyne et al. (2022), the narrative measures identify the columns of A_0 corresponding to tax shocks through the moment conditions:

$$E[m_t\varepsilon_{tax,t}'] = \Phi \neq 0, \quad E[m_t\varepsilon_{other,t}'] = 0$$

where $m_t = (m_t^{PI}, m_t^{CI})'$, $\varepsilon_{tax,t}$ denotes the two structural tax shocks, and $\varepsilon_{other,t}$ denotes all other structural disturbances. The first condition requires the instruments to correlate with the tax innovations; the second requires them to be uncorrelated with non-tax shocks.[10]

Because Congress often adjusts personal and corporate taxes in the same legislation, the two narrative instruments are contemporaneously correlated. This correlation means they identify the two-dimensional subspace spanned by the structural tax shocks but do not uniquely separate personal from corporate shocks within it. To isolate each shock, I apply a Cholesky decomposition to the identified subspace, ordering the shock of interest first. When estimating responses to the personal tax shock, I order the personal tax rate first, imposing that the personal shock has no contemporaneous effect on the corporate rate; when estimating responses to the corporate shock, I reverse the ordering. Each column of the elasticity matrix is estimated from a separate single-shock experiment, and the alignment metric requires only the individual revenue gradients—not a joint structural interpretation of both shocks.

The identification rests on the narrative exogeneity criterion: that the instruments are uncorrelated with non-tax shocks. Excluding reforms implemented more than 90 days after announcement mitigates anticipation effects that operate between enactment and implementation, but does not address anticipation that precedes enactment itself. For example, it was widely expected that President Reagan would pursue tax reform well before specific legislation was introduced, and

10. Cloyne et al. (2022) shows that both instruments are strong.

similarly that President Trump would push for a large corporate tax cut in 2017. If agents adjusted behavior in advance of these reforms, the narrative shock—which is dated at enactment—would be preceded by movements in the tax bases. Figure B.1 provides a direct test by estimating impulse responses at negative horizons $h = -8, \ldots, -1$, which measure whether narrative tax shocks predict *prior* movements in tax bases. Under valid identification, these responses should be zero. Pre-period responses are uniformly small and statistically indistinguishable from zero for all four base-shock pairs, with the dynamic response emerging only at impact.

3.4 Estimation

I estimate impulse responses using the two-stage Bayesian local projection framework developed by Cloyne et al. (2022). The first stage estimates a Bayesian VAR with hierarchical priors following Giannone, Lenza, and Primiceri (2015). Hyperparameters governing overall shrinkage are treated as random and sampled via Metropolis-Hastings. For each draw from the BVAR posterior, I apply the identification procedure described above to recover a draw of A_0. This delivers the causal impact of each tax shock on all variables in the system at horizon zero.

The second stage estimates local projections at each horizon $h \geq 1$:

$$Z_{t+h} = c^{(h)} + B_1^{(h)} Z_{t-1} + \sum_{j=2}^{p} b_j^{(h)} Z_{t-j} + u_{t+h}, \quad \text{var}(u_{t+h}) = \Omega_h \tag{18}$$

where $p = 4$ lags. At horizon zero, the residuals u_t relate to the structural shocks via $u_t = A_0 \varepsilon_t$. The impulse response at horizon h is $IRF_h = B_1^{(h-1)} A_0$. I estimate equation (18) with Bayesian methods using a Minnesota prior and Student-t errors to accommodate fat tails. Appendix B provides complete details on prior specification and the MCMC algorithm.

Bayesian estimation offers three advantages: it propagates identification uncertainty from A_0 into the impulse response posterior, it improves precision at long horizons where LP variance is high (Li, Plagborg-Møller, and Wolf 2024), and it produces posterior draws for straightforward inference on the alignment metrics. I use local projections rather than propagating shocks through a VAR because the present-value gradient requires reliable estimates over extended horizons, where VARs can suffer from misspecification bias due to lag truncation. However, I also show that the main results are robust to a variety of alternative estimation methods including standard local projections, penalized local projections, smooth local projections, Bayesian VARs, and proxy SVARs.

3.5 Present-Value Elasticities

Table 1 reports present-value elasticities under three specifications, all estimated by Bayesian local projections. The baseline specification includes only tax rates and bases. The diagonal tells a familiar story: both own-elasticities are positive, with the corporate base considerably more responsive ($\varepsilon_{CC} \approx 2.6$) than the personal base ($\varepsilon_{LL} \approx 1.0$). The striking feature is the off-diagonal asymmetry. A one percent reduction in personal taxes expands the corporate base by roughly eight percent ($\varepsilon_{CL} \approx 7.6$), while corporate tax cuts have essentially no effect on the personal base ($\varepsilon_{LC} \approx -0.1$). Personal tax policy generates enormous fiscal externalities on corporate revenue; the reverse channel is absent. These estimates are robust across estimation methods within the four-variable specification. Appendix C reports elasticities under frequentist local projections (Jorda 2005), smooth local projections (Barnichon and Brownlees 2019), bias-corrected local projections (Herbst and Johannsen 2024), BVAR and SVAR-IV methodologies, LP-IV, and time-varying BVAR (Primiceri 2005). Where relevant, I add linear trends as a second robustness check. All methods produce similar elasticity estimates, and agree that the cross-elasticity ε_{CL} is large.

Table 1: Present Value Elasticities of Tax Bases with Respect to Retention Rates

	Corporate Tax Shock		Personal Tax Shock	
Specification	ε_{CC}	ε_{LC}	ε_{CL}	ε_{LL}
Baseline	2.64	-0.12	7.55	0.96
	(1.88, 3.64)	(-0.31, 0.08)	(5.73, 9.58)	(0.56, 1.39)
S-Corp Reallocation	2.08	-0.34	7.39	1.39
	(1.46, 2.89)	(-0.50, -0.20)	(4.60, 11.62)	(0.82, 1.93)
Seven-Variable	2.8	0.5	9.3	0.8
	(1.9, 4.3)	(0.2, 0.9)	(5.5, 19.2)	(-0.0, 2.8)

Note: Each elasticity ε_{ij} measures the cumulative discounted percent change in base i per one percent increase in retention rate $(1 - \tau_j)$, with 90% credible intervals in parentheses. All specifications estimated by Bayesian local projections. The baseline includes four variables: personal and corporate tax rates and bases. The S-corp reallocation moves S-corporation income from the corporate to the personal base using IRS Statistics of Income data (Appendix F). The seven-variable specification adds output, government spending, and debt following Mertens and Ravn (2013a). Appendix Tables 1, F.1, and E.1 report estimates across alternative estimation methods.

The baseline corporate base is BEA corporate profits, which includes both C-corporations and S-corporations. S-corporations are pass-through entities taxed at personal rates, so when personal taxes fall, S-corporation activity expands and appears in the measured corporate base even though it does not generate corporate tax revenue. This channel is quantitatively important: S-corporations grew from a negligible share of business income in the 1970s to roughly half by the 2010s (Dyrda and Pugsley 2024; Smith et al. 2022), with growth accelerating after the Tax Reform Act of 1986 lowered the top personal rate below the corporate rate for the first time. To address this concern, I construct adjusted bases that reallocate S-corporation income from the corporate to the personal base using IRS Statistics of Income data (Appendix F). The own-elasticity ε_{CC} falls from 2.6 to 2.1, confirming that the baseline captures some organizational form arbitrage. But the cross-elasticity ε_{CL} is virtually unchanged at 7.4: even when S-corporation income is stripped from the corporate base entirely, personal tax cuts still generate substantial expansions in C-corporation profits. Appendix Table F.1 confirms this finding is robust across estimation methods.

A separate concern is that the four-variable specification omits macroeconomic aggregates that could absorb part of the estimated cross-elasticity. The seven-variable specification adds output, government spending, and debt to the conditioning set, following Mertens and Ravn (2013a). If the baseline cross-elasticity were inflated by omitting these controls, conditioning on them would shrink it. The opposite occurs: ε_{CL} rises to 9.3, with wider credible intervals reflecting the additional parameters. In other words, the four-variable baseline is conservative. Appendix Table E.1 reports the seven-variable estimates across all methods.

How do these estimates compare to the literature? The comparison is somewhat difficult because the elasticities differ conceptually from the taxable income elasticities in the public finance literature. That literature tends to identify static partial equilibrium elasticities, whereas the elasticities here incorporate general equilibrium adjustments in wages, interest rates, and organizational form, cumulated over the medium run. That is why the Saez, Slemrod, and Giertz (2012) survey of elasticities of taxable income finds relatively small estimates ranging from $\varepsilon_{LL} \approx 0.12$ to 0.40. On the other hand, Mertens and Montiel Olea (2018), using narrative identification similar to this study, find an aggregate elasticity $\varepsilon_{LL} \approx 1.2$. Similarly, Vergara and Swonder (2025) surveys estimates of ε_{CC} and finds values in the U.S. are typically less than one—far smaller than my $\varepsilon_{CC} \approx 2.6$. Moreover, the baseline's negligible estimate of the effect of corporate tax cuts on labor income is inconsistent with some partial equilibrium estimates (Garrett, Ohrn, and Suárez Serrato 2020; Curtis et al. 2021). This is especially puzzling because it would require income effects to substantially dominate substitution effects in general equilibrium, though the seven-variable specification yields a larger $\varepsilon_{LC} = 0.5$, suggesting the four-variable baseline may understate this channel. But there is no existing evidence on the large cross-elasticity from personal tax shocks to corporate income, which I discuss subsequently.

Rationalizing the Cross-Elasticity. The cross-elasticity from personal taxes to the corporate base is large. The explanation is operating leverage. Corporate profits are a residual claim on revenue minus costs. Some of these costs, including capital payments, debt service, leases, and overhead, adjust slowly or are entirely predetermined. When the cost structure of firms is slow to adjust to shocks, profits absorb more of the change in revenue. Consider a firm with \$100 in revenue, \$85 in costs, and \$15 in profits. If revenue rises 1 percent to \$101 but costs do not immediately adjust, profits rise to \$16. In other words, with sluggish adjustment, profits rise by $1/0.15 \approx 6.7$ percent in response to a small shock.

The same arithmetic applies to the aggregate corporate sector. Corporate value added decomposes into wages wL, profits π, and other costs F that are plausibly predetermined. When output rises by dY and wages absorb their proportional share $\gamma\, dY$, the remainder $(1-\gamma)\, dY$ accrues entirely to profits. Because profits are only a s_π share of output, this translates into an elasticity of $(1-\gamma)/s_\pi$. This ratio is directly observable from the National Income and Product Accounts and ranges from approximately 2 to 3.5 over the postwar period (Appendix Figure C.2).

The cross-elasticity is then the product of this leverage and the output response to personal tax cuts: $\varepsilon_{CL} \approx [(1-\gamma)/s_\pi] \times \varepsilon_{YL}$. When we add GDP to the specification in Appendix E, the output elasticity with respect to the log personal retention rate is $\varepsilon_{YL} \approx 1.4$. Combining a postwar average leverage of 2.6 with this output elasticity predicts $\varepsilon_{CL} \approx 3.6$, roughly half the baseline estimate of 7.6. This prediction assumes frictionless labor: wages absorb their full share of output movements so that only non-labor costs are predetermined. It is a lower bound. When wages are also slow to adjust, a larger share of output gains accrues to profits, raising effective leverage above the frictionless-labor benchmark.

Table 2: Frictionless-Labor Predictions vs. Data

	Frictionless Labor	Data (4-var)	Data (7-var)
ε_{CC}	$2.6 \times 0.7 \approx 1.8$	2.6	2.8
ε_{CL}	$2.6 \times 1.4 \approx 3.6$	7.6	9.3
ε_{LC}	$1 \times 0.7 \approx 0.7$	−0.11	0.5
ε_{LL}	$1 \times 1.4 \approx 1.4$	0.98	0.8

Note: Frictionless-labor predictions use operating leverage $(1-\gamma)/s_\pi \approx 2.6$ from the postwar mean of NIPA Table 1.14, applied to output elasticities $\varepsilon_{YC} \approx 0.7$ and $\varepsilon_{YL} \approx 1.4$ from the seven-variable specification. Personal base passthrough is assumed to be one under full labor flexibility. Data columns report present-value elasticities from the four- and seven-variable Bayesian LP specifications (Tables 1 and E.1).

Table 2 extends the decomposition to the full elasticity matrix. The frictionless-labor benchmark tends to underpredict corporate entries and overpredict personal entries, which is exactly what one would expect when wages are slow to adjust. Appendix C.1 develops the argument more fully.

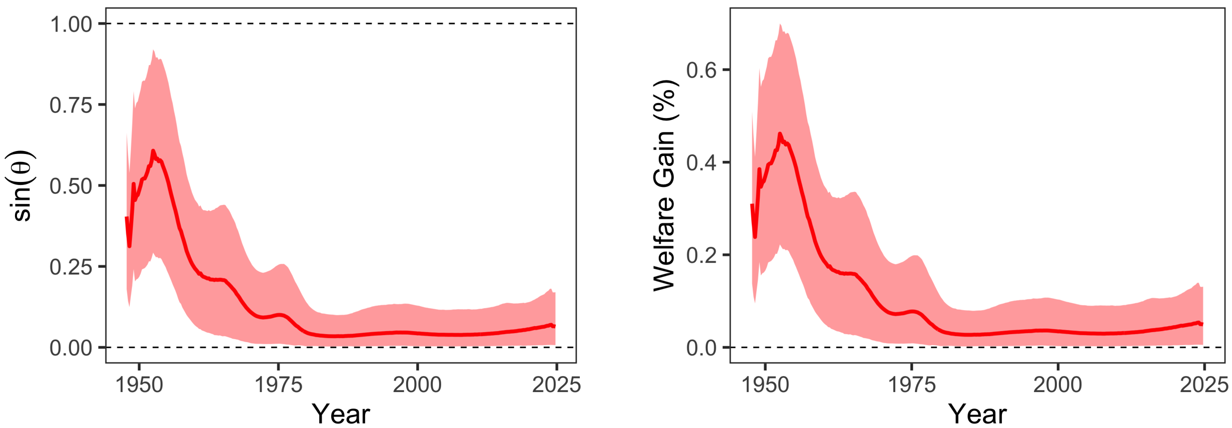
4 The Postwar Evolution of Tax System Efficiency

With both gradients in hand, I now evaluate U.S. federal income tax policy over the postwar era. The welfare gradient uses equal-incidence weights, so that a dollar of tax burden is valued equally regardless of the claimant. The revenue gradient combines the fixed elasticities from Section 3.5 with time-varying tax rates and base composition, measured using the long-run components of the data after removing business-cycle frequencies with a Christiano and Fitzgerald (2003) band-pass filter. Section 4.2 discusses the sensitivity of the results to both of these choices.

4.1 Results

Figure 4 presents the central finding. Panel (a) plots $\sin\theta$, which measures the scope for welfare-improving revenue-neutral reform. A value of zero means no rebalancing of tax instruments can improve welfare; higher values mean larger gains are available from reshuffling the tax mix. As long as $\cos\theta \geq 0$, as Figure D.8 shows it is, then it is sufficient to plot $\sin\theta$. Panel (b) translates this into dollars, plotting the welfare gain from a 1 percentage point reform in the optimal revenue-neutral direction as a percent of GDP. In the early 1950s, $\sin\theta$ peaked around 0.6 and the corresponding welfare gain exceeded 0.4% of GDP. By 1980, both had fallen close to zero, and they have remained there since. The tax system moved from substantial misalignment to near-optimality on the personal-corporate margin over roughly three decades. The two panels have nearly identical shapes because total tax bases as a share of GDP are roughly stable over the postwar period, so the decline in welfare gains is driven entirely by improved alignment, not by a shrinking tax system.

Figure 4: The Evolution of Tax System Efficiency

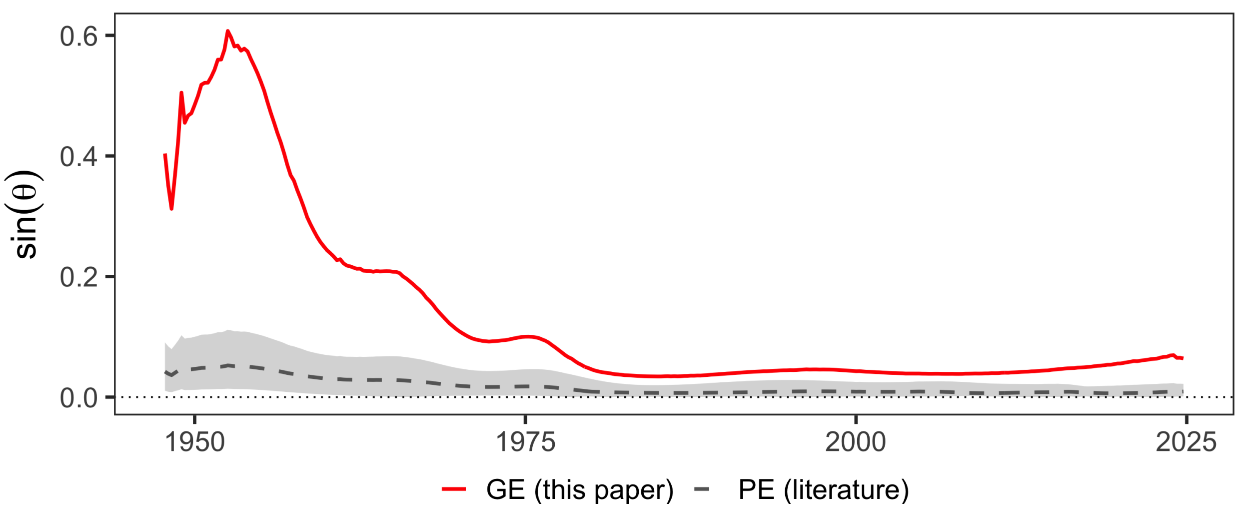


(a) Revenue-neutral potential $\sin\theta$ (b) Welfare gain from optimal reform (% of GDP)

Note: Panel (a) plots $\sin\theta$, the share of potential welfare gains achievable through revenue-neutral reform. Panel (b) plots the welfare gain from a 1 percentage point reform in the optimal revenue-neutral direction, expressed as a percent of GDP. Welfare gains are computed as $\Delta W = \alpha \|g\| \sin\theta$, where $\alpha = 0.01$ is the reform size, $\|g\|$ is the norm of the welfare gradient (the tax bases), and $\sin\theta$ measures the share of welfare gains achievable under revenue neutrality. Both panels use long-run components of tax rates and bases after removing business-cycle frequencies with a Christiano and Fitzgerald (2003) band-pass filter. Shaded regions show 90% credible intervals.

What did this misalignment cost? In the early 1950s, a 1 percentage point reform in the optimal revenue-neutral direction would have improved welfare by approximately 0.4% of GDP per year—roughly $120 billion in 2024 dollars. By 2020, the same reform would yield only 0.05% of GDP. In other words, the large efficiency gains have already been realized. Contemporary debates over corporate versus personal taxation, including the 2017 Tax Cuts and Jobs Act and the 2025 One Big Beautiful Bill Act, are not irrelevant, but the efficiency stakes are an order of magnitude smaller than they were two generations ago. If large welfare gains remain available, they lie in the level of taxation rather than its composition. Near-perfect alignment means the entire welfare gradient now points along the revenue direction: the remaining welfare stake is in how much revenue the system collects, not how it divides the burden.

Figure 5: Welfare Gains from Reform in Partial and General Equilibrium



Note: Solid red line shows $\sin\theta$ gains from general equilibrium estimates with cross-base spillovers. Dashed gray line and shaded band show $\sin\theta$ gains using partial equilibrium own-elasticities from the literature with cross-elasticities set to zero. The PE band spans all combinations of $\varepsilon_{CC} \in [0.2, 0.91]$ and $\varepsilon_{LL} \in [0, 0.58]$.

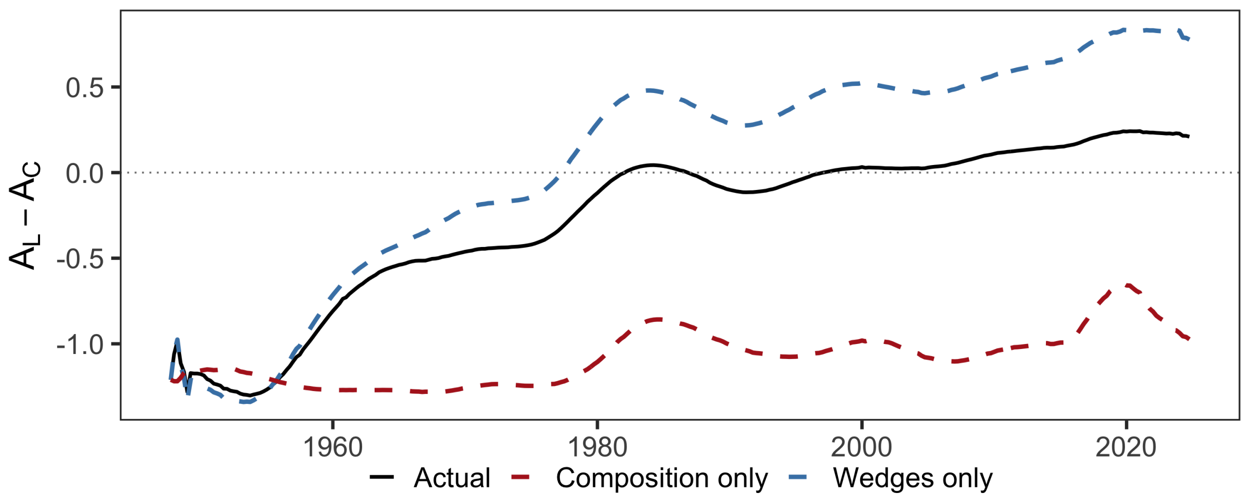
What conclusion would we have drawn had we relied on partial equilibrium estimates from the literature? Vergara and Swonder (2025) and Saez, Slemrod, and Giertz (2012) survey the literature and find own-base elasticities within the United States to be small and typically less than 1.[11] Setting cross-base elasticities to zero, $\varepsilon_{LL} \in [0, 0.58]$ and $\varepsilon_{CC} \in [0.2, 0.91]$, I compare the evolution of $\sin\theta$ generated by those partial equilibrium elasticities to my estimates. Figure 5 shows that $\sin\theta$ is close to zero under the range of PE estimates for the full sample. A policymaker relying on partial equilibrium evidence in the 1950s would have concluded the tax system was roughly efficient, missing hundreds of billions of dollars in available welfare gains. Equally, an analyst evaluating the postwar era in retrospect would find no welfare contribution from the shift away from corporate taxation—one of the largest changes in the structure of federal revenue in American history would appear to have been inconsequential.

What drove the improvement that partial equilibrium would have missed? Figure 1 shows two trends moving together: both corporate tax rates and the corporate share of revenue declined. Both could explain the convergence with the Ramsey optimum. Lower corporate tax wedges directly reduce the fiscal externality $A_C = \eta_C \tilde{\varepsilon}_C$ on corporate taxes. But a shrinking corporate revenue share also matters: it enters the effective elasticity $\tilde{\varepsilon}_L = \varepsilon_{LL} + (R_C/R_L)\varepsilon_{CL}$, so as corporate revenue fell relative to personal revenue, the cross-base spillover that made personal taxes

11. The only documented large ε_{CC} in partial equilibrium are from studies of Costa Rica (Bachas and Soto 2021), Germany pre-unification (Buettner 2003), and Slovakia (Bukovina et al. 2025). I exclude those studies from the comparison.

look cheap became less important. The question is which channel dominated. The theoretical framework offers a natural decomposition. Corollary 2 shows that alignment requires equalizing marginal fiscal externalities $A_i = \eta_i \tilde{\varepsilon}_i$ across instruments, where $\eta_i = \tau_i/(1-\tau_i)$ is the tax wedge and $\tilde{\varepsilon}_i$ is the effective elasticity incorporating cross-base spillovers. Changes in alignment can therefore be attributed to two channels: tax wedges η_i, which policy directly controls, and revenue composition R_C/R_L, which enters the effective elasticities and reflects both policy choices and structural shifts in the economy.

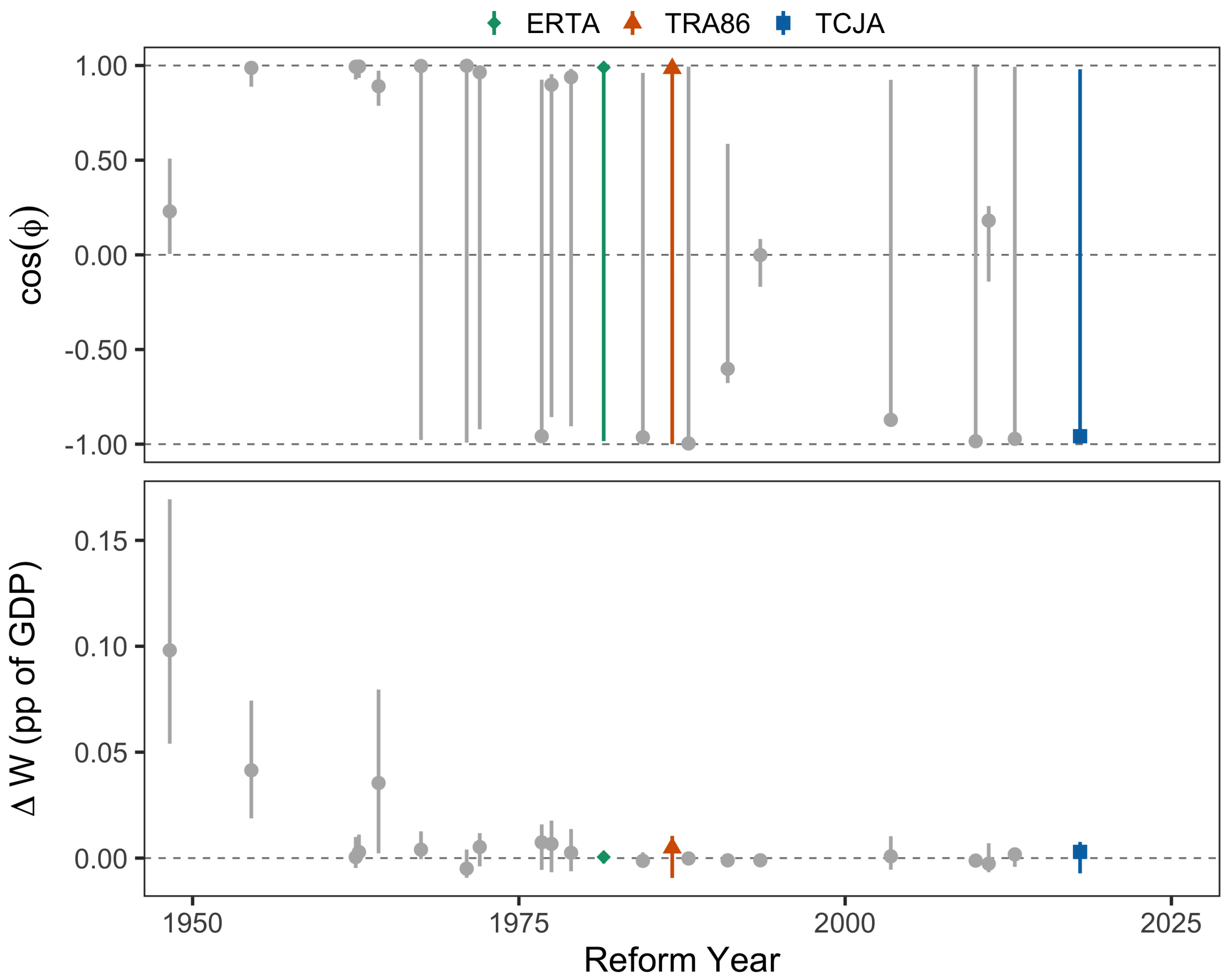
Figure 6: Decomposing the Convergence of Fiscal Externalities



Note: The solid black line plots the gap in marginal fiscal externalities $A_L - A_C$. The dashed blue line holds revenue composition fixed at 1950 values; the dashed red line holds tax wedges fixed at 1950 values. Alignment requires $A_L = A_C$, i.e., the gap equals zero.

Figure 6 decomposes the convergence into these two channels. The solid black line plots the gap in marginal fiscal externalities $A_L - A_C$, which starts around -1 in 1950 and converges to zero by 1980. The dashed blue line holds revenue composition fixed at 1950 values and allows only tax wedges to vary; the dashed red line does the reverse. The wedge channel accounts for essentially all of the convergence: the blue line tracks the actual gap almost perfectly, while the red line remains flat throughout the sample. Tax policy changes that reduced effective corporate tax rates—investment credits, accelerated depreciation, and eventually statutory rate cuts—drove the efficiency gains. Shifts in the composition of the tax base, including the rise of pass-through entities, contributed little.

Figure 7: Reform Direction and Welfare Contribution



Note: Top panel: $\cos\phi$ measures the directional alignment between each actual reform and the optimal revenue-neutral reform; $\cos\phi = 1$ indicates the reform moved in the welfare-improving direction. Bottom panel: welfare gain attributable to each reform, computed as the difference in ΔW between a no-reform counterfactual and actual outcomes, averaged over 20 quarters following enactment. Counterfactuals remove each reform's effects on tax rates and bases using the impulse responses from Section 3.5. Bars show 90% credible intervals.

Which reforms contributed to this convergence? Figure 7 evaluates each major tax reform on two dimensions: the directional alignment $\cos\phi$ between the actual reform and the optimal revenue-neutral reform (top panel), and the welfare gain attributable to each reform relative to a no-reform counterfactual (bottom panel). Before 1980, most reforms moved in the right direction and captured substantial welfare gains. The Revenue Act of 1948, passed over President Truman's veto, improved welfare more than any other reform. After 1980, credible intervals on $\cos\phi$ widen dramatically, spanning the full range from -1 to 1. But this apparent imprecision is inconsequential: every post-1980 reform has a welfare contribution indistinguishable from zero. The wide intervals reflect geometric degeneracy at near-optimality. When the welfare and revenue gradients are nearly collinear, the optimal reform direction is poorly identified, and small perturbations swing $\cos\phi$ across its entire range. Direction ceases to matter when there is nowhere left to go.

Among the reforms with welfare contributions indistinguishable from zero are the Economic Recovery Tax Act of 1981, the Tax Reform Act of 1986, and the Tax Cuts and Jobs Act of 2017. These reforms were not necessarily poorly designed; they arrived after the efficiency problem had already been largely solved on the personal-corporate margin. Whatever case exists for them must rest on considerations beyond compositional efficiency: the distribution of the tax burden, the timing of taxation through deficit finance, or margins outside the personal and corporate income tax system entirely. The same applies to the 2025 One Big Beautiful Bill Act. But the alignment gains are fragile. Because they were driven by tax wedge reductions rather than structural change, they could be reversed by policy: eliminating bonus depreciation, lengthening cost recovery, or raising statutory corporate rates would push the system back toward misalignment.

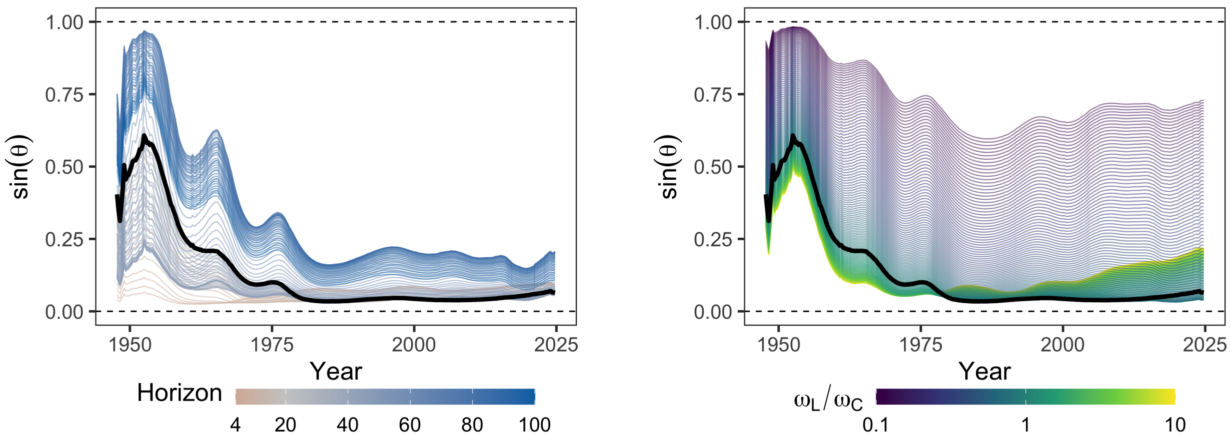
4.2 Robustness

The main results rely on present-value general equilibrium elasticities generated by a Bayesian local projection with equal-incidence welfare weights, a particular truncation horizon and discount rate, and elasticities held fixed over time. How sensitive are the findings to these choices? I partition the discussion into two parts. First, I discuss robustness to choices made within the Bayesian local projections framework. Then I discuss *external* robustness to alternative methodologies and specifications.

Internal Robustness

Figure 8 varies the two main choices in the alignment calculation. Panel (a) varies the truncation horizon H over which present-value elasticities are cumulated. At short horizons, fiscal externalities have little time to compound and early misalignment is modest. As the horizon lengthens, cross-base spillovers accumulate and $\sin\theta$ in the early postwar period rises toward one. This is a central point of the paper: dynamics are what make the early tax system look so inefficient. A static analyst using the same cross-elasticities but only contemporaneous responses would have understated the welfare stakes by a large margin. The baseline horizon of $H = 20$ quarters is conservative; longer horizons strengthen the main finding.

Figure 8: Sensitivity of $\sin\theta$ to Horizon and Welfare Weights



(a) Truncation horizon H

(b) Welfare weight ratio ω_L/ω_C

Note: Panel (a) varies the truncation horizon H used to compute present-value elasticities, with color indicating the number of quarters. Panel (b) varies the ratio of welfare weights on personal to corporate income tax claimants, with color on a log scale. In both panels, the black line denotes the baseline specification ($H = 20$, $\omega_L/\omega_C = 1$).

Panel (b) varies the ratio of welfare weights on personal to corporate income claimants. The results are not only robust but reveal an interesting pattern. When $\omega_L/\omega_C > 1$—the distributionally natural assumption, since corporate income claimants are generally wealthier—early-period misalignment actually falls below the baseline. This occurs because perfect alignment requires $\omega_L/\omega_C = (1 - A_L)/(1 - A_C)$, and in the early postwar period corporate fiscal externalities far exceeded personal ones ($A_C \gg A_L$), pushing the alignment-minimizing weight ratio above one. A redistributive planner would therefore have seen less misalignment than the equal-incidence baseline reports. By today, fiscal externalities are nearly equalized, so the minimizing ratio is close to one and the baseline is approximately optimal. Only when the planner places substantially more weight on corporate income claimants ($\omega_L/\omega_C \ll 1$) does the system appear meaningfully misaligned today, consistent with Vergara and Swonder (2025), who find that rationalizing the post-2017 corporate tax rate requires placing substantial value on the utility of corporate owners. The equal-incidence baseline is conservative: any plausible redistributive criterion would strengthen the main finding.

Appendix Figure D.2 varies β from 0.95 to 1; all specifications show the same trajectory. Appendix Figure D.4 varies the band-pass filter cutoff from 8 to 80 quarters and also shows unfiltered data; the transition from misalignment to near-optimality is present at every cutoff.

External Robustness

The alignment results are robust across estimation methods. Figure D.1 compares the evolution of $\sin\theta$ under ten estimation approaches, including frequentist local projections, smooth local projections (Barnichon and Brownlees 2019), bias-corrected local projections (Herbst and Johannsen 2024), Bayesian VARs, proxy SVARs, and three LP-IV variants. The LP-IV specifications are particularly informative because they bypass the invertibility assumption that all other methods rely on to estimate the impact matrix A_0, instrumenting the log retention rate directly with the narrative measure at each horizon. All methods show the same qualitative trajectory from misalignment to near-optimality, and the baseline BLP is among the most conservative. Dashed lines show that including a linear trend generally lowers $\sin\theta$ throughout the sample without changing the trajectory. Figure D.3 confirms that reversing the Cholesky ordering used to separate personal from corporate shocks has no effect, consistent with Mertens and Ravn (2013a).

The baseline holds elasticities fixed at full-sample estimates. How plausible is this? The cross-elasticity ε_{CL} is the product of operating leverage and the output response to personal tax cuts. Operating leverage is directly observable from the national accounts and varies only modestly over the postwar period, between 2 and 3.5, so time variation in ε_{CL} is bounded by changes in the output multiplier alone. Figure C.1 fails to confirm this reasoning. A TVP-BVAR with stochastic volatility (Primiceri 2005), which allows the full transmission mechanism to drift over time, estimates ε_{CL} remains flat around 2.5 throughout the postwar era. There is similarly little variation in the other elasticities. Feeding the time-varying elasticities into the alignment calculation (Figure D.1, TVP-BVAR line) yields a series that starts closer to optimal than the baseline fixed-coefficient baseline but reaches the same destination: near-optimality by the 2010s.[12] Figure D.5 provides a complementary check within the fixed-coefficient framework, re-estimating the BLP dropping each narrative shock in turn. No single reform drives the results, though the Tax Reform Act of 1986 is the most influential observation.

Section 3.5 showed that reallocating S-corporation income from the corporate to the personal base leaves ε_{CL} essentially unchanged, and that the seven-variable specification yields a larger cross-elasticity than the baseline. Appendices F and E show the implications for alignment. Across estimation methodologies, both confirm the central finding: substantial improvement in tax system efficiency over the postwar period. However, neither converges as close to zero as the four-variable baseline. Under S-corp reallocation, $\sin\theta$ settles around 0.15 by the 2010s (see Figures F.3 and F.4); the seven-variable specification is similar, with wider credible intervals reflecting the additional parameters (see Figures E.1 and E.2). The robust conclusion across all

12. The lack of time variation is somewhat at odds with Mumtaz and Petrova (2023) and De Wind (2014), who find only a moderate decline in the aggregate tax multiplier.

specifications is that the tax system improved dramatically from the early postwar period. The stronger claim that the system is near-optimal today is specific to the theoretically preferable four-variable baseline.

5 Tariffs and Tax Reform

Section 4 showed that the income tax mix has converged toward the Ramsey optimum, leaving little room for welfare-improving revenue-neutral reform between personal and corporate taxes. But the government has access to additional revenue instruments. The 2025 tariff increases were justified in part as a way to offset revenue lost from the corporate tax cuts in the One Big Beautiful Bill Act, effectively substituting tariff revenue for income tax revenue. Whether such substitutions improve the efficiency of the tax system is a question the geometric framework can answer, since it extends naturally to any number of instruments. This section applies the three-instrument version to evaluate whether tariffs belong in the efficient federal tax mix.

5.1 Extending the Empirical Framework

I extend the four-variable system to a six-variable VAR by adding the log tariff retention rate $-\log(1 + \tau_T)$ and the log tariff base (imports of goods). The tariff rate is customs duties divided by total imports of goods, constructed from NIPA data as described in Appendix A.1. The sample, lag length, and estimation procedure are otherwise identical to the baseline. The identification strategy is the same as in Section 3.3, with the narrative tariff shocks of Besten and Känzig (2026) serving as the third instrument. Their series classifies postwar tariff changes as exogenous when the legislative record indicates motivations unrelated to contemporaneous economic conditions, primarily GATT negotiating rounds and the 2018–2019 tariff increases. I make two modifications to improve instrument strength. First, I assign each shock to the implementation quarter rather than distributing it across adjacent years. Second, I replace their sign-only coding with magnitudes, using tariff rate changes from Bown and Irwin (2015) for the pre-Trump period and from Amiti, Redding, and Weinstein (2020) for the Trump-era tariffs. Appendix Figure A.1 plots the resulting shock series alongside the average tariff rate.

Table 3: Present-Value Elasticities of Tax Bases

	Personal Tax Shock	Corporate Tax Shock	Tariff Shock
$\varepsilon_{L\cdot}$ (Personal Base)	0.68	-0.29	-0.57
	(0.07, 1.37)	(-0.54, -0.05)	(-1.48, 0.31)
$\varepsilon_{C\cdot}$ (Corporate Base)	7.50	2.59	0.76
	(4.69, 10.40)	(1.81, 3.51)	(-2.80, 4.17)
$\varepsilon_{T\cdot}$ (Tariff Base)	-3.32	0.43	12.91
	(-7.51, 0.08)	(-1.00, 2.14)	(10.78, 15.45)

Note: Each elasticity ε_{ij} measures the cumulative discounted percent change in base i per one percent increase in the retention rate of instrument j, estimated by Bayesian local projection with 90% credible intervals in parentheses. The tariff retention rate enters the VAR as $-\log(1+\tau_T)$, so all elasticities are with respect to tax cuts. The personal and corporate elasticities are largely unchanged from the two-instrument baseline (Table 1). Appendix Table G.1 reports estimates across alternative estimation methods.

As in the two-instrument case, the revenue gradient for each instrument depends on the full matrix of present-value elasticities, which now includes cross-base spillovers between income taxes and tariffs. Table 3 reports the 3×3 matrix under the Bayesian local projection. Three features stand out. First, adding tariffs does not change the income tax block of the matrix. Second, the own-elasticity of the tariff base is large ($\varepsilon_{TT} \approx 13$). The most directly comparable estimate comes from Besten and Känzig (2026). In their postwar sample, a one percentage point tariff increase reduces real imports by approximately 10 percent on impact, with the response persisting at 5–6 percent through year five. Converting their impulse responses into a permanent-equivalent elasticity comparable to ε_{TT} yields a value of approximately 10.[13] Third, the cross-elasticity of the tariff base with respect to personal taxes is large and nearly significant ($\varepsilon_{TL} = -3.3$), consistent with higher after-tax income raising demand for imports. Fourth, the remaining cross-elasticities involving tariff shocks are imprecisely estimated, though the signs are economically coherent. $\varepsilon_{LT} = -0.57$ implies that tariff increases expand domestic labor income as import protection boosts domestic production and employment. $\varepsilon_{CT} = 0.76$ implies that tariff cuts expand corporate profits, consistent with evidence that firms are slow to adjust prices in response to cost

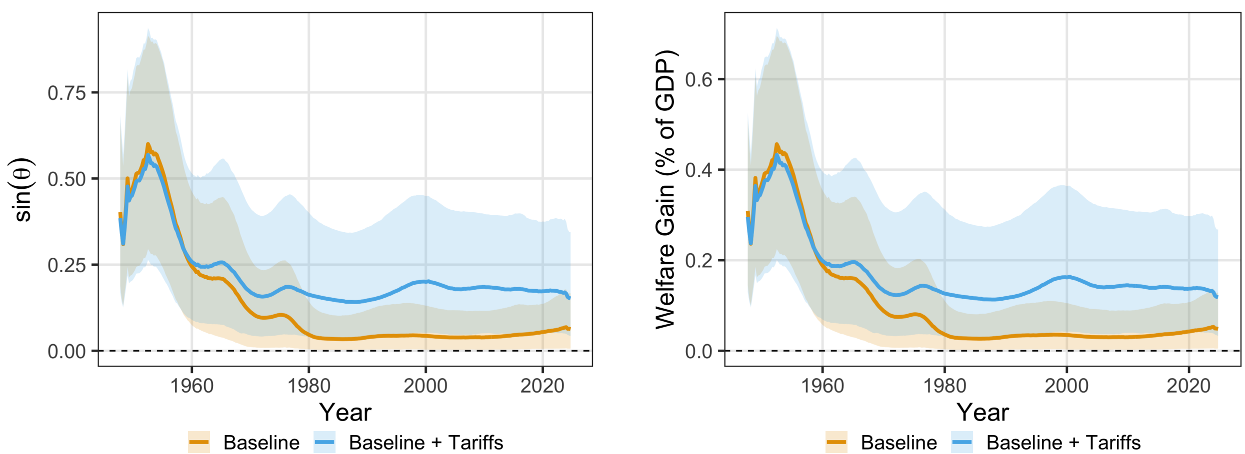
13. The trade elasticity literature typically estimates substitution elasticities—either across foreign varieties of a product or between domestic and foreign goods—rather than the response of the aggregate import base to an aggregate tariff change. The former captures reallocation across suppliers, while ε_{TT} additionally reflects income effects, general equilibrium feedbacks, and tariff avoidance, compounded over the five-year horizon. See Boehm, Levchenko, and Pandalai-Nayar (2023) for a comprehensive discussion of how concept, horizon, and identification shape trade elasticity estimates.

shocks, leading to profit expansions (Cavallo, Llamas, and Vazquez 2025). Appendix Table G.1 reports the full set of estimates across six estimation methods.

While the elasticity matrix determines the revenue gradient, the welfare gradient for tariffs requires an assumption about incidence. For income taxes, the burden falls entirely on domestic agents, so the welfare gradient is simply the discounted tax base under equal incidence weights. For tariffs, the incidence is split between domestic consumers and foreign exporters, with the division depending on terms-of-trade effects. Recent evidence on the 2018-19 trade war with China indicates that the United States bore around 60-80 percent of the incidence on the low end (Ganapati and Hottman 2026; Mejia 2026), while a significant number of studies show full passthrough (Amiti, Redding, and Weinstein 2019, 2020; Cavallo et al. 2021; Fajgelbaum et al. 2020). At baseline, I assume that domestic consumers bear the full burden of tariffs and show throughout the effect of that assumption.

5.2 Tariffs in the Postwar Tax System

Figure 9: The Evolution of Tax System Efficiency



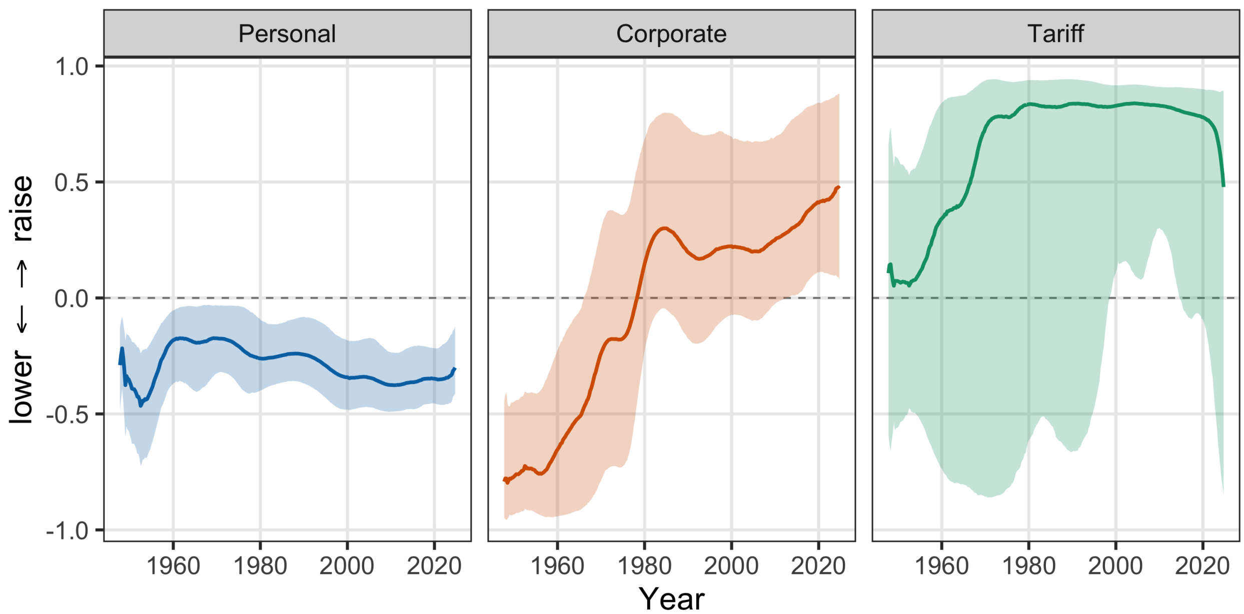
(a) Revenue-neutral potential $\sin\theta$

(b) Welfare gain from optimal reform (% of GDP)

Note: Panel (a) plots $\sin\theta$, the share of potential welfare gains achievable through revenue-neutral reform. Panel (b) plots the welfare gain from a 1 percentage point reform in the optimal revenue-neutral direction, expressed as a percent of GDP. Welfare gains are computed as $\Delta W = \alpha \|g\| \sin\theta$, where $\alpha = 0.01$ is the reform size, $\|g\|$ is the norm of the welfare gradient (the tax bases), and $\sin\theta$ measures the share of welfare gains achievable under revenue neutrality. Shaded regions show 90% credible intervals.

With the elasticity matrix and welfare weights in hand, I compute $\sin\theta$ for the three-instrument system and compare it to the two-instrument baseline. Figure 9a shows the result. The two series track closely through the 1960s, then diverge. The divergence is significant; the 90% credible interval for the three-instrument alignment does not overlap with the point estimate for the two-instrument system. By the 2010s the baseline $\sin\theta$ is near zero while the three-instrument system plateaus around 0.18. This indicates that while the income tax composition is roughly right, the full tax system retains meaningful misalignment on a margin the two-instrument analysis could not detect.

Figure 10: Optimal Reform Direction by Instrument

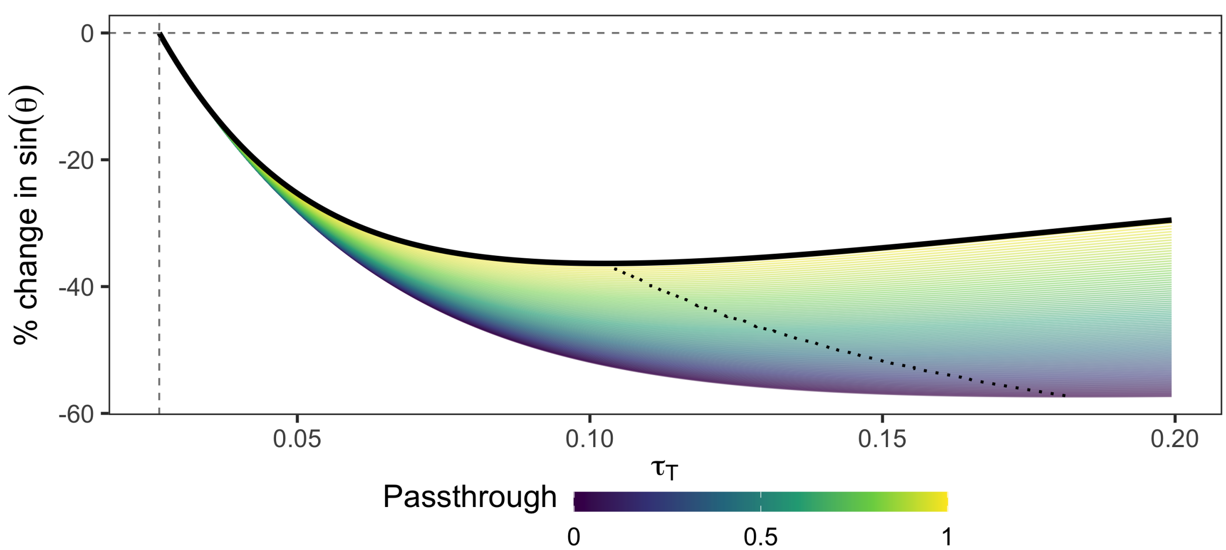


Note: Each panel plots the component of the optimal revenue-neutral reform direction d^* for one instrument, with 90% credible intervals. Positive values indicate the instrument should be raised; negative values indicate it should be lowered. The personal component is consistently negative and precisely estimated. The corporate component crosses zero around 1980, shifting from "lower" to "raise." The tariff component is consistently positive from the 1970s onward, though with wide credible intervals reflecting the imprecision of the tariff elasticities.

Where should the system move to close the remaining gap? Figure 10 decomposes the optimal revenue-neutral reform direction d^* into its three components. The direction d^* is the unit vector in tax-rate space that maximally reduces misalignment while holding revenue constant; each panel shows one component, bounded between -1 (lower the instrument as much as possible) and 1 (raise it as much as possible). The personal component is consistently negative and precisely estimated: lower personal taxes throughout the postwar period. The corporate component crosses zero around 1980, consistent with the Section 4 finding that corporate taxes shifted from too high to roughly right. The tariff component is consistently positive from the 1970s onward, with wide credible intervals but a stable sign. This explains the divergence in Figure 9a: before 1980, both the income tax and tariff margins called for reform, and the two systems improved together. After 1980, the income tax margin was roughly optimized but the tariff prescription persisted, so the three-instrument system plateaued while the baseline continued to converge.

The reform direction reflects the marginal value of public funds of each instrument. At current rates and bases, the MVPF is approximately 1.6 for personal taxes, 1.0 for corporate taxes, and 0.8 for tariffs (Appendix Figure G.4). Tariffs are the cheapest margin despite having the most elastic base. The reason is that the tariff base is small relative to the personal base. Imports are roughly one-seventh of personal income, so the tariff rate is low and the fiscal externality on tariff revenue is modest even at $\varepsilon_{TT} \approx 13$. At the same time, the size asymmetry means that even modest cross-base spillovers from tariffs onto the personal base generate large revenue gains in absolute terms. A cross-elasticity of $\varepsilon_{LT} = -0.57$ applied to a base seven times larger than the tariff base generates more additional revenue than the own-base distortion costs. This ordering has been stable since the 1970s: the income tax MVPFs converged as corporate reforms brought the corporate MVPF down, but the gap between personal taxes and tariffs persisted.

Figure 11: Counterfactual Tariff-for-Corporate-Tax Swap



Note: This figure plots the percent change in $\sin\theta$ from a revenue-neutral swap of tariffs for corporate taxes, starting from 2024Q4 rates and bases, using median BLP elasticities. The x-axis is the counterfactual tariff rate τ_T; the dashed vertical line marks the 2024Q4 rate. Each curve corresponds to a different passthrough assumption. The dashed curve traces the optimum across passthrough values. Appendix Figure G.2 shows the full posterior uncertainty under full passthrough.

The persistent optimality of raising tariffs prompts a concrete question motivated by recent policy: what would happen to alignment if the government swapped corporate tax revenue for tariff revenue? The 2025 tariff increases were justified in part as offsetting the revenue cost of corporate tax cuts in the One Big Beautiful Bill Act, making this more than a hypothetical. Figure 11 evaluates such swaps starting from 2024Q4 rates and bases, varying the tariff rate along the horizontal axis and adjusting the corporate rate to maintain revenue neutrality. Under full domestic passthrough, the swap reduces $\sin\theta$ by approximately 25 percent at the optimum of $\tau_T \approx 11\%$. The gains are mechanically larger under lower passthrough, which makes tariffs cheaper from a domestic welfare perspective and shifts the optimum rightward. This has little real impact, however, because the precisely estimated own-elasticity constrains the optimal rate to a narrow range. Even with no passthrough, the optimal tariff rate is only 18%.[14] The posterior uncertainty is nonetheless substantial: Appendix Figure G.2 shows the improvement is statistically indistinguishable from zero at the 68 percent level. The point estimates indicate that the general direction of the policy may be welfare-improving, but it is difficult to draw that conclusion robustly.[15] What is robust is that the income-tax-only analysis misses a margin on which the postwar tax system appears persistently misaligned, and the direction of improvement consistently points toward somewhat higher tariffs.

6 Conclusion

This paper develops a geometric framework for evaluating tax reform in general equilibrium. The angle between the welfare gradient and the revenue gradient measures alignment with the Ramsey optimum: when these vectors are collinear, no revenue-neutral reform can improve welfare; when they are orthogonal, the system is maximally inefficient. Computing this angle requires only two sufficient statistics, tax bases and the general equilibrium response of revenue to tax changes, without specifying preferences, production, or market structure.

Applying this framework to the U.S. federal income tax system from 1950 to 2025 reveals that alignment improved dramatically over the postwar period. In the early 1950s, welfare and revenue gradients were substantially misaligned, and a small reform in the optimal direction would have yielded large welfare gains. By the 2010s, the system had moved to near-optimality on the personal-corporate margin. The decomposition shows that tax policy changes, including investment incentives, accelerated depreciation, and eventually statutory rate cuts, drove this convergence, not structural shifts in the economy. A corollary is that every major reform since 1980, including the Tax Reform Act of 1986 and the Tax Cuts and Jobs Act, has had no measurable welfare effect on this margin. Whatever case exists for those reforms must rest on distribution,

14. The counterfactual holds elasticities fixed at values estimated from historical variation around low tariff rates. The horizon robustness in Appendix Figure G.3a shows that the three-instrument results are more sensitive to the truncation horizon than the baseline, reflecting the non-monotone dynamics of the tariff base impulse response.

15. Alessandria et al. (2025) reach a similar conclusion in a structural general equilibrium framework, finding that tariff-for-investment-subsidy swaps can be welfare-improving.

timing, or margins outside the personal and corporate income tax system.

Extending the framework to include tariffs reveals that the federal tax system remains persistently misaligned on a margin the income-tax analysis cannot detect, with the optimal direction pointing toward somewhat higher tariffs since the 1970s. The point estimates suggest the recent tariff-for-corporate-tax swap may improve efficiency, though the posterior uncertainty is substantial.

Near-perfect alignment on the income tax margin means the entire welfare gradient now points along the revenue direction: the remaining question is not how to divide the tax burden between personal and corporate income, but how much total revenue to collect. The framework quantifies the welfare stake in that question but cannot resolve it without spending-side evidence. The efficiency gains documented here are also not permanent. They emerged from policy choices that reduced effective corporate taxation, and they can be reversed by policy choices that raise it. U.S. debt-to-GDP is high, and stabilizing it may eventually require raising revenue, cutting spending, or both. If the adjustment falls disproportionately on corporate taxes, whether through rate increases, base broadening, or the expiration of investment incentives, the alignment gains of the past 70 years could unwind rapidly. Alternatively, the U.S. could turn to other margins to raise revenue. Consumption taxes, payroll taxes, and wealth taxes all lie outside the current empirical analysis, but the framework extends naturally to these settings, as it does to state and local tax systems, environmental policy, or social insurance: anywhere policymakers face multi-instrument problems with general equilibrium spillovers. The sufficient statistics it requires, welfare costs and revenue responses, can be estimated from reduced-form evidence in any of these domains, diagnosing both the composition and the scale of remaining inefficiency.

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Appendices

Online Appendix

A Data

The variables and their construction are an extension of Mertens and Ravn (2013a) and Cloyne et al. (2022) from 1947Q1-2019Q4:

- **Personal Income Tax Base.** The personal income tax base is National Income and Product Accounts (NIPA) personal income (Table 2.1 Line 1) plus contributions for government social insurance (3.2 Line 11) less transfers (2.1 Line 17).
- **Personal Income Tax Revenue.** This comes from summing Table 3.2 Lines 3 and 11.
- **Average Personal Income Tax Rate.** The APITR is revenue divided by the lagged base.
- **Corporate Income Tax Base.** The corporate tax base is table 1.2 Line 13 (corporate profits with inventory valuation and capital consumption adjustments) less Federal Reserve Bank profits (Historical Tables 6.16 B-C-D).
- **Corporate Income Tax Revenue.** Corporate revenue is federal taxes on corporate income excluding Federal Reserve banks (Table 3.2 Line 8).
- **Average Corporate Income Tax Rate.** The ACITR is revenue divided by the lagged base.
- **Government Spending.** Real federal government consumption expenditures and gross investment (1.1.3 Line 23).
- **GDP.** Real GDP is from NIPA Table 1.1.3 Line 1.
- **Debt.** Debt is spliced together using Federal debt held by the public from Favero and Giavazzi (2012) and the series FYGFDPUN from FRED.
- **Narrative Instruments.** See Cloyne et al. (2022) for an extension to the Mertens and Ravn (2013a) corporate and personal income tax shock series.

All series in levels are deflated by the GDP deflator (NIPA Table 1.1.9 Line 1) and population, then log-transformed. Population is total population over age 16 from the Bureau of Labor Statistics (CNP16OV in FRED). Because this series starts in 1948, we use a simple linear extrapolation to extend back four quarters to 1947Q1.

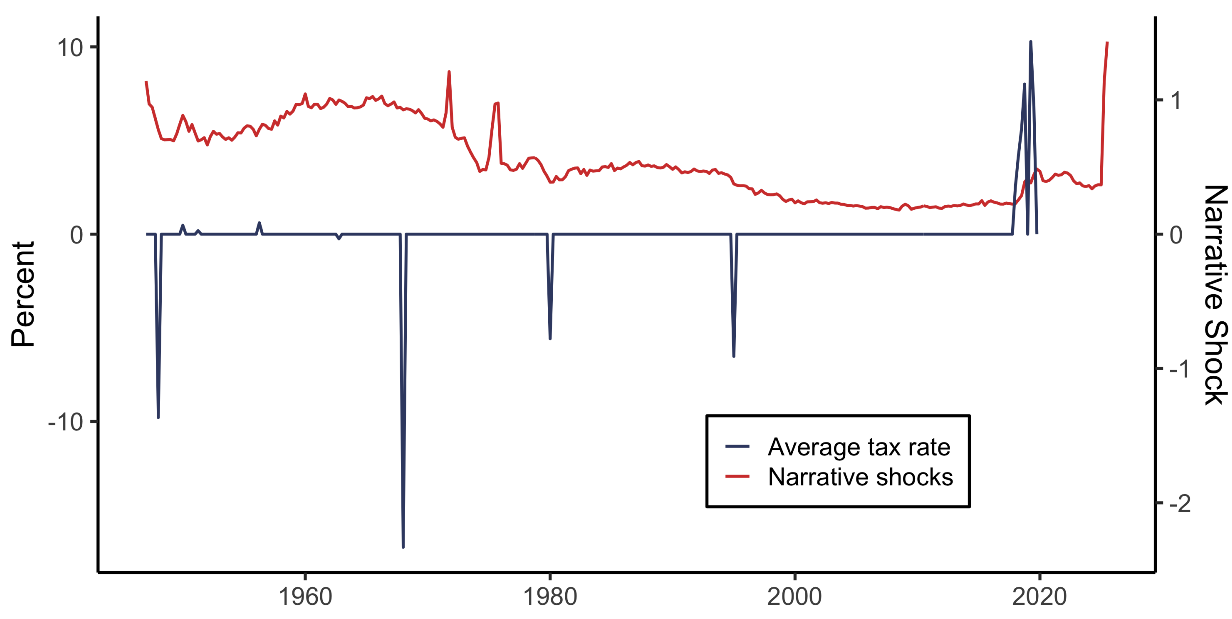
A.1 Tariff Data Construction

The three-instrument extension requires quarterly series for the tariff rate and tariff base from 1947Q1 though 2019Q4.

Customs Duties. Quarterly federal customs duty receipts are reported in NIPA Table 3.2 Line 6 starting in 1959Q1. For the period 1947Q1-1958Q4, I construct quarterly customs duties by combining annual duties from NIPA Table 3.2 Line 6 with a monthly indicator series. The indicator is monthly Treasury customs receipts from the NBER Macrohistory Database (FRED Series M15001USM144NNBR), seasonally adjusted using the X-13ARIMA-SEATS procedure and aggregated to quarterly frequency. I then distribute the annual NIPA customs duty totals across quarters using the Denton-Cholette proportional benchmarking method, which finds the quarterly path that sums to each annual total while preserving the within-year pattern of the monthly indicator. All quarterly values are expressed at seasonally adjusted annual rates to match the NIPA convention. The pre-1959 and post-1959 series are spliced at 1959Q1.

Tariff Rate. The tariff base is imports of goods from NIPA Table 4.1 Line 19. The average aggregate tariff rate is customs duties divided by total imports of goods. The tariff rate enters the VAR as $-\log(1+\tau)$. For small shocks, that means $-\log(1+\tau) \approx -\tau \approx \log(1-\tau)$.

Figure A.1: Average Tariff Rate and Shocks



Note: This figure plots the average tariff rate on the left axis and the demeaned shock series on the right axis.

Narrative shocks. As a starting point for identification, I take the narrative series from Besten and Känzig (2026). Following similar criteria to Romer and Romer (2010), they narratively identify exogenous tariff shocks, many of which come from the General Agreement on Tariffs and Trade in the postwar era. Their shocks are annual and are constrained to -1 or +1 (sign-only). To improve the instrument strength, I take two steps. First, I assign the shock to the implementation quarter, and let all other quarters take a value of zero. Second, I give the shocks magnitudes. In the pre-Trump years, I multiply the prevailing tariff rate by the percentage reduction in tariffs from Bown and Irwin (2015). In the Trump years, I use Amiti, Redding, and Weinstein (2020) to assign both the implementation quarter and magnitude of the shocks.

B Estimation Details

This appendix provides details on the two-stage Bayesian local projection procedure used to estimate impulse response functions. Many of the estimation details follow Cloyne et al. (2022), Appendix G.

B.1 Stage 1: Bayesian VAR with Hierarchical Priors

The first stage estimates a Bayesian VAR to obtain draws from the posterior distribution of the impact matrix A_0. The VAR is:

$$Z_t = c + B_1 Z_{t-1} + \cdots + B_p Z_{t-p} + u_t, \qquad u_t \sim N(0, \Sigma) \tag{A.1}$$

where Z_t is the vector of endogenous variables and $p = 4$ lags.

Priors. Following Giannone, Lenza, and Primiceri (2015), I treat the hyperparameters governing prior tightness as random and estimate them jointly with the VAR coefficients. The prior combines three components:

- *Minnesota prior*: The prior mean for the coefficient on the first own lag is set to the AR(1) coefficient from a preliminary regression; all other coefficients have prior mean zero. The prior variance for the coefficient on lag ℓ of variable j in equation i is:

  $$V_{ij\ell} = \frac{\lambda^2}{\ell^2} \cdot \frac{s_i^2}{s_j^2} \tag{A.2}$$

  where s_i and s_j are residual standard deviations from preliminary AR(1) regressions and λ controls overall tightness. The prior on λ is $\lambda \sim N^+(0.2, 0.4^2)$ truncated to $[10^{-4}, 5]$.

- *Sum-of-coefficients prior*: This prior, introduced by Doan, Litterman, and Sims (1984), shrinks the sum of lag coefficients toward unity for persistent variables, helping capture unit root behavior.

- *Single-unit-root prior*: This prior implements the dummy initial observation approach of Sims (1993), providing additional shrinkage toward a random walk.

The hyperparameters governing the sum-of-coefficients and single-unit-root priors are also treated as random, with priors $\Gamma(1, 1)$ truncated to $[10^{-4}, 50]$.

Posterior simulation. The posterior is approximated via Markov chain Monte Carlo. Conditional on hyperparameters, the posterior for VAR coefficients and the covariance matrix Σ is Normal-inverse-Wishart and can be sampled directly. The hyperparameters are updated using a Metropolis-Hastings step. I use 20,000 iterations with 10,000 discarded as burn-in.

B.2 Stage 1b: Proxy Identification of A_0

For each draw of the VAR coefficients and covariance matrix from the Stage 1 posterior, I apply the proxy identification procedure of Mertens and Ravn (2013a) to obtain a draw of the impact matrix A_0.

Let $u_t = A_0 \varepsilon_t$ where ε_t contains the structural shocks with $\text{Var}(\varepsilon_t) = I$. Partition the variables so that the first k correspond to the tax rates (instrumented by the narrative measures m_t) and the remaining $n - k$ are non-tax variables. The reduced-form covariance matrix $\Sigma = A_0 A_0'$ can be partitioned conformably:

$$\Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \tag{A.3}$$

The proxy relevance condition $E[m_t \varepsilon_{1,t}'] = \Phi \neq 0$ and exogeneity condition $E[m_t \varepsilon_{2,t}'] = 0$ identify the subspace spanned by the tax shocks. Regressing the non-tax residuals on the tax residuals, using the proxies as instruments, recovers the ratio $\beta_{21}\beta_{11}^{-1}$. The remaining elements of A_0 corresponding to the tax shocks follow from the algebra in Mertens and Ravn (2013a).

To separate the personal and corporate tax shocks within the identified subspace, I apply a Cholesky factorization. When estimating the impulse response to shock j, I order tax j last. This imposes that shock j has no direct contemporaneous effect on the other tax rate.

Each draw of A_0 is normalized so that the impact effect on the relevant tax rate equals −1 percentage point (a tax cut).

B.3 Stage 2: Bayesian Local Projections

The second stage estimates local projections at each horizon $h \geq 1$. For outcome variable y (e.g., a tax rate or log tax base), the local projection is:

$$y_{t+h} = c^{(h)} + B_1^{(h)} Z_{t-1} + \sum_{\ell=2}^{p} b_\ell^{(h)} Z_{t-\ell} + u_{t+h} \tag{A.4}$$

where $p = 4$ lags. Let $\omega^{(h)}$ collect the coefficients and let $X_t = (Z_{t-1}', \ldots, Z_{t-p}', 1)'$ denote the regressors.

Priors. The prior for $\omega^{(h)}$ is Normal with mean ω_0 and variance S_0. The prior mean implies that the outcome variable follows an AR(1) process: the element corresponding to the first own lag equals the AR(1) coefficient from a preliminary regression, with all other elements set to zero. The prior variance takes the Minnesota form:

$$S_{0,j\ell} = \frac{\lambda^2}{\ell^2} \cdot \frac{s_y^2}{s_j^2} \tag{A.5}$$

where s_y is the standard deviation of the outcome and s_j is the standard deviation of regressor j. The tightness parameter is $\lambda = 10$, reflecting weak prior information. The prior variance on the intercept is 100.

Non-Gaussian errors. As discussed in Jorda (2005), local projection residuals are generally heteroskedastic when $h > 0$. To accommodate this, I model the errors as a scale mixture of normals. Specifically:

$$u_{t+h} | \phi_t \sim N(0, \sigma^2/\phi_t), \qquad \phi_t \sim \Gamma(\nu/2, \nu/2) \tag{A.6}$$

Integrating out the mixing variable ϕ_t yields a Student-t distribution for u_{t+h} with ν degrees of freedom (Geweke 1993). This formulation accommodates heteroskedasticity of unknown form without imposing a specific structure.

The degrees of freedom ν receives an exponential prior:

$$p(\nu) \propto \exp(-\nu/\nu_0) \tag{A.7}$$

with $\nu_0 = 10$. Small values of ν allow heavy tails; the prior places mass on moderate tail thickness while permitting approximate normality.

Posterior simulation. I approximate the posterior using Gibbs sampling with a Metropolis-Hastings step for ν. The sampler iterates over the following blocks, conditioning on all other parameters:

1. *Mixing weights* ϕ_t: Given the current residuals and ν, each ϕ_t is drawn independently from a Gamma distribution with shape $(\nu + 1)/2$ and rate $(u^2_{t+h}/\sigma^2 + \nu)/2$.

2. *Degrees of freedom* ν: The conditional posterior is nonstandard. I update ν via random walk Metropolis-Hastings, proposing $\nu' = \nu + \eta$ where $\eta \sim N(0, c)$. The proposal is accepted with the usual Metropolis probability. The step size c is tuned to achieve an acceptance rate between 30% and 50%.

3. *Error variance* σ^2: Defining transformed residuals $\tilde{u}_{t+h} = u_{t+h}\sqrt{\phi_t}$, the conditional posterior for σ^2 is inverse Gamma with shape $T/2$ and scale $\sum_t \tilde{u}^2_{t+h}/2$.

4. *Coefficients* $\omega^{(h)}$: Applying the same transformation to the dependent variable and regressors, $\tilde{y}_{t+h} = y_{t+h}\sqrt{\phi_t}$ and $\tilde{X}_t = X_t\sqrt{\phi_t}$, yields a standard conjugate updating formula. The conditional posterior is Normal with variance
$$V = \left(S_0^{-1} + \sigma^{-2}\tilde{X}'\tilde{X}\right)^{-1} \tag{A.8}$$
and mean
$$\bar{\omega} = V\left(S_0^{-1}\omega_0 + \sigma^{-2}\tilde{X}'\tilde{y}\right) \tag{A.9}$$

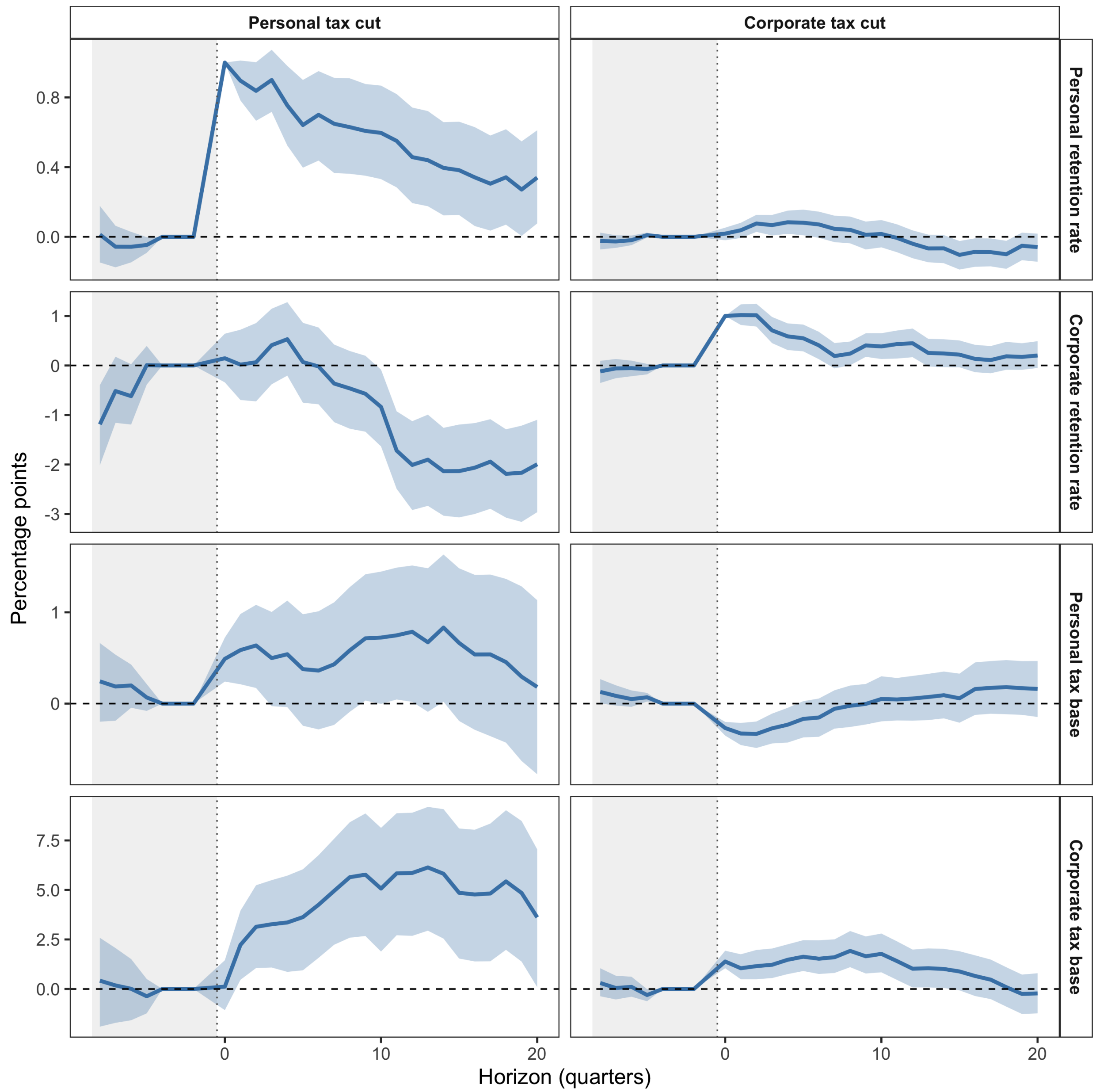
I run 12,000 iterations, discarding the first 3,000 as burn-in and retaining every third draw, yielding 3,000 posterior draws for inference.

Impulse response construction. At horizon $h = 0$, the impulse response equals the A_0 draw from Stage 1b. At horizons $h \geq 1$, the impulse response is constructed following Jorda (2005):

$$IRF_h = B_1^{(h-1)} A_0 \tag{A.10}$$

where $B_1^{(h)}$ contains the coefficients on Z_{t-1} from the horizon-h local projection. For each posterior draw, I pair the A_0 draw from Stage 1 with the $B_1^{(h)}$ draw from Stage 2, ensuring that uncertainty in the impact matrix propagates through to longer horizons. By construction, at $h = 0$, $IRF_0 = A_0$.

Figure B.1: Shock Pretrends for Bayesian LP



Note: IRFs are plotted from $H = -8$ quarters prior to the shock to 20 quarters after the shock under the baseline Bayesian LP specification.

Revenue gradients. As discussed in Section 3, the present-value revenue gradient is constructed from the impulse responses of retention rates and tax bases. For tax instrument j, equation (16) gives:

$$r_j^{PV} = B_{j,0} \cdot A \cdot \left[1 - \frac{\tau_j}{1-\tau_j}\varepsilon_{jj} - \frac{\tau_k}{1-\tau_j}\frac{B_k}{B_j}\varepsilon_{kj}\right], \tag{A.11}$$

where $A = \sum_{h=0}^{H} \beta^h$ is the annuity factor. Since the elasticities ε_{ij} are permanent-equivalent (cumulative base responses divided by the cumulative retention rate response κ_j), the revenue gradient uses the common annuity factor rather than instrument-specific persistence. This ensures that g^{PV} and r^{PV} correspond to the same policy experiment: a permanent marginal reform. This calculation is performed for each posterior draw, yielding a posterior distribution for the revenue gradient.

C Elasticity Robustness

This appendix reports robustness of the present-value elasticities from Table 1 to alternative estimation methods. Table C.1 reports the full set of elasticities across six methods, with and without linear trends. Table C.2 reports LP-IV estimates that bypass the invertibility assumption. Figure C.1 reports time-varying elasticities from a TVP-BVAR with stochastic volatility.

1. **Frequentist local projections.** Standard local projections using the Mertens-Ravn structural shocks identified from the BVAR as regressors. Uncertainty about A_0 is propagated using the wild bootstrap proxy SVAR of Mertens and Ravn (2013a).

2. **Smooth local projections.** Frequentist local projections with a penalty on deviations across adjacent horizons following Barnichon and Brownlees (2019), which reduces the lumpiness of standard LP estimates at the cost of imposing smoothness.

3. **Bias-corrected local projections.** Local projections with the bias correction of Herbst and Johannsen (2024), which adjusts for the finite-sample bias that arises from projecting on persistent regressors.

4. **Bayesian VAR.** A four-variable BVAR with four lags and hierarchical Minnesota priors following Giannone, Lenza, and Primiceri (2015). Impulse responses are computed by propagating shocks through the VAR companion form rather than estimating separate regressions at each horizon. Point estimates are qualitatively similar to the BLP baseline but somewhat larger, reflecting the longer effective horizon of VAR propagation.

5. **Proxy SVAR.** A frequentist structural VAR using the narrative measures as external instruments following Mertens and Ravn (2013a), extended to 2019Q4. Confidence intervals are constructed using the wild bootstrap. Estimates are less precise but quantitatively similar to the baseline.

6. **Linear trends.** Each of the above methods (where applicable) is re-estimated with a linear trend added to the specification. Results are reported in the lower panel of Table C.1.

7. **LP-IV.** At each horizon h, the log retention rate is instrumented directly with the narrative measure, controlling for four lags of all variables. This specification is robust to non-invertibility of the VAR representation by construction. Three variants are reported in Table C.2: plain LP-IV, smoothed LP-IV following Barnichon and Brownlees (2019), and bias-corrected LP-IV following Herbst and Johannsen (2024). All produce elasticities quantitatively similar to the baseline.

8. **TVP-BVAR.** A four-variable time-varying parameter VAR with stochastic volatility following Primiceri (2005), which allows the full transmission mechanism to drift over time. Figure C.1 reports the resulting present-value elasticities.

Table C.1: Present Value Elasticities of Tax Bases with Respect to Retention Rates

	Corporate Tax Shock		Personal Tax Shock	
Method	ε_{CC}	ε_{LC}	ε_{CL}	ε_{LL}
BLP	2.61	-0.11	7.55	0.98
	(1.89, 3.60)	(-0.31, 0.07)	(5.66, 9.62)	(0.57, 1.40)
LP	4.02	-0.04	11.85	1.05
	(2.71, 6.19)	(-0.63, 0.49)	(6.40, 24.28)	(0.13, 2.16)
SLP	4.04	-0.04	11.79	1.05
	(2.73, 6.20)	(-0.64, 0.49)	(6.32, 24.12)	(0.13, 2.17)
LP-BC	4.01	-0.06	11.54	1.01
	(2.70, 6.23)	(-0.64, 0.46)	(6.22, 23.57)	(0.09, 2.07)
BVAR	4.53	-0.46	3.66	1.25
	(2.83, 7.31)	(-0.74, -0.25)	(0.34, 7.20)	(0.59, 2.10)
SVAR-IV	5.56	-0.24	13.88	1.39
	(-10.90, 36.06)	(-3.82, 2.37)	(7.22, 29.51)	(-0.59, 5.60)
Linear Trends				
BLP	4.10	-0.42	8.49	0.81
	(2.71, 6.13)	(-0.75, -0.09)	(6.68, 10.53)	(0.35, 1.29)
LP	5.47	-0.21	12.37	0.90
	(3.75, 8.36)	(-0.92, 0.63)	(8.99, 19.17)	(0.02, 2.09)
SLP	5.49	-0.21	12.34	0.90
	(3.77, 8.38)	(-0.92, 0.62)	(8.99, 19.17)	(0.02, 2.09)
LP-BC	5.44	-0.23	12.06	0.88
	(3.75, 8.39)	(-0.91, 0.57)	(8.74, 18.65)	(0.00, 2.04)
SVAR-IV	7.25	-0.64	13.65	1.13
	(1.93, 35.95)	(-3.90, 1.65)	(7.97, 30.38)	(-0.67, 6.38)

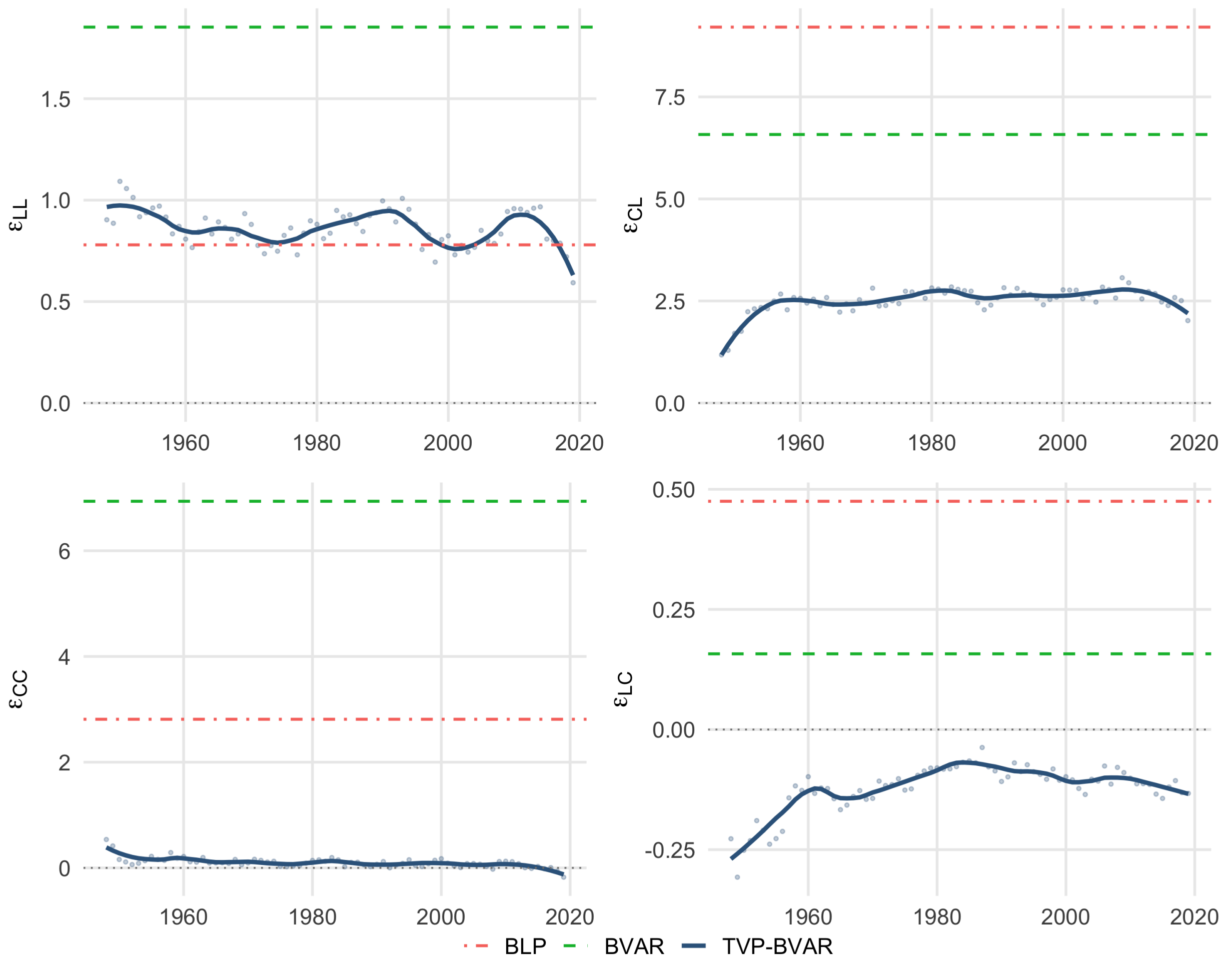
Note: Each elasticity ε_{ij} measures the percent change in base i per one percent increase in retention rate $(1 - \tau_j)$. Bayesian specifications have 90% credible intervals in parentheses, while frequentist ones are 90% confidence intervals. This table reports elasticities for the four-variable system comprised solely of tax bases and rates.

Table C.2: Present Value Elasticities of Tax Bases with Respect to Retention Rates (LP-IV)

	Corporate Tax Shock		Personal Tax Shock	
Method	ε_{CC}	ε_{LC}	ε_{CL}	ε_{LL}
LP-IV	3.75	-0.69	6.46	-0.82
	(-26.08, 34.34)	(-5.58, 3.51)	(1.88, 12.46)	(-1.94, 0.41)
LP-IV (Smooth)	3.77	-0.69	6.47	-0.82
	(-25.05, 34.77)	(-5.54, 3.52)	(1.88, 12.46)	(-1.93, 0.41)
LP-IV (BC)	3.67	-0.71	6.10	-0.82
	(-21.44, 33.20)	(-5.61, 3.20)	(1.69, 11.59)	(-1.90, 0.35)

Note: Each elasticity ε_{ij} measures the percent change in base i per one percent increase in retention rate $(1-\tau_j)$. 90% CI in parentheses.

Figure C.1: Time-Varying Present-Value Elasticities (TVP-VAR)



Note: Posterior median present-value elasticities from a four-variable TVP-VAR with stochastic volatility (Primiceri 2005) estimated over 1947Q1–2019Q4. The reduced-form state vector is initialized at the posterior median of the fixed-coefficient BVAR, with the full sample used for estimation. Structural shocks are identified at each date via the proxy SVAR procedure of Mertens and Ravn (2013b). Draws with explosive companion-matrix eigenvalues are discarded. Elasticities are computed as the ratio of discounted cumulative IRFs. Solid lines show loess-smoothed posterior medians; faint dots show unsmoothed values. Dashed lines indicate fixed-coefficient BVAR and BLP point estimates from Table 1.

C.1 Rationalizing the Magnitude of ε_{CL}

The main text argues that the cross-elasticity from personal taxes to the corporate base reflects operating leverage amplifying the output response to tax cuts. This appendix develops the argument formally and shows it accounts for the full pattern of elasticities in Table C.1.

Operating Leverage. Corporate gross value added decomposes as

$$Y = wL + \pi + F,$$

where wL is the wage bill, π is pre-tax corporate profits, and F is the remainder of costs, including interest payments, taxes, and depreciation. Define the labor share as $\gamma = wL/Y$ and the profit share as $s_\pi = \pi/Y$. When output changes by dY and F does not adjust, the change in profits is $d\pi = dY - d(wL)$. If wages are fully flexible and the labor share is roughly constant at the margin, then $d(wL) = \gamma\, dY$ and

$$d \log \pi = \frac{1-\gamma}{s_\pi}\, d \log Y.$$

The ratio $(1-\gamma)/s_\pi$ is the frictionless-labor operating leverage. It exceeds one whenever $F > 0$, since non-labor costs must be covered before profits are earned.[16]

This is a lower bound on effective leverage because it assumes wages absorb their full share of output movements. When wages are sticky, compensation adjusts by less than $\gamma\, dY$, so a larger share of any output gain accrues to profits. This raises effective leverage above $(1-\gamma)/s_\pi$ and simultaneously attenuates the passthrough of output to the personal tax base below one. The same friction that makes corporate base elasticities larger than the frictionless-labor prediction makes personal base elasticities smaller.

Tax Cuts and the Response of Output. The leverage result shows that the elasticity of the corporate base with respect to any shock decomposes into the output response amplified by $(1-\gamma)/s_\pi$:

$$\varepsilon_{CL} = \frac{1-\gamma}{s_\pi} \times \varepsilon_{YL} \qquad \text{and} \qquad \varepsilon_{CC} = \frac{1-\gamma}{s_\pi} \times \varepsilon_{YC}$$

where ε_{YL} and ε_{YC} are the elasticities of output with respect to the personal and corporate retention rates. These are not directly estimated in the baseline four-variable specification, but the seven-variable specification in Appendix E includes GDP and allows direct estimation. Under the Bayesian local projection, $\varepsilon_{YL} \approx 1.4$ and $\varepsilon_{YC} \approx 0.7$. The asymmetry is intuitive: personal tax cuts raise labor supply through a higher after-tax wage, which immediately raises output, while corporate tax cuts lower the user cost of capital, but capital accumulation is slow and the direct effect is confined to the investment margin.

For the personal base, there is no leverage since wages scale with output under frictionless labor:

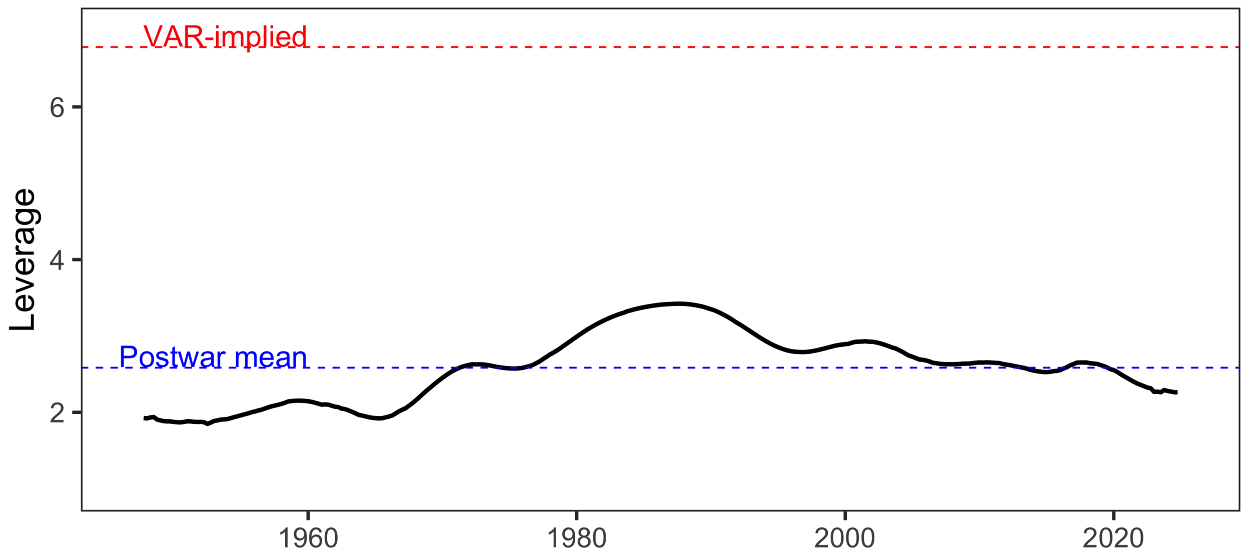
$$\varepsilon_{LL} \approx \varepsilon_{YL} \qquad \text{and} \qquad \varepsilon_{LC} \approx \varepsilon_{YC}.$$

Both γ and s_π are directly observable from NIPA Table 1.14 (Gross Value Added of Domestic Corporate Business): γ is compensation of employees divided by GVA, and s_π is corporate profits with inventory valuation and capital consumption adjustments divided by GVA. Figure C.2 plots the leverage series over time. It ranges from approximately 2 to 3.5, with a postwar mean of 2.6. Empirical leverage from the seven-variable specification—$\varepsilon_{CL}/\varepsilon_{YL} \approx 6.6$—lies above the frictionless-labor benchmark at every date, consistent with wage stickiness raising effective leverage.

16. All that we require is that some portion of the cost structure is predetermined. This is almost trivially satisfied. Capital is subject to fixed adjustment costs, with the majority of firms holding their capital stock fixed in any given quarter (Winberry 2021; Koby and Wolf 2020). Employment adjusts in infrequent, discrete changes rather than continuously (Caballero, Engel, and Haltiwanger 1997). Leases, debt service, and administrative overhead are fixed by construction.

Combining the postwar average leverage of 2.6 with the output elasticities yields the frictionless-labor predictions in Table 2. The benchmark reproduces the qualitative ordering $\varepsilon_{CL} > \varepsilon_{CC} > \varepsilon_{LL} > \varepsilon_{LC}$ and accounts for roughly half the baseline cross-elasticity (3.6 versus 7.6). The systematic residual pattern—underpredicting every corporate entry and overpredicting every personal entry—confirms the wage stickiness channel. The baseline estimate of 7.6 sits between the frictionless-labor floor of 3.6 and the sticky-wage ceiling implied by the seven-variable specification (9.3), exactly where partial wage adjustment would place it. The frictionless-labor benchmark is not intended as a complete account; it shows that the cross-elasticity is a predictable consequence of operating leverage and the output multiplier asymmetry, not of exotic behavioral responses.

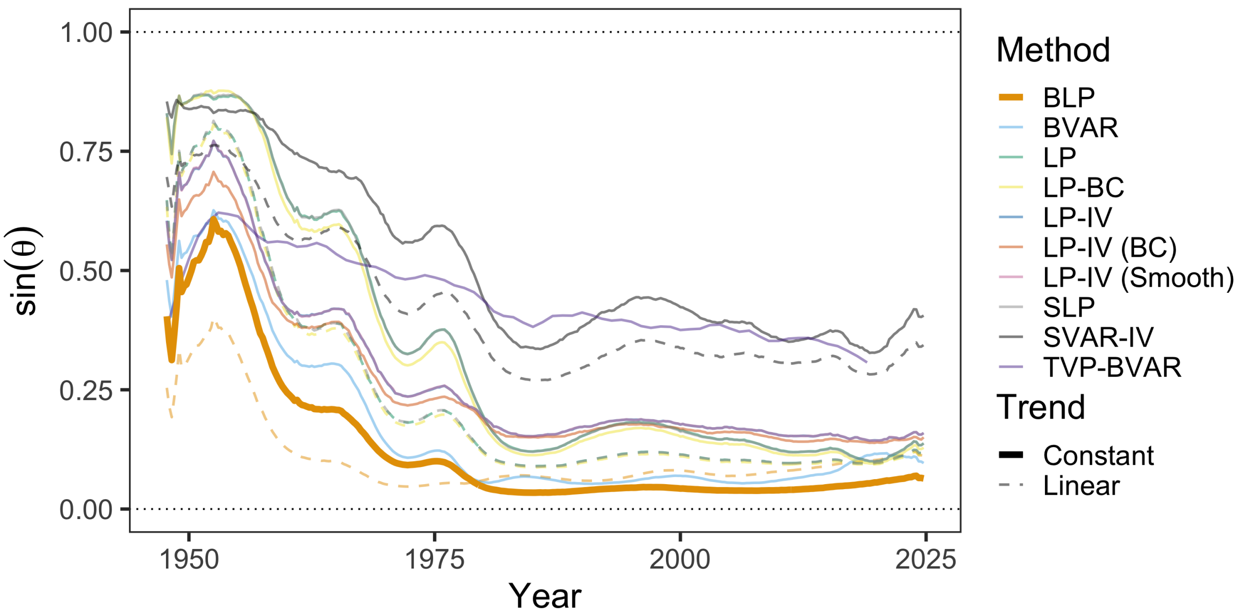
Figure C.2: Leverage in the National Income and Product Accounts



Note: Leverage is computed using Table 1.14 from the National Income and Product Accounts. I compute the compensation share γ as compensation of employees (line 7) divided by gross value added of corporate business (line 1), and the profit share s_π as corporate profits with inventory valuation and capital consumption adjustments (line 14) divided by gross value added of corporate business. Raw leverage is $(1-\gamma)/s_\pi$. I remove fluctuations at business cycle frequencies (2–40 quarters) using the Christiano and Fitzgerald (2003) filter and plot the resulting trend.

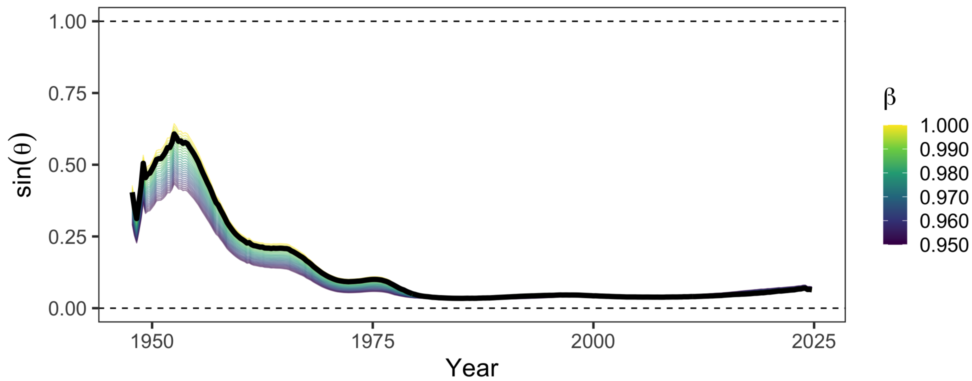
D Additional Alignment Results and Robustness

Figure D.1: Robustness Across Estimation Methodologies



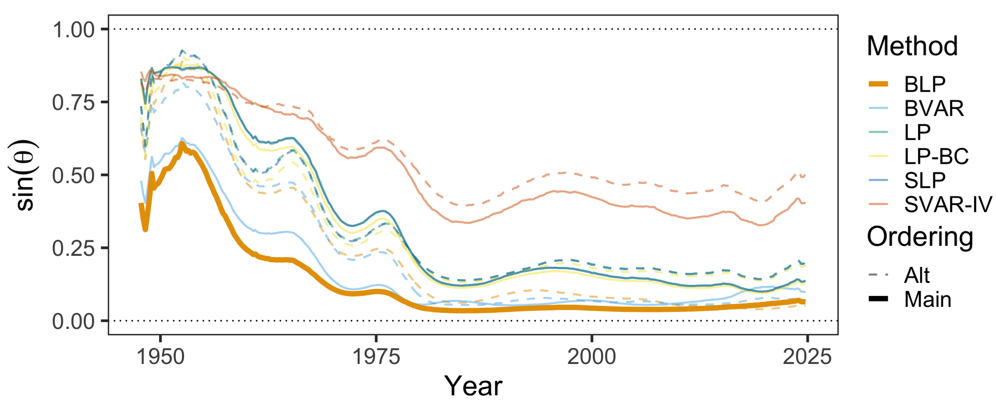
Note: This figure plots $\sin\theta$ across estimation methodologies, including Bayesian local projections (BLP, thick line), frequentist local projections (LP), smooth local projections (SLP), bias-corrected local projections (LP-BC), LP-IV variants, Bayesian VAR (BVAR), proxy SVAR (SVAR-IV), and TVP-BVAR. Solid lines use a constant only. Dashed lines include a linear trend. The baseline BLP is among the most conservative specifications. All methods show the same qualitative trajectory from misalignment to near-optimality.

Figure D.2: Robustness Across Discount Factors



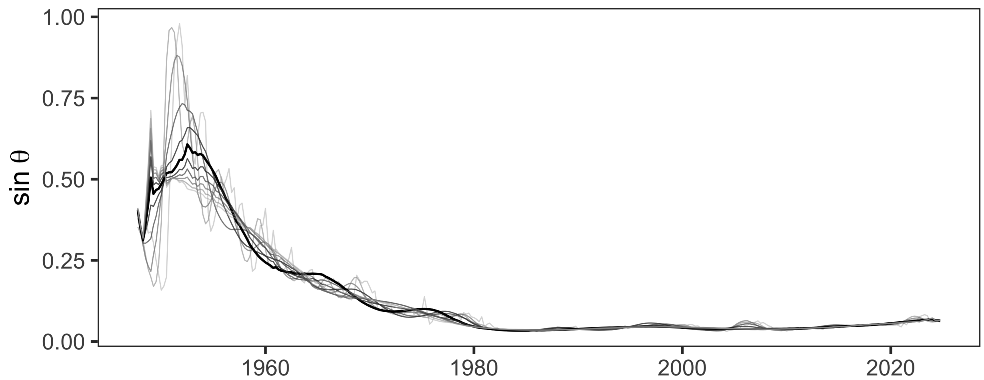
Note: This figure plots $\sin\theta$ across discount factors β ranging from 0.95 to 1. Higher β places more weight on long-horizon responses where fiscal externalities cumulate, increasing measured misalignment in the early postwar period. The baseline uses $\beta = 0.9926$ (approximately 3% annual discount rate). All specifications show the same qualitative trajectory from misalignment to near-optimality.

Figure D.3: Alignment with Reversed Cholesky Ordering



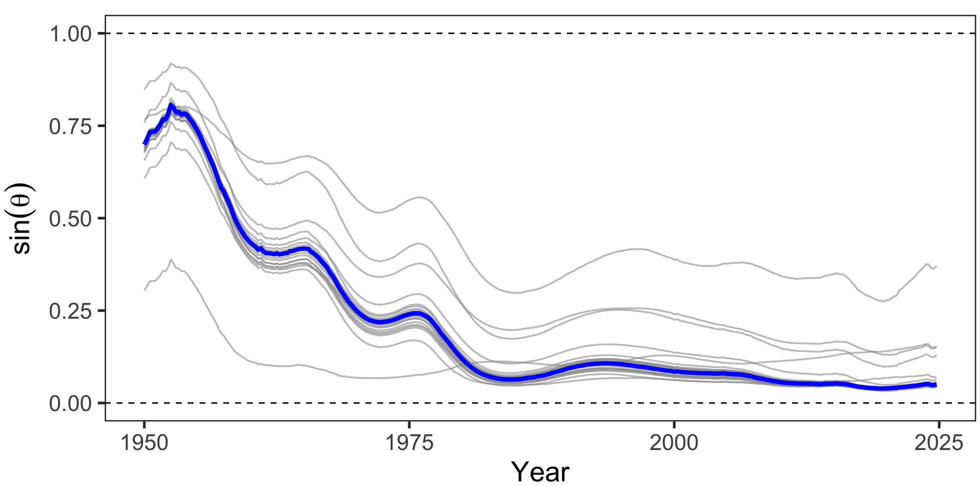
Note: The baseline identifies each tax shock by ordering the shock of interest first in the Cholesky decomposition of the proxy-identified subspace. This figure plots $\sin\theta$ under both the baseline ordering (solid lines) and the reversed ordering (dashed lines) for each estimation method. Results are nearly identical across all methods, indicating the contemporaneous correlation between the two structural tax shocks is small and the identifying restriction is not driving the results.

Figure D.4: Robustness to Filtering Choice



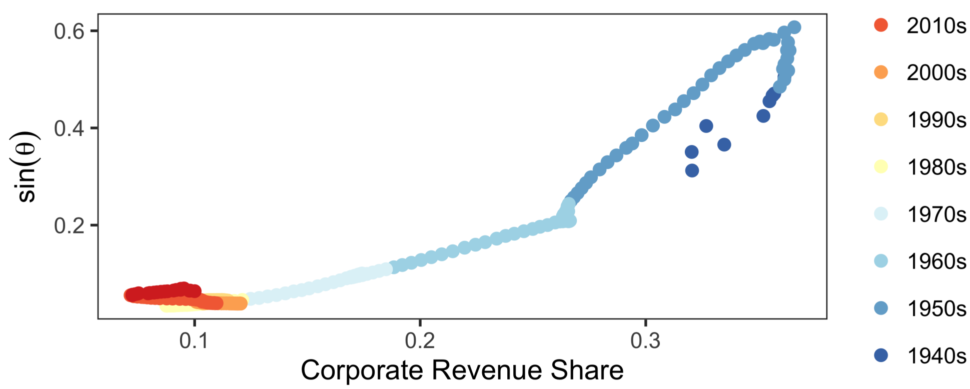
Note: This figure plots $\sin\theta$ under alternative Christiano-Fitzgerald band-pass filter cutoffs applied to tax rates and bases. Filtering removes business-cycle fluctuations that are orthogonal to the permanent tax structure the framework evaluates. "Raw" uses unfiltered quarterly data. Other series retain variation at periodicities above the indicated cutoff in quarters. The baseline specification (40Q) is highlighted in black. Shorter cutoffs produce noisier series reflecting the cyclical sensitivity of corporate profits, but all specifications show the same trajectory and converge to the same level.

Figure D.5: Leave-One-Out Estimator for Bayesian LP



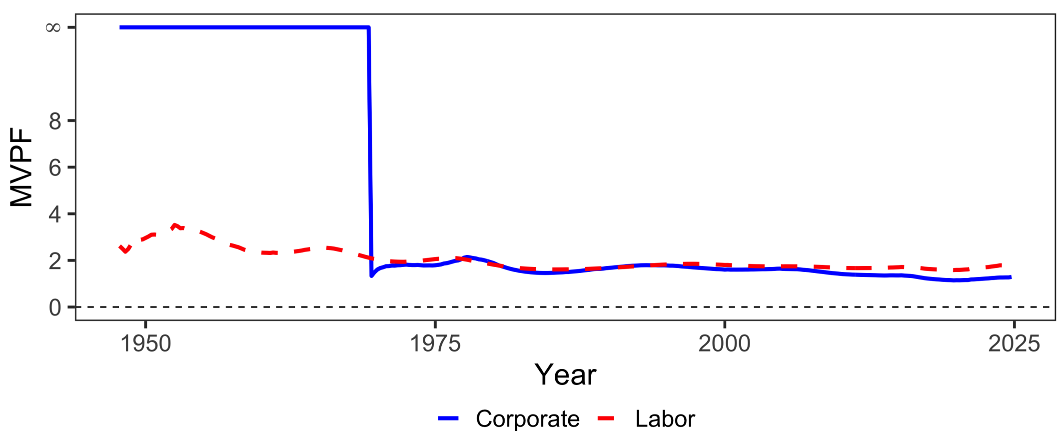
Note: This figure plots $\sin\theta$ over time, re-estimating the baseline Bayesian LP specification dropping each narrative shock in turn. Each line represents the alignment series with one reform excluded from the instrument set. The exercise tests whether any single reform drives the baseline results. The most influential observation is the Tax Reform Act of 1986, whose exclusion shifts the transition period but preserves the broad pattern of improvement.

Figure D.6: System Efficiency and the Corporate Revenue Share



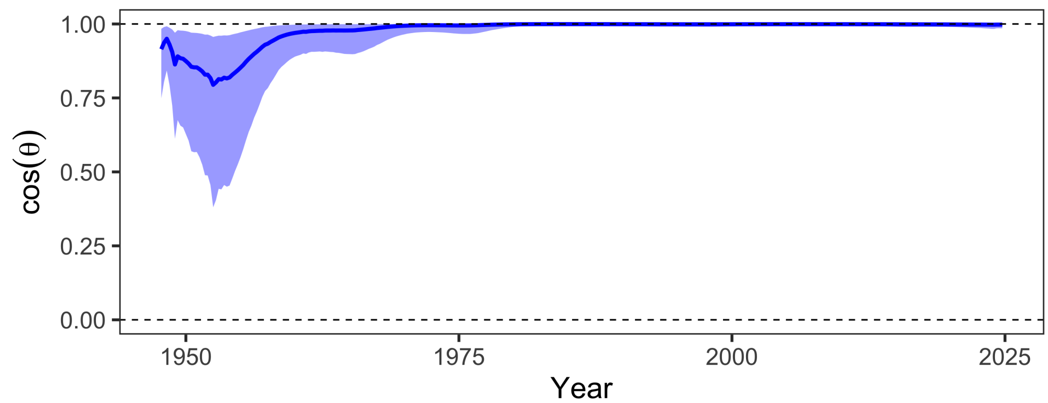
Note: Each point represents a quarter, colored by decade. The horizontal axis plots the corporate share of federal income tax revenue; the vertical axis plots $\sin\theta$, the alignment between welfare and revenue gradients. Higher values indicate greater alignment with the Ramsey optimum.

Figure D.7: The Marginal Value of Public Funds



Note: This figure plots the marginal value of public funds for corporate and personal income taxes. At the Ramsey optimum, MVPFs are equalized across instruments, so convergence in the figure corresponds to the alignment improvement documented in Section 4. The MVPF is undefined when the instrument is beyond the peak of its Laffer curve, following the convention in Finkelstein and Hendren (2020). The corporate MVPF starts undefined in the early postwar period, indicating corporate taxes were generating negative net revenue at the margin, and falls toward convergence with the personal MVPF by the 1980s.

Figure D.8: The Evolution of Tax System Efficiency



Note: $\cos\theta$ is the alignment between welfare and revenue gradients; $\cos\theta = 1$ indicates the Ramsey optimum. Shaded region shows 90% credible interval.

E Seven-Variable System Robustness

The baseline specification uses four variables: personal and corporate tax rates and their corresponding bases. This appendix explains why the four-variable specification is appropriate for estimating revenue gradients, and shows robustness to a seven-variable specification that adds output, government spending, and debt.

The revenue gradient in the projection framework is the total derivative of present-value revenue with respect to the tax rate:

$$r_j = \frac{\partial}{\partial \tau_j} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t R_t,$$

where $R_t = \tau_L B_L + \tau_C B_C$ is total income tax revenue. The key object is $\frac{\partial B_k}{\partial \tau_j}$, the total derivative of base k with respect to tax rate j. This total derivative includes all channels through which a tax change affects the base: direct behavioral responses, income shifting between bases, and general equilibrium effects through output, investment, and employment. The theory does not ask for $\frac{\partial B_k}{\partial \tau_j}\big|_Y$, the partial derivative holding output constant. It asks for the full response that a policymaker would observe if they changed the tax rate.

The four-variable specification delivers this total derivative. When we estimate the impulse response of the corporate base to a personal tax shock, the response captures all channels, including any effects operating through GDP. We are not conditioning on GDP, so the full reduced-form relationship between tax shocks and bases is preserved.

A seven-variable specification that includes GDP, government spending, and debt estimates a different object. By including lagged GDP as a regressor in the local projection, we ask: what is the effect of a tax shock on the corporate base, controlling for GDP? This partials out GDP-mediated effects. If a personal tax cut raises GDP, and higher GDP raises corporate profits, then including GDP in the system absorbs some of the variation we want to attribute to the tax shock. The resulting elasticity is a conditional object that understates the total revenue response.

One might worry that omitting GDP leaves the tax shock correlated with other structural disturbances, biasing the estimates. But narrative identification does not require controlling for other variables. The Mertens-Ravn proxy SVAR achieves identification through moment conditions requiring that the narrative instruments correlate with structural tax shocks and are uncorrelated with other structural shocks. These conditions are satisfied by construction: Romer and Romer (2010) selected tax changes legislated for reasons unrelated to current economic conditions. This selection is what ensures exogeneity, not the inclusion of control variables. Adding variables to the VAR does not improve identification; it changes what object is being estimated.

Mertens and Ravn (2013a) included output, government spending, and debt because they were interested in output multipliers and fiscal sustainability. For their research question, the seven-variable specification was appropriate. For the present question—what are the revenue effects of tax changes for evaluating Ramsey alignment?—the total elasticity is required, and the additional variables are unnecessary.

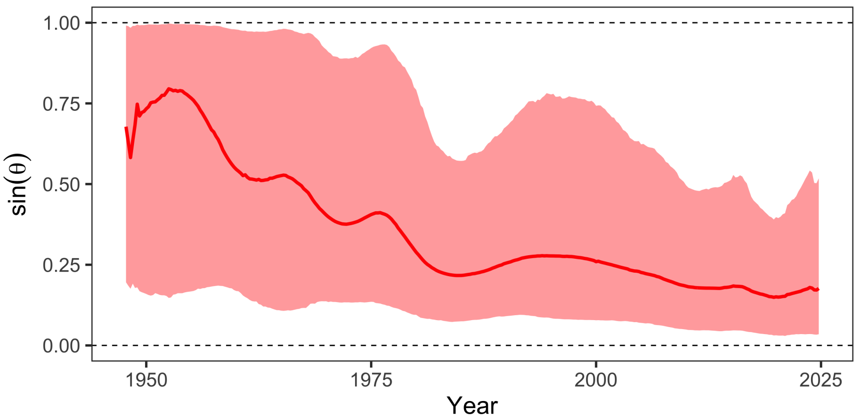
Table E.1 reports elasticities from both specifications. The seven-variable specification yields qualitatively similar point estimates but wider credible intervals, as expected given the additional parameters. The alignment metrics are robust to this choice. The four-variable specification is the appropriate baseline because it estimates the object the theory requires; the seven-variable specification serves as a robustness check demonstrating that results are not sensitive to the conditioning set.

Table E.1: Present Value Elasticities of Tax Bases with Respect to Retention Rates (Seven Variable System)

	Corporate Tax Shock			Personal Tax Shock		
Method	ε_{CC}	ε_{LC}	ε_{CY}	ε_{CL}	ε_{LL}	ε_{LY}
BLP	2.8	0.5	0.7	9.3	0.8	1.4
	(1.9, 4.3)	(0.2, 0.9)	(0.4, 1.2)	(5.5, 19.2)	(-0.0, 2.8)	(0.4, 4.0)
LP	3.9	0.9	1.1	5.7	-0.1	0.4
	(2.5, 6.0)	(0.4, 1.6)	(0.7, 1.9)	(-4.5, 18.4)	(-2.6, 2.8)	(-2.6, 4.1)
SLP	3.9	0.9	1.1	5.6	-0.1	0.3
	(2.6, 6.0)	(0.5, 1.6)	(0.7, 1.9)	(-4.7, 18.1)	(-2.6, 2.8)	(-2.7, 4.1)
LP-BC	3.9	0.8	1.1	5.4	-0.1	0.4
	(2.6, 6.1)	(0.4, 1.5)	(0.7, 1.8)	(-5.2, 17.8)	(-2.4, 2.6)	(-2.6, 4.0)
BVAR	6.9	0.2	1.3	6.6	1.9	2.1
	(2.3, 31.0)	(-0.7, 2.2)	(0.3, 6.4)	(0.9, 21.3)	(0.8, 4.2)	(0.8, 5.8)
SVAR-IV	6.1	1.3	1.7	7.6	1.0	1.4
	(-5.6, 30.3)	(-0.8, 6.6)	(-1.3, 8.3)	(-17.1, 39.9)	(-3.7, 11.0)	(-4.9, 12.6)

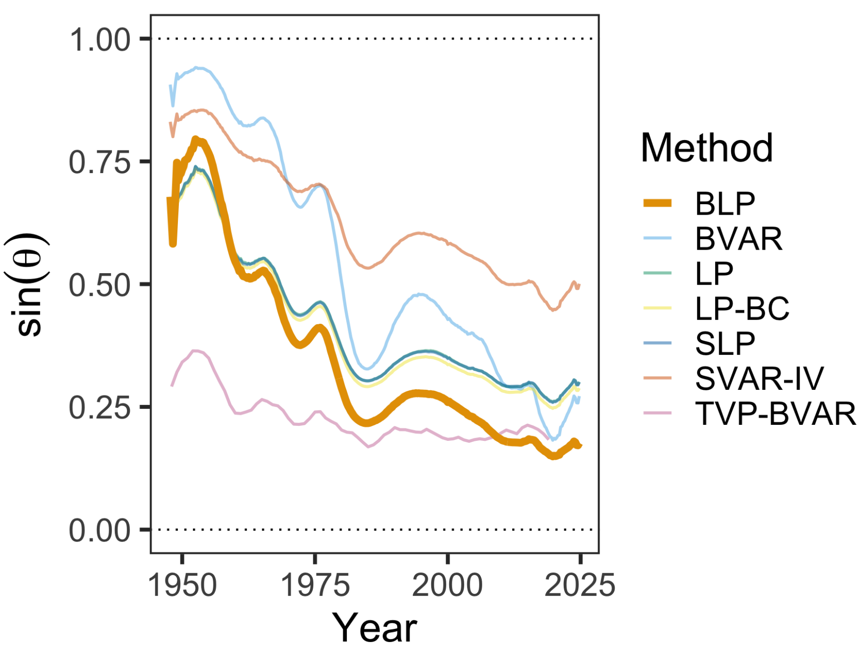
Note: Each elasticity ε_{ij} measures the percent change in base i per one percent increase in retention rate $(1 - \tau_j)$. Bayesian specifications have 90% credible intervals in parentheses, while frequentist ones are 90% confidence intervals. This table reports elasticities for the seven-variable system, which adds government spending, debt, and GDP to the four-variable system. The output elasticities ε_{CY} and ε_{LY} are used in the operating leverage decomposition in Section 3.5 and Appendix C.1.

Figure E.1: Evolution of Tax System Efficiency



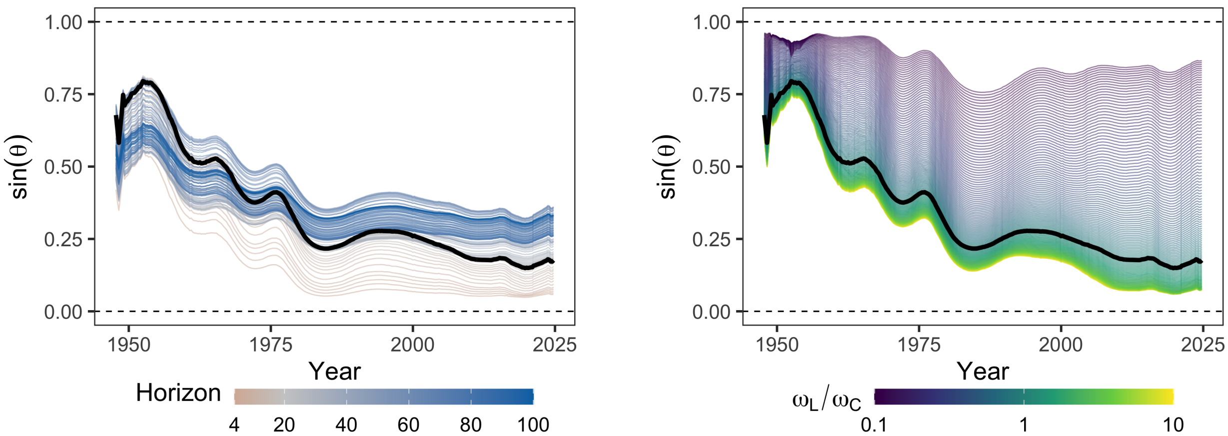
Note: This figure plots $\sin\theta$ for the seven-variable specification, which adds output, government spending, and debt to the baseline four-variable system. Shaded region shows 90% credible intervals from the Bayesian local projection.

Figure E.2: Robustness Across Estimation Methodologies



Note: This figure plots $\sin\theta$ across estimation methodologies for the seven-variable specification, which adds output, government spending, and debt to the baseline. Methods include Bayesian local projections (BLP, thick line), frequentist local projections (LP), smooth local projections (SLP), bias-corrected local projections (LP-BC), Bayesian VAR (BVAR), and proxy SVAR (SVAR-IV).

Figure E.3: Sensitivity of $\sin\theta$ to Horizon and Welfare Weights



(a) Truncation horizon H

(b) Welfare weight ratio ω_L/ω_C

Note: Panel (a) varies the truncation horizon H used to compute present-value elasticities, with color indicating the number of quarters. Panel (b) varies the ratio of welfare weights on personal to corporate income tax claimants, with color on a log scale. In both panels, the black line denotes the baseline specification ($H = 20$, $\omega_L/\omega_C = 1$). These figures use the seven-variable specification.

F Robustness to S-Corporation Treatment

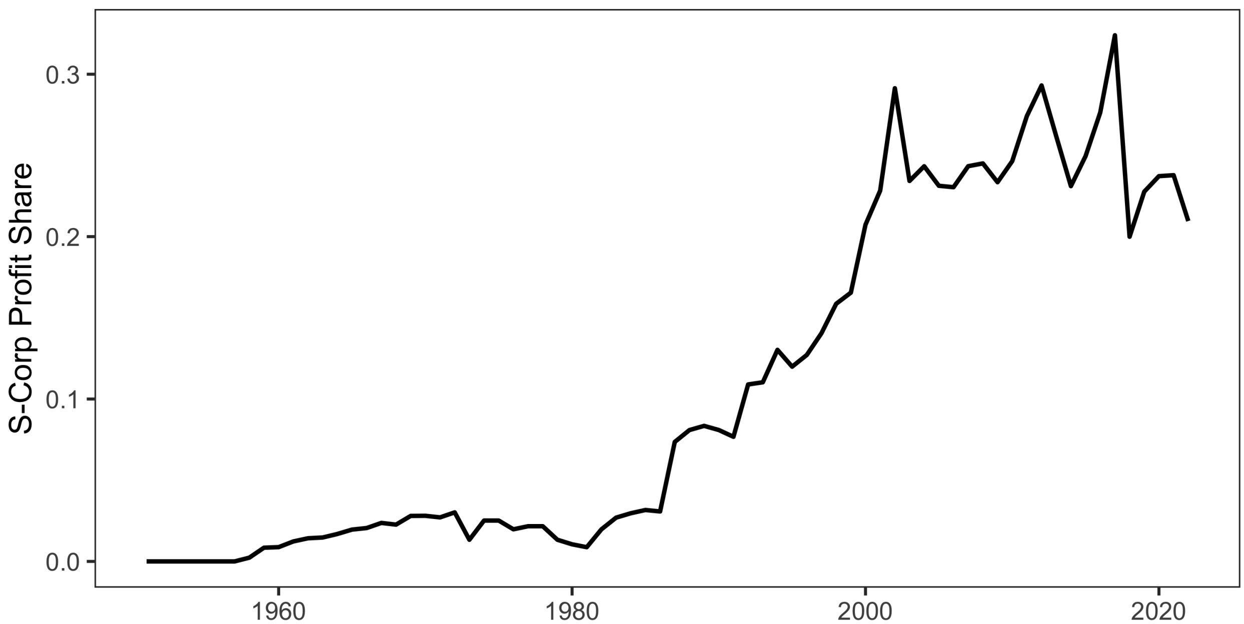
The baseline specification uses NIPA corporate profits as the corporate tax base, which includes both C-corporation and S-corporation income. Since S-corporations are pass-through entities whose income is taxed at the shareholder level rather than at the entity, one might be concerned that the results are driven by organizational form arbitrage rather than traditional general equilibrium channels.

To address this concern, we construct adjusted tax bases that reallocate S-corporation income from the corporate to the personal base. I use the Annual Corporate Reports from the Statistics of Income (SOI) to compile a time series of the profits of C-corporations and S-corporations. I measure profits for the former using the SOI's measure of "Income Subject to Tax" for form 1120 filers (excluding 1120-REIT, 1120-RIC, and 1120-S filers) and profits for the latter as "Net Income (Less Deficit)" for form 1120-S filers. Since Subchapter S was enacted in 1958, allowing qualifying corporations to elect passthrough treatment, I do this for every year from 1958-2020. Toward reallocating NIPA corporate income for S-corps, I compute the S-corporation share of total corporate income as:

$$s_t = \frac{\text{S-corp net income}_t}{\text{S-corp net income}_t + \text{C-corp net income}_t} \tag{A.12}$$

using SOI data for both numerator and denominator to maintain consistency. The resulting series is plotted in Figure F.1. As is well-documented elsewhere, s_t is negligible until after the Tax Reform Act of 1986; since then, S-corps have been large relative to C-corps (Dyrda and Pugsley 2024).

Figure F.1: S-Corporate Share of Corporate Profits



Note: The S-corporate share of corporate profits is computed from the Statistics of Income's Annual Corporate Reports.

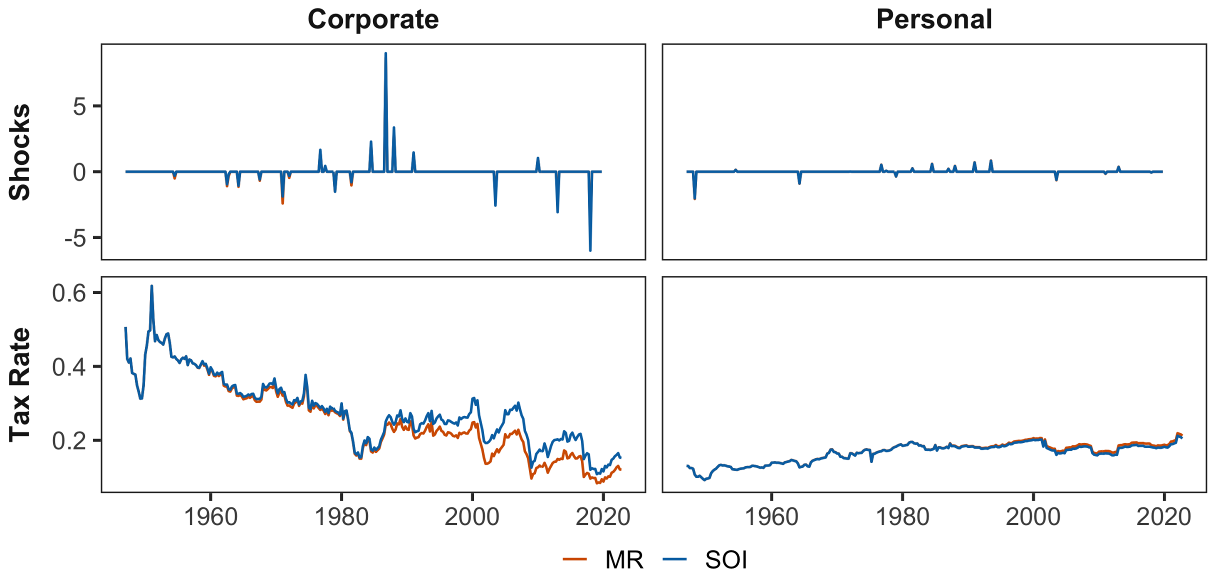
NIPA applies a number of conceptual adjustments to corporate profits which are sensible economically, but are not present in the SOI data. To maintain consistency, I assume that these adjustments are proportional, and get quarterly tax bases by applying s_t in year t to NIPA data in every quarter q of that year as:

$$\text{Corporate base}_{q,t}^{adj} = \text{NIPA corporate profits}_{q,t} \times (1 - s_t) \tag{A.13}$$

$$\text{Personal base}_{q,t}^{adj} = \text{NIPA personal income}_{q,t} + \text{NIPA corporate profits}_{q,t} \times s_t \tag{A.14}$$

Table F.1 reports the estimated elasticities under the adjusted bases. Several patterns are notable. First, the own-elasticity of the corporate base falls from 2.61 to 2.08 under BLP estimation. This is consistent with the baseline estimate capturing some organizational form arbitrage: when personal taxes change, firms shift between C-corp and S-corp status, and this response inflates the measured elasticity of NIPA corporate profits. Isolating C-corporation income removes this margin, yielding a smaller but still substantial own-elasticity.

Figure F.2: Comparing Mertens-Ravn and Reallocated S-Corp Tax Rates and Shocks



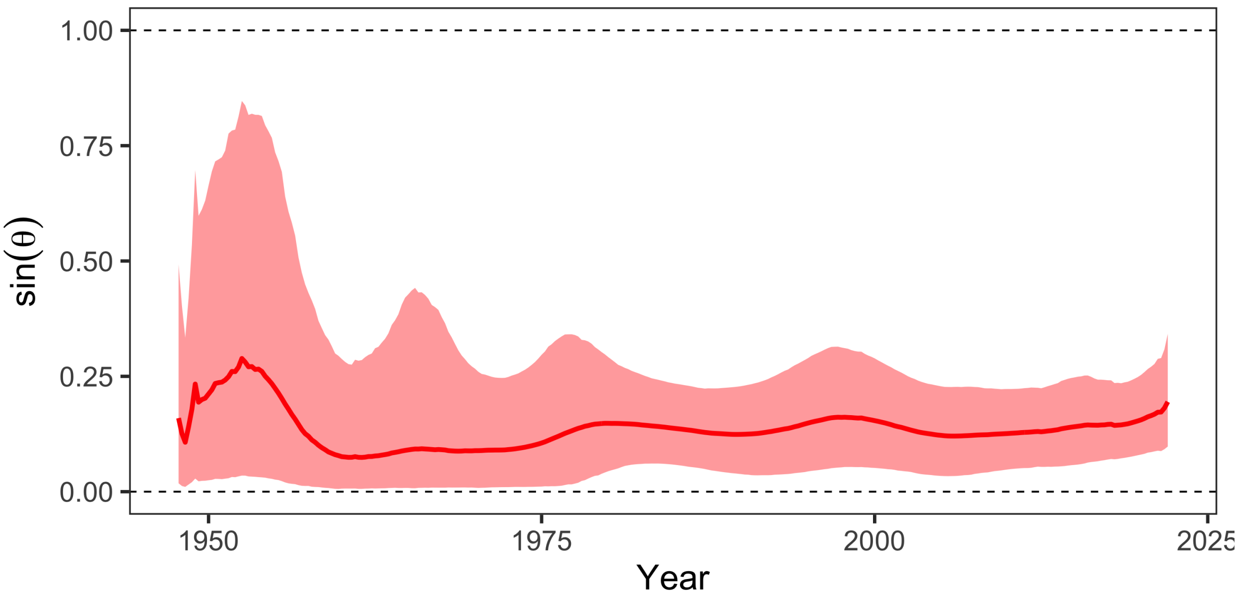
Note: After reallocating S-corps to the personal tax base, I recompute average tax rates and tax shocks (as shown above). There is a somewhat sizable difference for the average corporate tax rate, but the results are similar otherwise.

Second, the cross-elasticity ε_{CL} remains large at 7.39 under BLP, virtually unchanged from the baseline 7.55. This is the key finding: the large cross-base spillover from personal to corporate taxes is not an artifact of S-corporation dynamics. Even when S-corporation income is stripped from the corporate base and added to the personal base, personal tax cuts still generate substantial expansions in C-corporation profits. This suggests the cross-elasticity reflects genuine general equilibrium channels—demand effects, labor-capital complementarity, investment responses—rather than mere organizational form shifting.

Third, the cross-elasticity ε_{LC} becomes more negative (-0.34 versus -0.11 in the baseline), and the personal own-elasticity ε_{LL} rises modestly (1.40 versus 0.98). Both patterns are consistent with the reallocation: S-corporation income, which responds positively to personal tax cuts, has been moved from the corporate base (where it attenuated negative ε_{LC}) to the personal base (where it amplifies positive ε_{LL}).

The qualitative pattern—large asymmetric cross-elasticity, positive own-elasticities—is robust across all estimation methods. The frequentist LP variants yield somewhat larger ε_{CL} estimates (12–13) with wider confidence intervals, while BVAR produces estimates similar to BLP. The key finding that $\varepsilon_{CL} \gg |\varepsilon_{LC}|$ holds uniformly.

Figure F.3: Tax System Alignment with S-Corp Reallocation



Note: This figure plots $\sin\theta$ for the S-corp adjusted specification, which reallocates S-corporation income from the corporate to the personal base using IRS Statistics of Income data. Shaded region shows 90% credible interval from the Bayesian local projection. The transition from misalignment to near-optimality occurs over the same period as in the baseline specification.

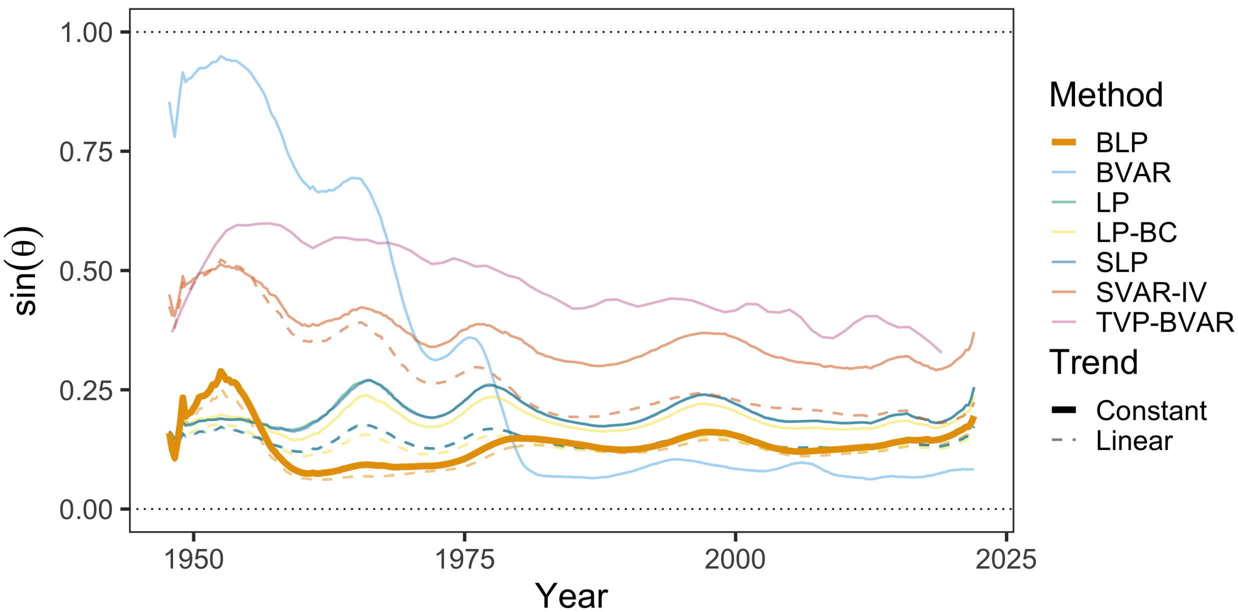
Table F.1: Present Value Elasticities of Tax Bases with Respect to Retention Rates (S-Corp Adjusted)

	Corporate Tax Shock		Personal Tax Shock	
Method	ε_{CC}	ε_{LC}	ε_{CL}	ε_{LL}
BLP	2.08	-0.34	7.39	1.40
	(1.45, 2.88)	(-0.50, -0.20)	(4.62, 11.62)	(0.83, 1.92)
LP	1.87	-0.36	10.94	1.28
	(1.39, 2.49)	(-0.55, -0.19)	(5.13, 30.09)	(-0.07, 2.27)
SLP	1.87	-0.36	10.89	1.29
	(1.40, 2.49)	(-0.55, -0.19)	(5.12, 29.46)	(-0.03, 2.26)
LP-BC	1.95	-0.37	10.69	1.21
	(1.47, 2.61)	(-0.56, -0.20)	(5.06, 28.18)	(-0.13, 2.16)
BVAR	7.58	-0.64	6.76	1.07
	(4.82, 14.23)	(-1.09, -0.35)	(2.15, 12.15)	(0.39, 1.88)
SVAR-IV	2.76	-0.65	11.68	1.18
	(0.81, 10.82)	(-3.17, 0.14)	(6.47, 27.48)	(-0.84, 4.32)
With Linear Trend				
BLP	2.19	-0.40	6.84	1.10
	(1.54, 3.05)	(-0.55, -0.26)	(4.10, 11.18)	(0.52, 1.63)
LP	1.87	-0.36	10.66	0.83
	(1.39, 2.49)	(-0.52, -0.21)	(3.54, 35.60)	(-1.12, 2.02)
SLP	1.88	-0.36	10.66	0.83
	(1.39, 2.50)	(-0.52, -0.21)	(3.53, 35.73)	(-1.14, 2.02)
LP-BC	1.96	-0.37	10.47	0.78
	(1.47, 2.60)	(-0.53, -0.22)	(3.51, 33.13)	(-1.10, 1.95)
SVAR-IV	0.84	-0.50	10.59	0.41
	(-1.66, 8.33)	(-1.52, 0.07)	(4.55, 29.43)	(-1.65, 3.68)

Note: Each elasticity ε_{ij} measures the percent change in base i per one percent increase in retention rate $(1 - \tau_j)$. Bayesian specifications have 90% credible intervals in parentheses, while frequentist ones are 90% confidence intervals. This table reports elasticities for the four-variable system with an adjustment for moving s-corp income from the corporate to the personal base.

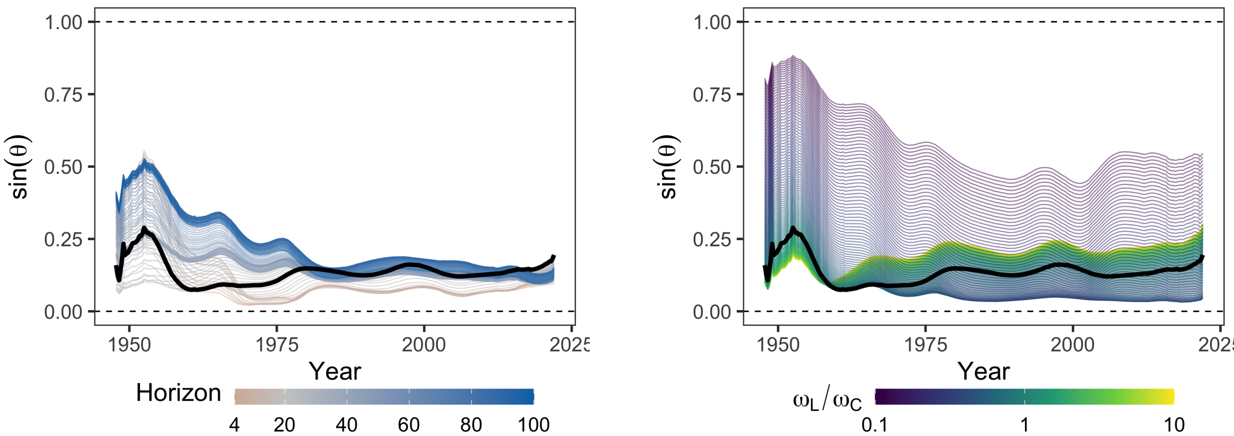
Figure F.3 shows that the alignment results are similarly robust. The transition from misalignment to near-optimality occurs over the same period as in the baseline specification. $\sin\theta$ settles around 0.15 by the 2010s, somewhat above the four-variable baseline but confirming that the efficiency gains documented in Section 4 are not an artifact of organizational form shifting.

Figure F.4: Evolution of $\sin\theta$ Across Methods



Note: This figure plots $\sin\theta$ over time for the S-corp adjusted specification across estimation methodologies, including Bayesian local projections (BLP, thick line), frequentist local projections (LP), smooth local projections (SLP), bias-corrected local projections (LP-BC), Bayesian VAR (BVAR), and proxy SVAR (SVAR-IV). Estimation details for each method are described in Appendix C.

Figure F.5: Sensitivity of $\sin\theta$ to Horizon and Welfare Weights



(a) Truncation horizon H (b) Welfare weight ratio ω_L/ω_C

Note: Panel (a) varies the truncation horizon H used to compute present-value elasticities, with color indicating the number of quarters. Panel (b) varies the ratio of welfare weights on personal to corporate income tax claimants, with color on a log scale. In both panels, the black line denotes the baseline specification ($H = 20$, $\omega_L/\omega_C = 1$). These figures use the S-corp adjusted specification.

G Three-Instrument Extension

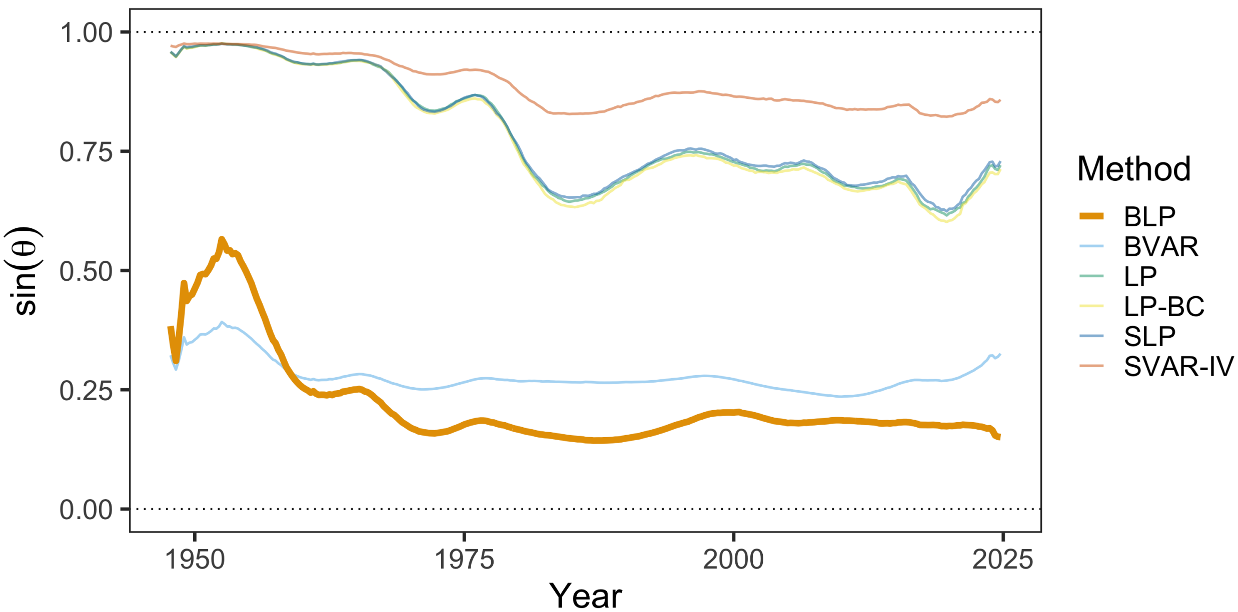
G.1 Baseline

Table G.1: Present-Value Elasticities of Tax Bases with Respect to Retention Rates (Three-Instrument System)

	Corporate Tax Shock			Personal Tax Shock			Tariff Shock		
Method	ε_{CC}	ε_{LC}	ε_{TC}	ε_{CL}	ε_{LL}	ε_{TL}	ε_{CT}	ε_{LT}	ε_{TT}
BLP	2.52	-0.29	0.54	7.52	0.67	-3.09	0.98	-0.60	14.24
	(1.81, 3.49)	(-0.55, -0.05)	(-0.92, 2.32)	(4.53, 10.39)	(0.04, 1.38)	(-7.31, 0.51)	(-2.86, 4.56)	(-1.59, 0.35)	(11.93, 17.03)
LP	4.30	-0.11	0.45	7.00	0.64	-1.19	18.56	-0.26	17.34
	(1.09, 8.50)	(-1.11, 0.86)	(-3.52, 6.29)	(-22.12, 38.36)	(-1.77, 3.40)	(-10.20, 8.20)	(-128.49, 194.07)	(-14.30, 14.48)	(-18.76, 54.67)
SLP	4.31	-0.11	0.46	6.98	0.64	-1.22	18.41	-0.27	17.28
	(1.14, 8.74)	(-1.10, 0.86)	(-3.51, 6.39)	(-21.88, 37.53)	(-1.78, 3.38)	(-9.99, 8.09)	(-129.88, 186.54)	(-14.12, 14.76)	(-19.24, 56.67)
LP-BC	4.27	-0.14	0.35	6.65	0.62	-1.27	18.99	-0.27	16.94
	(0.91, 9.45)	(-1.18, 0.80)	(-3.52, 6.36)	(-23.87, 34.75)	(-1.67, 3.39)	(-9.89, 7.58)	(-132.88, 202.51)	(-14.86, 13.85)	(-16.10, 52.94)
BVAR	2.80	-0.41	-0.09	1.09	0.51	-1.78	0.30	1.16	14.90
	(1.68, 5.20)	(-0.68, -0.24)	(-1.19, 1.80)	(-1.60, 3.55)	(-0.06, 1.20)	(-4.85, 0.87)	(-6.76, 8.81)	(-0.53, 2.87)	(9.43, 23.68)
SVAR-IV	4.39	-0.38	-1.64	10.54	1.07	-1.23	17.41	0.51	13.73
	(-20.36, 35.82)	(-5.85, 3.56)	(-15.86, 11.58)	(-21.53, 42.52)	(-6.09, 10.77)	(-18.33, 22.51)	(-162.05, 202.65)	(-28.71, 23.18)	(-39.50, 58.60)

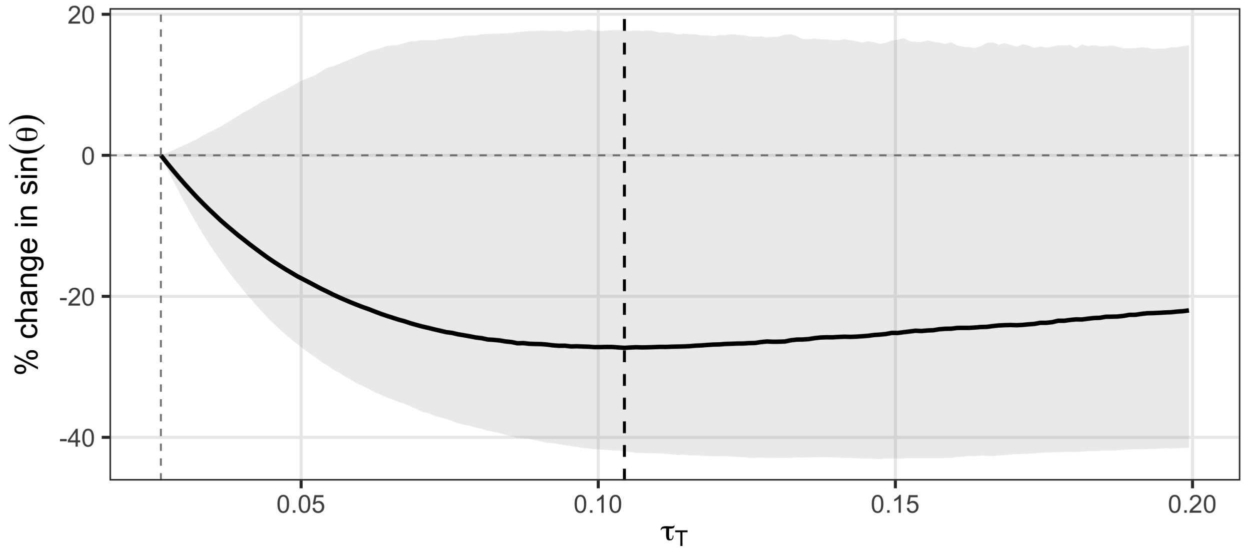
Note: Each elasticity ε_{ij} measures the cumulative discounted percent change in base i per one percent increase in the retention rate of instrument j. Bayesian specifications report 90% credible intervals in parentheses, while frequentist specifications report 90% confidence intervals.

Figure G.1: Three-Instrument $\sin\theta$ Across Estimation Methods



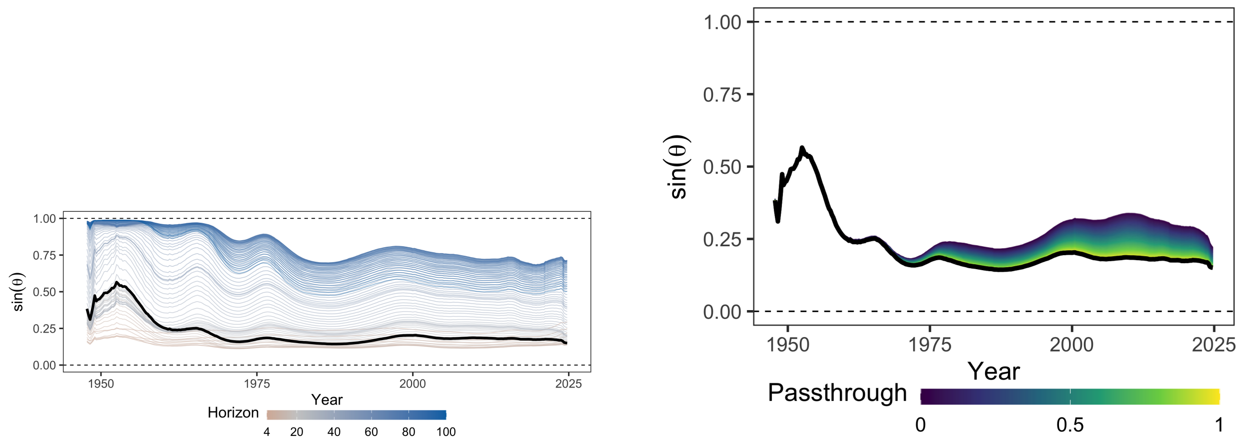
Note: This figure plots $\sin\theta$ for the three-instrument system (personal taxes, corporate taxes, and tariffs) across estimation methodologies. The BLP (thick line) and BVAR produce informative alignment series that show improvement over the postwar period, settling around 0.2 by the 2010s. The frequentist methods (LP, LP-BC, SVAR-IV) produce $\sin\theta$ near or above 0.75 throughout the sample, reflecting the imprecision of tariff elasticity estimates under those methods rather than a substantive disagreement about the direction of alignment. This pattern motivates the Bayesian approach for the three-instrument extension.

Figure G.2: Counterfactual Tariff-for-Corporate-Tax Swap: Posterior Uncertainty



Note: This figure plots the percent change in $\sin\theta$ from a revenue-neutral swap of tariffs for corporate taxes, starting from 2024Q4 rates and bases, under full domestic passthrough. The solid line uses median BLP elasticities; the shaded region shows the 90% credible interval from the full posterior. The thin dashed vertical line marks the 2024Q4 tariff rate; the thick dashed vertical line marks the optimum at median elasticities. The improvement is statistically indistinguishable from zero at the 90% level throughout the grid.

Figure G.3: Sensitivity of Three-Instrument $\sin\theta$ to Horizon and Tariff Passthrough

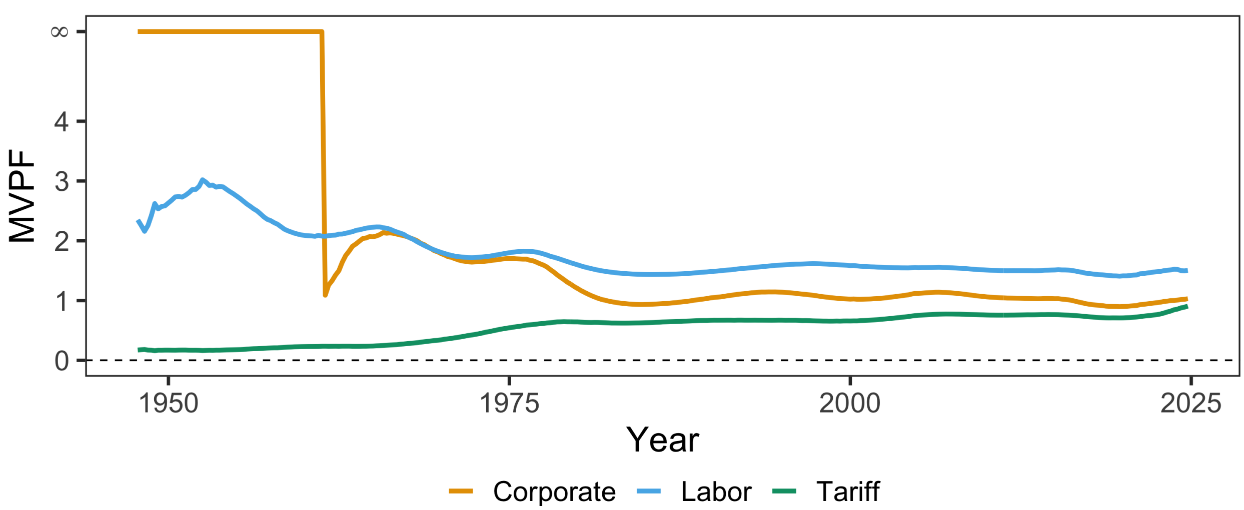


(a) Truncation horizon H

(b) Tariff passthrough

Note: Both panels plot $\sin\theta$ for the three-instrument system (personal taxes, corporate taxes, and tariffs). Panel (a) varies the truncation horizon H used to compute present-value elasticities, with color indicating the number of quarters. Panel (b) varies the share of tariff incidence borne by domestic consumers, with color on a continuous scale from zero (foreign exporters bear the full burden) to one (full domestic passthrough). In both panels, the black line denotes the baseline specification ($H = 20$, full domestic passthrough).

Figure G.4: Marginal Value of Public Funds by Instrument



Note: This figure plots the marginal value of public funds (MVPF) for each instrument in the three-instrument system, computed from the baseline Bayesian local projection elasticities with full domestic tariff passthrough.

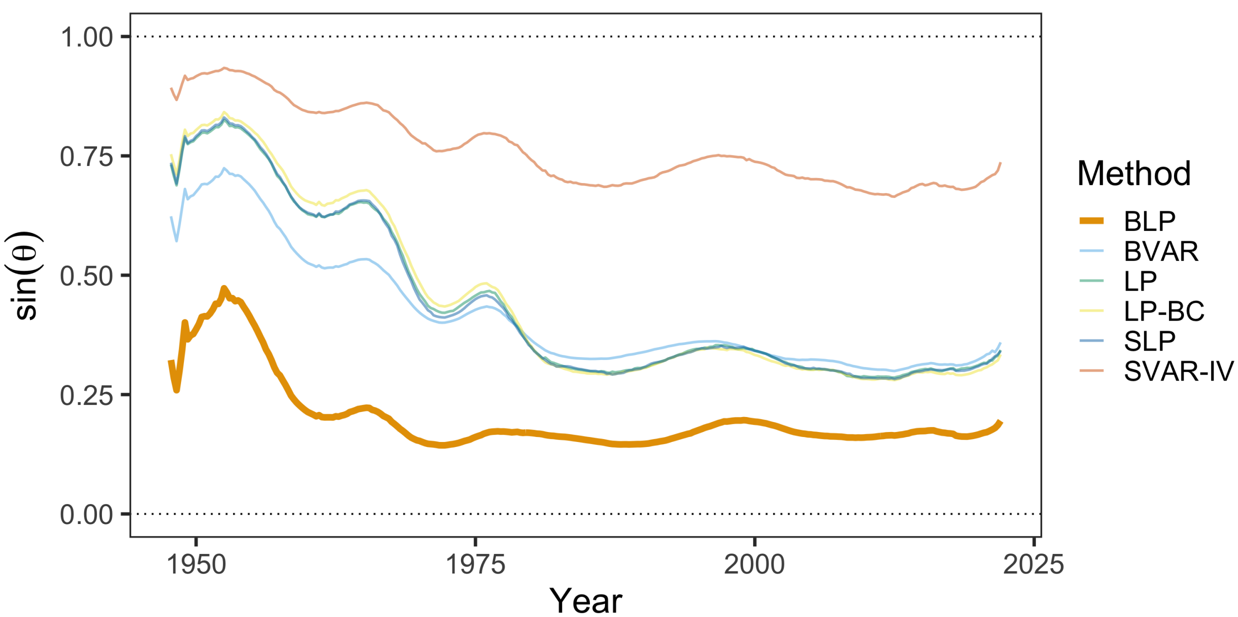
Table G.2: Present Value Elasticities of Tax Bases with Respect to Retention Rates

	Corporate Tax Shock			Personal Tax Shock			Tariff Shock		
Method	ε_{CC}	ε_{LC}	ε_{TC}	ε_{CL}	ε_{LL}	ε_{TL}	ε_{CT}	ε_{LT}	ε_{TT}
BLP	2.30	-0.31	1.02	6.97	1.13	-2.68	0.89	-0.32	13.56
	(1.68, 3.06)	(-0.50, -0.15)	(-0.30, 2.62)	(3.18, 11.80)	(0.40, 1.85)	(-7.47, 1.90)	(-2.68, 4.39)	(-1.29, 0.59)	(11.26, 16.25)
LP	2.45	-0.27	0.87	3.27	1.23	-3.67	13.50	-1.41	22.23
	(1.77, 3.42)	(-0.57, 0.03)	(-0.48, 2.62)	(-4.70, 14.47)	(0.08, 2.49)	(-8.43, 2.15)	(3.73, 49.74)	(-7.76, 1.19)	(16.40, 42.62)
SLP	2.47	-0.27	0.87	3.31	1.24	-3.71	13.37	-1.43	22.23
	(1.78, 3.41)	(-0.57, 0.03)	(-0.49, 2.63)	(-4.60, 14.43)	(0.08, 2.50)	(-8.46, 2.14)	(3.63, 49.95)	(-7.74, 1.19)	(16.35, 43.01)
LP-BC	2.57	-0.28	0.79	3.01	1.19	-3.62	14.45	-1.42	22.01
	(1.87, 3.61)	(-0.58, 0.01)	(-0.54, 2.58)	(-5.27, 13.83)	(0.05, 2.42)	(-8.39, 2.02)	(4.04, 53.67)	(-7.88, 1.16)	(16.26, 42.31)
BVAR	5.54	-0.40	0.52	1.83	0.44	-1.78	2.81	1.15	14.83
	(3.34, 10.89)	(-0.71, -0.18)	(-0.87, 3.22)	(-2.49, 5.69)	(-0.15, 1.13)	(-5.40, 1.13)	(-6.26, 14.90)	(-0.29, 2.65)	(9.81, 23.23)
SVAR-IV	2.76	-0.48	-0.86	5.06	1.41	-2.45	14.18	-0.58	15.19
	(0.58, 10.25)	(-2.22, 0.41)	(-4.70, 3.27)	(-8.84, 16.96)	(-0.75, 6.23)	(-10.14, 6.85)	(-49.59, 102.26)	(-16.30, 10.89)	(-5.17, 40.26)

Note: Each elasticity ε_{ij} measures the cumulative discounted percent change in base i per one percent increase in the retention rate of instrument j. Bayesian specifications report 90% credible intervals in parentheses, while frequentist specifications report 90% confidence intervals. This table reallocates s-corporations to the personal tax base.

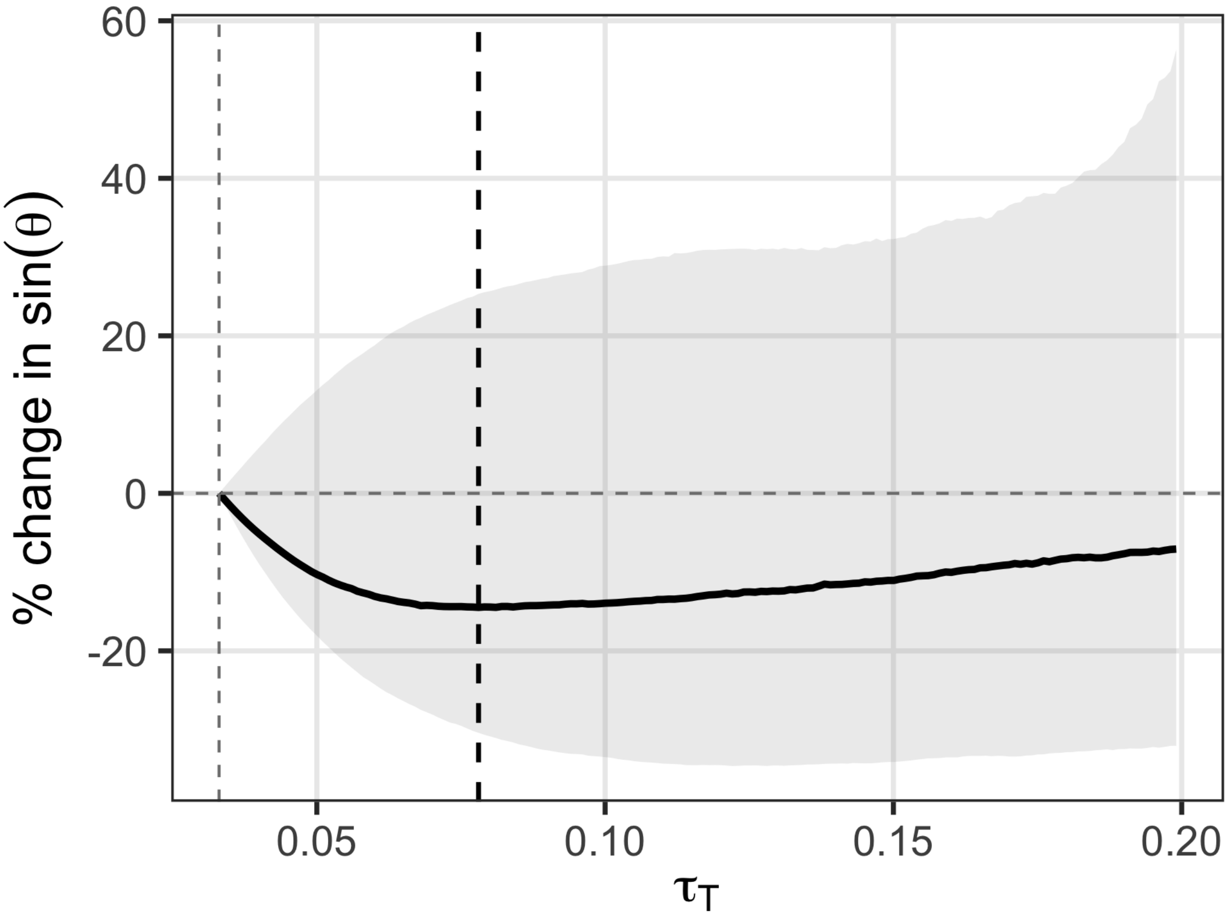
G.2 SOI-Adjusted

Figure G.5: Three-Instrument $\sin\theta$ Across Estimation Methods



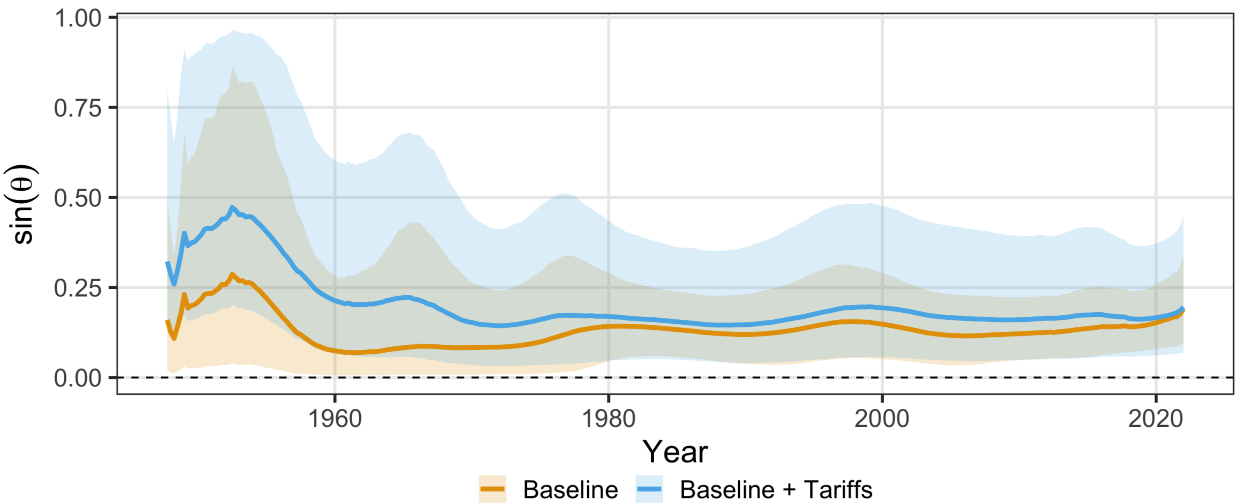
Note: This figure plots $\sin\theta$ for the three-instrument system (personal taxes, corporate taxes, and tariffs) across estimation methodologies. This figure reallocates s-corporates to the personal tax base.

Figure G.6: Counterfactual Tariff-for-Corporate-Tax Swap: Posterior Uncertainty



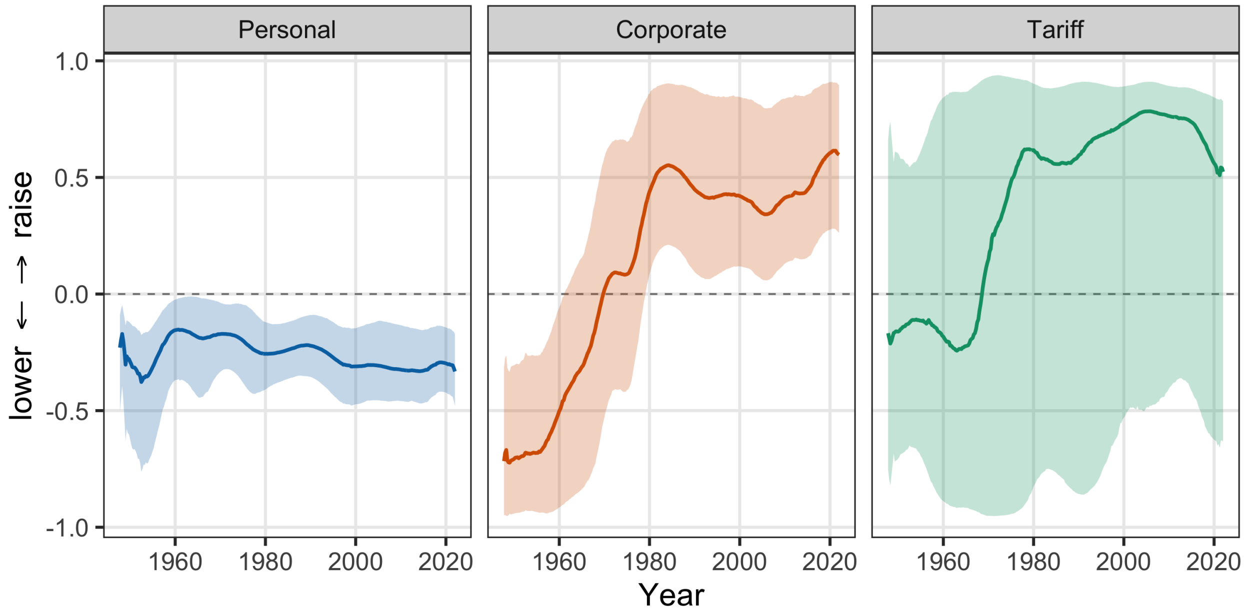
Note: This figure plots the percent change in sin θ from a revenue-neutral swap of tariffs for corporate taxes, starting from 2024Q4 rates and bases, under full domestic passthrough. The solid line uses median BLP elasticities; the shaded region shows the 90% credible interval from the full posterior. The thin dashed vertical line marks the 2024Q4 tariff rate; the thick dashed vertical line marks the optimum at median elasticities. This figure reallocates s-corporates to the personal tax base.

Figure G.7: Tax System Alignment: Two vs. Three Instruments



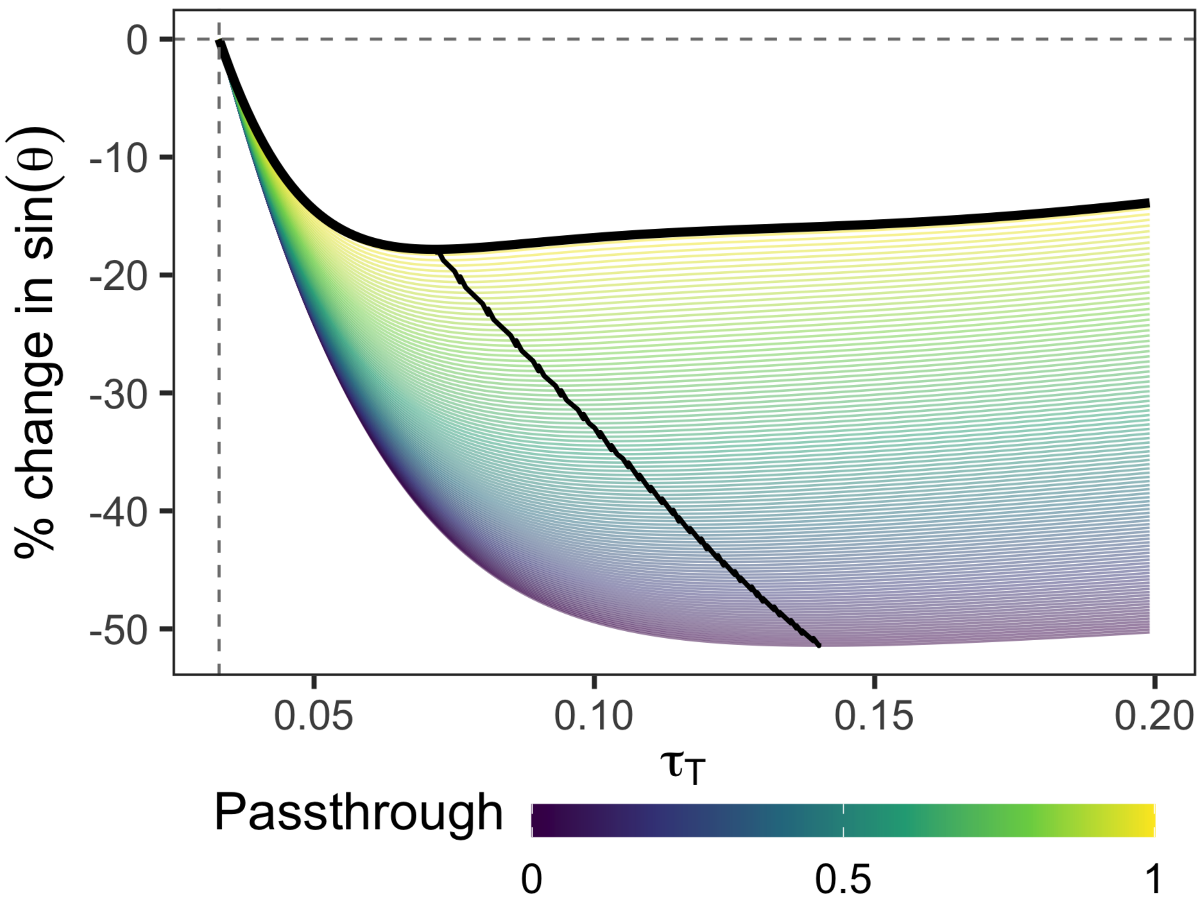
Note: This figure plots $\sin\theta$ for the two-instrument baseline (personal and corporate income taxes) and the three-instrument extension (adding tariffs) for the s-corp-adjusted sample. Both series use Bayesian local projection elasticities with 90% credible intervals.

Figure G.8: Optimal Reform Direction by Instrument



Note: Each panel plots the component of the optimal revenue-neutral reform direction d^* for one instrument, with 90% credible intervals. Positive values indicate the instrument should be raised; negative values indicate it should be lowered. The personal component is consistently negative and precisely estimated. The corporate component crosses zero around 1980, shifting from "lower" to "raise." The tariff component is consistently positive from the 1970s onward, though with wide credible intervals reflecting the imprecision of the tariff elasticities.

Figure G.9: Counterfactual Tariff-for-Corporate-Tax Swap



Note: This figure plots the percent change in $\sin\theta$ from a revenue-neutral swap of tariffs for corporate taxes, starting from 2024Q4 rates and bases, using median BLP elasticities from the SOI-adjusted sample. The x-axis is the counterfactual tariff rate τ_T; the dashed vertical line marks the 2024Q4 rate. Each curve corresponds to a different passthrough assumption. The dashed curve traces the optimum across passthrough values. Appendix Figure G.2 shows the full posterior uncertainty under full passthrough.

H Proofs

H.1 MVPF Equalization

The government chooses $\boldsymbol{\tau}$ to minimize social cost subject to a revenue requirement:

$$\min_{\boldsymbol{\tau}} W(\boldsymbol{\tau}) \quad \text{subject to} \quad R(\boldsymbol{\tau}) = G. \tag{A.15}$$

Proposition 2 (Ramsey first-order condition)**.** *Suppose an interior optimum $\boldsymbol{\tau}^* \in int(\mathcal{T})$ exists with $B_i(\boldsymbol{\tau}^*) > 0$ and $r_i(\boldsymbol{\tau}^*) > 0$ for all i, and constraint qualification holds, where i indexes components of $\boldsymbol{\tau} \in \mathbb{R}^{n(T+1)}$, i.e., instrument-date* *pairs. Then there exists $\lambda \in \mathbb{R}$ such that*

$$g(\boldsymbol{\tau}^*) = \lambda\, r(\boldsymbol{\tau}^*). \tag{A.16}$$

Equivalently, marginal values of public funds are equalized across instruments:

$$\mathit{MVPF}_i(\boldsymbol{\tau}^*) := \frac{g_i(\boldsymbol{\tau}^*)}{r_i(\boldsymbol{\tau}^*)} = \lambda \quad \textit{for all } i. \tag{A.17}$$

Proof. Form the Lagrangian $\mathcal{L}(\boldsymbol{\tau}, \lambda) = W(\boldsymbol{\tau}) - \lambda(R(\boldsymbol{\tau}) - G)$. Interior first-order conditions give $\nabla_{\boldsymbol{\tau}} \mathcal{L} = 0$, i.e., $g(\boldsymbol{\tau}^*) = \lambda\, r(\boldsymbol{\tau}^*)$. Dividing componentwise by $r_i(\boldsymbol{\tau}^*) > 0$ yields (A.17). □

H.2 Proof of Lemma 1

Proof. The problem is $\max_{d\boldsymbol{\tau}} -g^\top d\boldsymbol{\tau}$ subject to $r^\top d\boldsymbol{\tau} = 0$. Form the Lagrangian:

$$\mathcal{L}(d\boldsymbol{\tau}, \mu) = -g^\top d\boldsymbol{\tau} + \mu r^\top d\boldsymbol{\tau}. \tag{A.18}$$

First-order conditions yield $-g + \mu r = 0$. This cannot be satisfied for finite μ when MVPFs differ (which would require $g = \mu r$, i.e., g parallel to r). The constraint $r^\top d\boldsymbol{\tau} = 0$ implies $r^\top(-g + \mu r) = 0$, giving:

$$\mu = \frac{r^\top g}{r^\top r}. \tag{A.19}$$

Thus the optimal direction is:

$$d\boldsymbol{\tau}^* \propto -\left(g - \frac{r^\top g}{r^\top r} r\right) = -P_{\perp r}\, g. \tag{A.20}$$

Normalizing to a reform of size α (in percentage points) gives $d\boldsymbol{\tau}^* = -\alpha \frac{P_{\perp r} g}{\|P_{\perp r} g\|}$. □

H.3 Interpreting the Alignment Metric

Table H.1: Interpreting the Alignment Metric

$\cos\theta$	θ	$\sin\theta$	Interpretation
1.00	0°	0.00	Ramsey optimum
0.87	30°	0.50	Near optimum; 50% of gains RN
0.50	60°	0.87	Moderate misalignment; 87% of gains RN
0.00	90°	1.00	Orthogonal; all gains RN
−0.50	120°	0.87	Negative alignment; 87% of gains RN
−0.87	150°	0.50	Severe misalignment; 50% of gains RN
−1.00	180°	0.00	Perfect anti-alignment

Note: Revenue-neutral potential $\sin\theta$ is maximized at $\theta = 90^\circ$ and symmetric around it. For example, $\theta = 60^\circ$ and $\theta = 120^\circ$ both yield $\sin\theta \approx 0.87$ because the perpendicular component $\|g\|\sin\theta$ has equal magnitude in both cases.

I Evaluating Deficit-Financed Reforms

Real-world fiscal reforms rarely occur under strict revenue neutrality. Issuing government debt is equivalent to choosing the timing of taxation: the transversality condition ensures all debt must eventually be repaid, so deficit financing defers the tax burden rather than eliminating it. The question “should we issue more debt?” reduces to “should we lower taxes today and raise them tomorrow?” This appendix extends the projection framework to evaluate such intertemporal reforms.

I.1 The Dynamic Ramsey Problem with Dual Discounting

Imposing the transversality condition, the government chooses the tax path $\{\boldsymbol{\tau}_t\}_{t=0}^{\infty}$ to maximize:

$$\max_{\{\boldsymbol{\tau}_t\}} \sum_{t=0}^{\infty} \beta^t W_t(\boldsymbol{\tau}) \quad \text{subject to} \quad \sum_{t=0}^{\infty} q^t R_t(\boldsymbol{\tau}) = \sum_{t=0}^{\infty} q^t G_t + D_0, \tag{A.21}$$

where $\beta = (1+\rho)^{-1}$ is the social discount factor, $q = (1 + r_{\text{debt}})^{-1}$ is the fiscal discount factor, and D_0 is initial debt. The debt path $\{D_t\}$ is residual, determined by the budget identity $D_t = D_{t-1}(1 + r_{\text{debt}}) + G_t - R_t$. Debt is not a choice variable; the transversality condition eliminates it as a degree of freedom. The only choice is the tax path.

Because changing $\tau_{i,t}$ affects bases at all dates, the first-order condition involves the full intertemporal derivatives:

$$\underbrace{\sum_{s=0}^{\infty} \beta^s \frac{\partial W_s}{\partial \tau_{i,t}}}_{g_{i,t}} = \lambda \underbrace{\sum_{s=0}^{\infty} q^s \frac{\partial R_s}{\partial \tau_{i,t}}}_{\tilde{r}_{i,t}} \quad \text{for all } i, t, \tag{A.22}$$

where λ is the multiplier on the intertemporal budget constraint. The left-hand side is the welfare gradient $g_{i,t}$ from Section 2.1. The right-hand side is the revenue gradient discounted at the fiscal rate q rather than the social rate β. Rearranging:

$$\frac{g_{i,t}}{\tilde{r}_{i,t}} = \lambda \quad \text{for all } i, t. \tag{A.23}$$

When $r_{\text{debt}} = \rho$, $\tilde{r}_{i,t} = r_{i,t}$ and this reduces to MVPF equalization from Corollary 1. When $r_{\text{debt}} < \rho$ (as in recent decades), the fiscal constraint underweights future costs relative to social welfare, favoring back-loaded taxation. When $r_{\text{debt}} = \rho$, this recovers the tax smoothing result derived in Appendix J.2.

For local reforms around baseline $\bar{\boldsymbol{\tau}}$, the projection formula from Section 2.2 carries over with one modification: the revenue gradient must be discounted at q, not β.

$$d\boldsymbol{\tau}^* = -P_{\perp\tilde{r}}\, g, \tag{A.24}$$

where $P_{\perp\tilde{r}} = I - \tilde{r}(\tilde{r}^\top\tilde{r})^{-1}\tilde{r}^\top$. The welfare gradient uses social discounting because we evaluate welfare from society's perspective; the revenue constraint uses fiscal discounting because the government's ability to borrow and repay is governed by market interest rates. When the two rates coincide, equation (A.24) reduces to $d\boldsymbol{\tau}^* = -P_{\perp r}\, g$ from Section 2.

I.2 Bounding the Cost of Deferral

The Ramsey problem delivers the globally optimal tax path. In practice, we rarely observe complete reform paths—a deficit-financed tax cut specifies changes at $t = 0$ but leaves the repayment path implicit. Evaluating such incomplete reforms requires comparing the welfare cost of deferral against immediate revenue-neutral financing, without knowing when or how the debt will be repaid.

A deficit-financed reform at $t = 0$ creates a present-value fiscal gap $\text{FG} = \sum_{h=0}^{H} q^h [r_{C,h}\Delta\tau_C(h) + r_{L,h}\Delta\tau_L(h)]$. This gap must be closed by raising taxes in future periods. A repayment rule $\mathcal{P}$ specifies the path $\{\Delta\tau_{k,t}\}_{t\geq 1}$ satisfying:

$$\sum_{t\geq 1}\sum_{k} q^t r_{k,t}\Delta\tau_{k,t} = \text{FG}. \tag{A.25}$$

The marginal value of public funds for deferral under rule $\mathcal{P}$ is a weighted average of future MVPFs:

$$\text{MVPF}(\mathcal{P}) = \frac{\sum_{t\geq 1}\sum_{k}\beta^t g_{k,t}\Delta\tau_{k,t}}{\text{FG}} = \sum_{t,k}\pi_{t,k}(\mathcal{P})\left(\frac{\beta}{q}\right)^t \text{MVPF}_{k,t}, \tag{A.26}$$

where $\pi_{t,k}(\mathcal{P}) = q^t r_{k,t}\Delta\tau_{k,t}/\text{FG}$ are repayment shares summing to one and $\text{MVPF}_{k,t} = g_{k,t}/r_{k,t}$. The discount wedge $(\beta/q)^t$ appears because welfare and revenue are discounted at different rates. When $r_{\text{debt}} < \rho$, this wedge is below one for $t > 0$, mechanically lowering the effective cost of deferral: low borrowing rates make deferral cheap.

The repayment rule $\mathcal{P}$ is unobserved. But restricting attention to rules that only raise taxes ($\Delta\tau_{k,t} \geq 0$), the weights $\pi_{t,k}$ are non-negative and sum to one, so $\text{MVPF}(\mathcal{P})$ is a convex combination of the atoms $a_{t,k} := (\beta/q)^t\text{MVPF}_{k,t}$. For any repayment horizon $\mathcal{T}$ and set of instruments $\mathcal{K}$:

$$\min_{t\in\mathcal{T},k\in\mathcal{K}} a_{t,k} \;\leq\; \text{MVPF}(\mathcal{P}) \;\leq\; \max_{t\in\mathcal{T},k\in\mathcal{K}} a_{t,k}. \tag{A.27}$$

Let $\text{MVPF}_{j,0}$ denote the cost of the least distortionary instrument available for immediate revenue-neutral financing. If $\max_{t,k} a_{t,k} < \text{MVPF}_{j,0}$, deferral is unambiguously superior regardless of the repayment rule. If $\min_{t,k} a_{t,k} >$

$\text{MVPF}_{j,0}$, revenue-neutral financing dominates. Otherwise, the conclusion depends on the actual repayment path, and the width of the bounds $[\min a_{t,k}, \max a_{t,k}]$ measures how much that uncertainty matters.

J Classic Optimal Tax Results as Projections

This appendix demonstrates that the projection framework nests several canonical results from optimal tax theory. While the main text focuses on policy evaluation, these derivations show the framework is consistent with standard normative prescriptions under various structural assumptions. The first two subsections restate the Ramsey and Barro conditions in the notation and geometry of this paper. The final subsection shows how the zero-capital-tax result can be interpreted as a special configuration of the welfare and revenue gradients under its usual assumptions. The Atkinson–Stiglitz theorem admits a similar interpretation: under their assumptions, the welfare and revenue gradients are collinear within the commodity-tax subspace, so revenue-neutral commodity-tax reforms yield zero first-order welfare gains.

J.1 The Ramsey Inverse Elasticity Rule

The Ramsey problem seeks the tax structure that minimizes welfare costs subject to raising a fixed revenue requirement. At the optimum, the first-order conditions imply that marginal values of public funds are equalized across instruments. The projection framework recovers this result and clarifies its geometric interpretation.

Proposition 3 (Ramsey as Perfect Alignment)**.** *At a Ramsey optimum $\bar{\boldsymbol{\tau}}$, the welfare and revenue gradients are perfectly aligned:*

$$g(\bar{\boldsymbol{\tau}}) = \mu r(\bar{\boldsymbol{\tau}})$$

for some scalar $\mu > 0$. Equivalently, $\cos\theta = 1$ and the alignment is perfect.

Proof. The Ramsey problem is:

$$\min_{\boldsymbol{\tau}} W(\boldsymbol{\tau}) \quad \text{subject to} \quad R(\boldsymbol{\tau}) = \bar{R}.$$

The Lagrangian is $\mathcal{L} = W(\boldsymbol{\tau}) - \mu[R(\boldsymbol{\tau}) - \bar{R}]$. The first-order condition is:

$$\frac{\partial W}{\partial \tau_i} = \mu \frac{\partial R}{\partial \tau_i} \quad \text{for all } i,$$

which in vector form is $g(\bar{\boldsymbol{\tau}}) = \mu r(\bar{\boldsymbol{\tau}})$. Since g and r point in the same direction, the angle between them is zero: $\cos\theta = 1$. The sign $\mu > 0$ follows from the requirement that both $g_i > 0$ (positive tax bases) and $r_i > 0$ (the economy is below the Laffer peak) at the optimum. □

When gradients are perfectly aligned, the projection $P_{\perp r} g = 0$ is the zero vector: there is no component of the welfare gradient perpendicular to revenue. All welfare improvements require changing total revenue. This is the geometric interpretation of "no further revenue-neutral gains are available."

The Ramsey optimum also implies MVPF equalization. From $g_i = \mu r_i$, we have:

$$\text{MVPF}_i = \frac{g_i}{r_i} = \mu \quad \text{for all } i.$$

All MVPFs equal the Lagrange multiplier μ, the shadow cost of the revenue constraint.

The inverse elasticity rule. Under additional structure, the Ramsey formula yields the classic inverse elasticity result. Suppose:

1. Utilitarian welfare: $g_i = B_i$ (the tax base).
2. Ad valorem taxation with instrument-specific rates τ_i.
3. No cross-base spillovers: $\Lambda_i = 0$ (partial equilibrium).

Then from equation (3), the revenue gradient is:

$$r_i = B_i \left[1 - \frac{\tau_i}{1-\tau_i}\varepsilon_{ii}\right].$$

MVPF equalization $g_i/r_i = \mu$ implies:

$$\frac{B_i}{B_i\left[1 - \frac{\tau_i}{1-\tau_i}\varepsilon_{ii}\right]} = \mu \quad \Rightarrow \quad \frac{\tau_i}{1-\tau_i}\varepsilon_{ii} = 1 - \frac{1}{\mu}.$$

Since the right-hand side is constant across i, we have $\frac{\tau_i}{1-\tau_i}\varepsilon_{ii} = \text{constant}$. For small tax rates, this implies:

$$\tau_i \propto \frac{1}{\varepsilon_{ii}}.$$

Taxes should be inversely proportional to elasticities. Highly elastic bases receive low taxes; inelastic bases receive high taxes. This is Ramsey's (1927) classic result, recovered as the shadow of perfect alignment under no cross-base spillovers.

J.2 Barro Tax Smoothing

Barro (1979) shows that when the government's borrowing rate equals the social discount rate, optimal tax policy smooths distortions over time. This result emerges directly from the intertemporal projection framework in Section 2.

Proposition 4 (Barro as Temporal MVPF Equalization)**.** *Consider the dynamic Ramsey problem from equation (A.21). Suppose (1) the government's borrowing rate equals the social discount rate: $r_{debt} = \rho$, so $q = \beta$; and (2) tax bases and behavioral responses are stationary: $B_{i,t} = B_i$ and $\varepsilon_{ik,t} = \varepsilon_{ik}$ for all t.*

Then at the Ramsey optimum, MVPFs are constant over time:

$$MVPF_{i,t} = MVPF_{i,s} \quad \text{for all } t, s.$$

Proof. From equation (A.22), the first-order condition for the dynamic Ramsey problem is:

$$\beta^t \frac{\partial W_t}{\partial \tau_{i,t}} = \lambda q^t \frac{\partial R_t}{\partial \tau_{i,t}}.$$

When $q = \beta$, this simplifies to:

$$\frac{\partial W_t}{\partial \tau_{i,t}} = \lambda \frac{\partial R_t}{\partial \tau_{i,t}}.$$

The discount factors cancel. Under stationarity, $\partial W_t/\partial\tau_{i,t} = g_i$ and $\partial R_t/\partial\tau_{i,t} = r_i$ are constant across t. Therefore:

$$\text{MVPF}_{i,t} = \frac{g_i}{r_i} = \lambda \quad \text{for all } t.$$

MVPFs are equalized not only across instruments (the static Ramsey condition) but also across time. □

This is Barro's tax smoothing result: optimal policy equalizes distortions intertemporally. When $r_{\text{debt}} = \rho$, there is no wedge between fiscal and social accounting, so the government should smooth MVPFs over time just as it smooths them across instruments. Temporary shocks to revenue needs should be financed primarily by debt, with small permanent tax adjustments to service the debt, maintaining equalized MVPFs across time. Permanent shocks should be financed by permanent tax increases.

When $r_{\text{debt}} \neq \rho$, the result no longer holds. From equation (A.23):

$$\text{MVPF}_{i,t} = \lambda \left(\frac{1 + r_{\text{debt}}}{1 + \rho}\right)^t.$$

If $r_{\text{debt}} < \rho$ (as in recent decades), the government borrows cheaply relative to its social discount rate, creating an incentive to back-load taxation. MVPFs should fall over time at rate $(1 + r_{\text{debt}})/(1 + \rho) < 1$, reflecting lower marginal costs of future financing. Conversely, if $r_{\text{debt}} > \rho$, MVPFs should rise over time, favoring front-loaded taxation.

J.3 Zero Capital Taxation as a Limiting Case

Chamley (1986) and Chari, Nicolini, and Teles (2020) show that under standard macroeconomic assumptions (infinitely lived agents, perfect capital markets, homothetic preferences, additively separable preferences over time), the optimal long-run tax on capital is zero. In those environments, a permanent change in the capital tax rate induces such a large long-run response of the capital stock that the present-value revenue raised by the tax tends to zero. The projection framework makes this logic transparent by expressing it in terms of the present-value revenue gradient.

In the notation of Section 2, consider a permanent increase in the capital tax $\tau_{K,0}$. Its present-value revenue effect is

$$r_{K,0}^{PV} = \sum_{t=0}^{\infty} \beta^t \frac{\partial R_t}{\partial \tau_{K,0}},$$

and the present-value marginal value of public funds for capital taxation is

$$\text{MVPF}_K^{PV} = \frac{g_{K,0}^{PV}}{r_{K,0}^{PV}}.$$

In a Chamley-type environment, long-run capital supply is effectively infinitely elastic: a permanent increase in τ_K causes the capital tax base to shrink so much over time that

$$r_{K,0}^{PV} \to 0.$$

Given a finite welfare gradient $g_{K,0}^{PV}$, this implies $\text{MVPF}_K^{PV} \to \infty$. In the projection formula, instruments with very high MVPFs are pushed toward their lower bounds. When the present-value revenue gradient for capital taxation vanishes, the optimal reform direction drives τ_K toward zero. The familiar zero-capital-tax result thus appears as a limiting case of the general projection framework, corresponding to an extreme configuration of the present-value revenue gradient.

This limiting case is not generic. If the present-value elasticity of capital supply is large but finite—because of adjustment costs, firm-specific capital, borrowing constraints, or heterogeneous discount rates—then $r_{K,0}^{PV}$ remains strictly positive and so does MVPF_K^{PV}. Straub and Werning (2020) show that relaxing the standard assumptions can overturn the zero-capital-tax result. In the projection framework, this fragility appears as sensitivity of the optimal capital tax to the shape of r_K^{PV}: whenever $r_{K,0}^{PV}$ does not collapse to zero, the optimal reform sets a finite capital tax rate determined by the same MVPF equalization logic as for any other instrument.